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NACHIP Decision Support Tool – User Manual

Version 1.1.0

Decision support tool for modelling infrastructure of hydrogen-based renewable energy hubs.

November 9, 2026

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Preface

This is the documentation for version 1.1.0. (Version history). For download link and other useful information see the main project website.

This user manual introduces the **Decision Support Tool (DST)**, a practical and user-friendly solution for planning and sizing investments in integrated energy systems based on green hydrogen technologies. The tool enables users to evaluate how locally available renewable energy and other resources can be optimally converted into green hydrogen and efficiently integrated with electricity, heating, and water networks. By quantifying the costs, revenues, and optimal sizes of every component, it facilitates the monetization of new technologies and supports the calibration of the hydrogen value chain, from energy supply to end use.

The DST delivers techno-economic insights, including optimal system configuration, CAPEX/OPEX estimates, and key financial indicators such as NPV, IRR, and ROI. Designed for municipalities, energy planners, development agencies, and investors, the DST supports informed, data-driven decisions for developing cost-effective and future-ready hydrogen hubs.

Key Features of the DST:

- **Green hydrogen focus**
  Convert local and regional renewable energy surpluses into clean, versatile hydrogen.
- **Multi-grid integration**
  Seamlessly plan interactions across electricity, gas, heating, and water networks.
- **Comprehensive system modelling**
  Understand every component and how they interact within the overall system.
- **Techno-economic analysis**
  Get CAPEX/OPEX estimates and financial indicators like NPV, IRR, ROI, and payoff period.

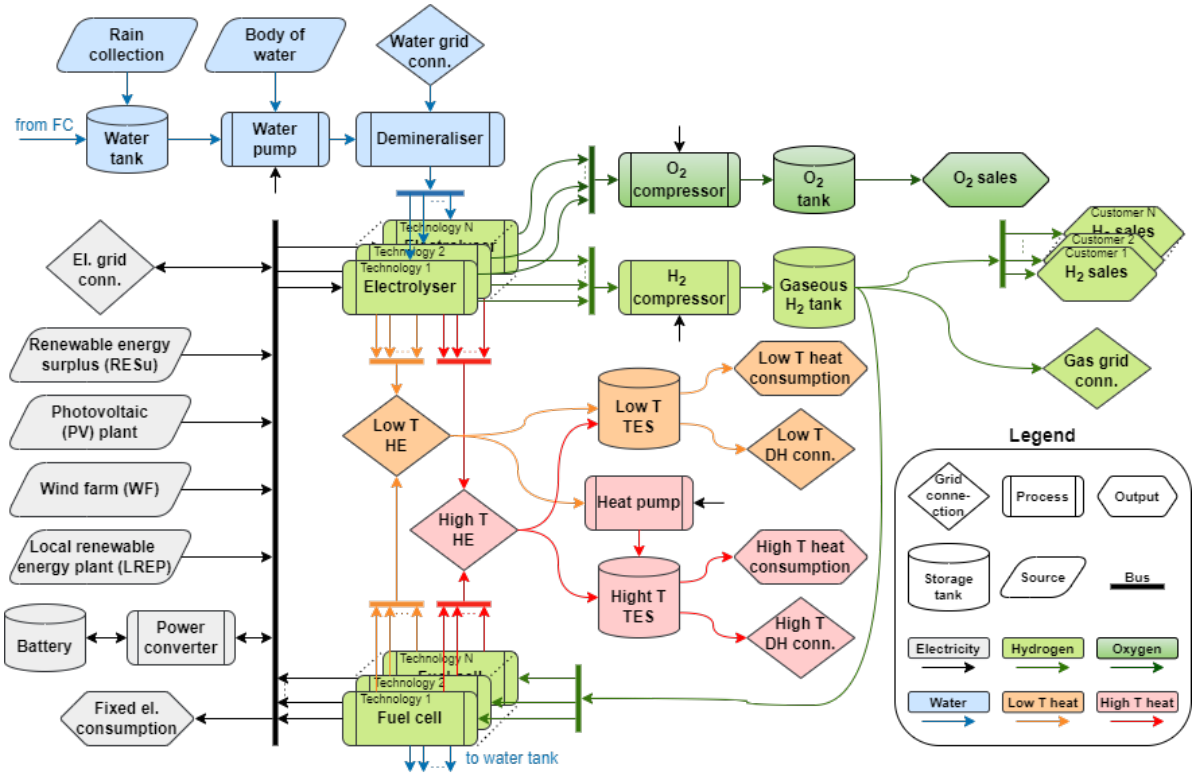


Figure 1: Diagram of the overall system configuration that can be modelled.

- **Optimal sizing & operation**
  Receive recommendations for sizing electrolysers, fuel cells, photovoltaics, wind farms, energy storage, and more, plus operation schedules.
- **Intuitive graphical interface**
  User-friendly GUI for easy input, visualization, and interpretation of results.
- **Built on proven optimization tools**
  Leverages and expands the DanuP-2-Gas Optimization Tool for advanced system planning.
- **Targeted user support**
  Designed for municipalities, energy planners, development agencies, and private investors.
- **Scenario analysis capability**
  Explore dynamic planning scenarios to optimize infrastructure, investment, and performance.

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Further development

To report issues, request changes, ask questions, or provide feedback, email us to: ivan.grabic@fer.hr; marko.kovacevic@fer.hr; filip.rukavina@fer.hr; mario.vasak@fer.hr, with indication whether it is about:

- A **bug** – the tool does not operate as it should. Make sure to:
  - tell us what you expected to happen,
  - tell us what actually happened,
  - tell us what version of the DST you use,
  - send us screenshots of the problem,
  - send us the project file (.NACHIP) where it happens.
- A **change** – you would like to see an improvement, whether it is to change an existing feature or adding a new one. Make sure you explain it in your email and tell us which version of the tool you are currently using.
- **Questions** – just ask if you have questions or need assistance with tool usage.
- **Documentation** – let us know if something is wrong or missing in the documentation.

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1 Introduction

The Decision Support Tool (DST) helps to design and evaluate future green energy systems. It answers practical questions such as:

- How large should a solar plant, battery, or electrolyser be?
- Is it financially viable to build a hydrogen hub at this location?
- What would the investment cost be?
- How long would it take to recover the investment?
- How would the system operate over the year?

The DST automatically calculates the **most cost-effective system design and operation** based on the technologies you select and the data you provide.

The tool focuses especially on hydrogen-based energy hubs (e.g., electrolysers, hydrogen storage, fuel cells), but it also includes renewable electricity generation (solar and wind), batteries, grid interaction, heat recovery, and water supply.

In addition to technical results, the DST provides a **full techno-economic evaluation**, including:

- Investment cost (CAPEX)
- Operational cost (OPEX)
- Payback period
- Net Present Value (NPV)
- Internal Rate of Return (IRR)
- Return on Investment (ROI)
- Levelized Cost of Electricity (LCOE)
- Levelized Cost of Hydrogen (LCOH)

The goal of the tool is to support informed investment and planning decisions.

1.1 What does “optimal size” mean?

An optimal size is the capacity of each system component that results in the lowest total cost while meeting all constraints.

Examples:

- For a PV plant, the size is its installed peak power [kW_p].
- For a battery, the size is its energy capacity [kWh].
- For a hydrogen tank, the size is the amount of hydrogen it can store [kg].

The DST determines the best size for each selected component. In some cases, the optimal size may be zero, meaning that the optimal solution does not include that component.

1.2 What is operational schedule?

An operation schedule describes how each component operates over time during the simulation period (typically one year).

For example:

- An electrolyser may run at full capacity during hours of low electricity prices, and at partial capacity during high electricity prices.
- A battery may charge during midday solar peaks and discharge in the evening.
- Hydrogen may be stored in summer and used in winter.

If the model uses hourly data, the operation schedule consists of 8,760 hourly values (24 hours $\times$ 365 days). This helps planners understand not only what to build, but also how the system will behave in practice.

1.3 What can the DST model?

The tool can model integrated energy systems consisting of interacting technologies. It allows you to combine and evaluate:

Electricity generation and supply

- Photovoltaic (PV) plants
- Wind farms
- Wave energy converters
- Grid connection
- Electricity supply
- Renewable energy surplus
- Fixed electricity demand

Storage

- Battery energy storage systems (BESS)
- Hydrogen storage tanks
- Oxygen storage tanks

- Thermal energy storage (TES)

Hydrogen production and use

- Electrolysers
- Fuel cells
- Hydrogen compressors
- Hydrogen sales (market and gas grid injection)
- Differentiation between green and non-green hydrogen

Oxygen as a byproduct

- Possibility of using oxygen from the electrolyzer or releasing it into the atmosphere
- Oxygen compressors and tanks
- Oxygen sales

Heat recovery and use

- Waste heat from electrolysers and fuel cells
- Heat exchangers
- Heat pumps
- District heating connection
- Local heat consumption
- Differentiation between low- and high-temperature heat

Water supply and treatment

- Water grid
- Rainwater
- Surface water
- Demineralization and pumping systems

The tool evaluates all selected technologies together and determines the most economically efficient system configuration.

1.4 Basic workflow

The DST is organized by projects and scenarios. Typical workflow:

1. Create a new project (or open an existing one)
2. Add a new scenario
3. Select system components and enter their parameters
4. Run the optimization
5. Review and analyze the results

6. Compare scenarios if needed

Scenarios allow you to compare different technology combinations, cost assumptions, or policy conditions.

1.5 How does the optimization work?

The DST minimizes the **total annual system cost**.

Total cost includes:

- **Investment costs (CAPEX)** Equipment purchase, construction, installation.
- **Maintenance and degradation costs** Inspections, servicing, insurance, component replacement, etc.
- **Operational costs** Electricity purchases, grid fees, water supply.
- **Revenue** Selling electricity, hydrogen, oxygen, heat.

The tool simultaneously determines:

- The optimal system structure
- The optimal size of each component
- The optimal hourly operation over the year

This ensures that investment and operation decisions are consistent.

1.6 Features

1.6.1 Investment and financial constraints

The user can:

- Set a maximum total investment budget
- Set a maximum acceptable payback period

The model ensures that solutions respect these limits. If the calculated payback period is shorter than the maximum allowed, it means the constraint was not binding and the project performs better than required.

1.6.2 Simulation settings

These settings allow balancing precision and computational effort.

- The default simulation period is one full year.
- The time resolution (e.g., hourly or daily) can be adjusted.
- A shorter simulation period or larger time steps reduce calculation time but may slightly reduce accuracy.

1.6.3 Existing infrastructure

If certain components already exist at the site (e.g., grid connection, PV plant, storage), the tool can:

- Keep the existing capacity
- Optimize only the expansion
- Optimize joint operation of existing and new components

1.6.4 Additional modeling features

- Gas pressure levels are considered throughout the hydrogen system.
- Multiple electrolyser and fuel cell technologies can be evaluated simultaneously.
- The tool distinguishes between **green and non-green electricity**.
- The tool distinguishes between **green and non-green hydrogen**.

2 Getting started

2.1 System requirements

The NACHIP Decision Support Tool (DST) is a Windows desktop application.

- **Operating system**: Windows 10 or newer (64-bit)
- **Processor (CPU)**: No strict requirements
- **Memory (RAM)**: Minimum 16 GB, 32 GB or more recommended
- **Graphics (GPU)**: Not required for computation

2.1.1 Network

An internet connection is required for:

- location-based data,
- PV production data,
- meteorological data (e.g., precipitation).

The application can run offline, but these features will be unavailable.

2.2 Simple example

This section guides you through creating a simple project with one scenario containing:

- A **Photovoltaic (PV) system**,
- A **Battery Energy Storage System (BESS)**, and
- A fixed electricity consumption.

By following these steps, you will learn the basic workflow of the DST: creating a project, configuring components, running optimization, and viewing results.

Step 1: Launch the application

1. Launch the NACHIP Decision Support Tool.
2. You will see the welcome window with options to create a new project, open an existing project, or check the user manual.

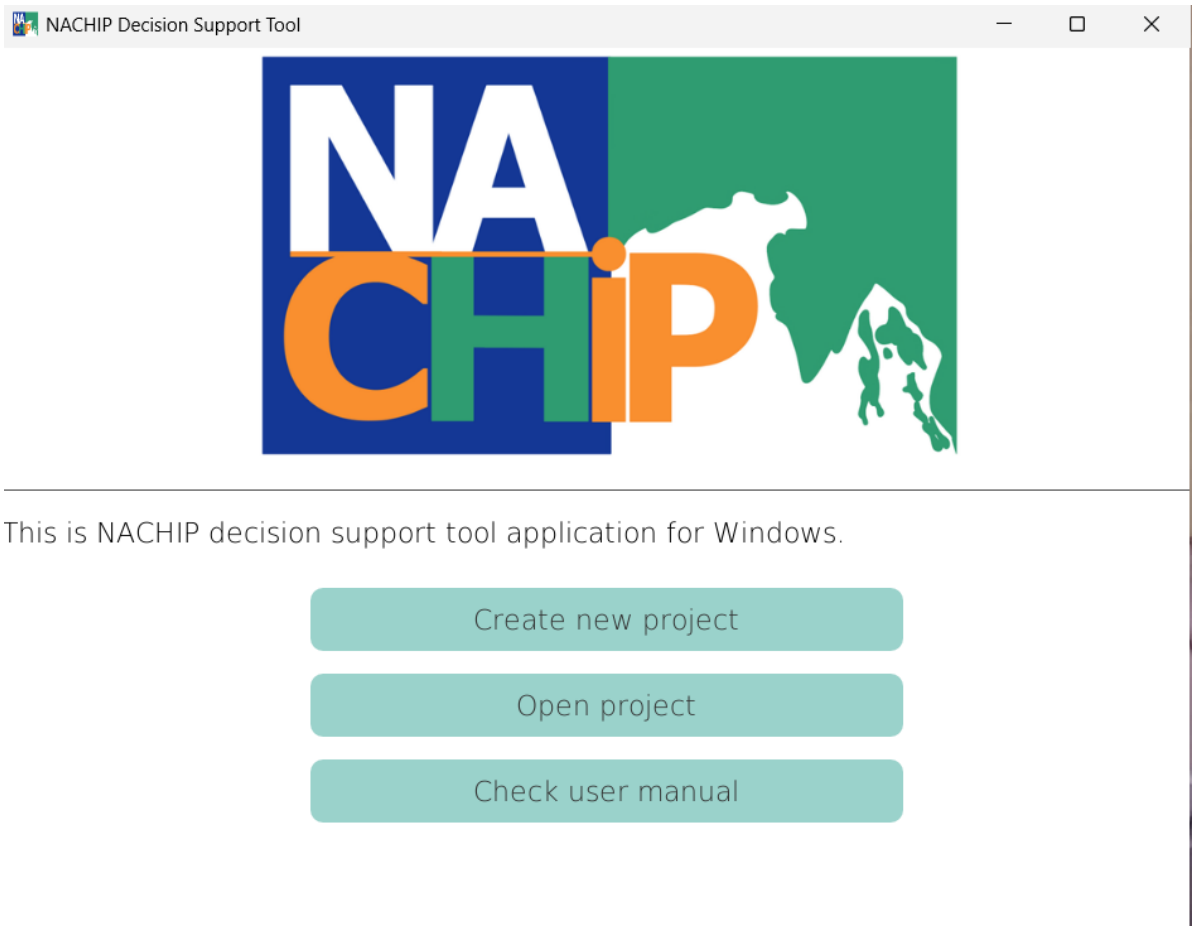


Figure 2.1: Welcome window

Step 2: Create a new project

1. Click **Create new project** in the welcome window.
2. The *New Project* popup window will appear.
3. Enter a project name (e.g., `My first project`).
4. Optionally, change the project location by clicking **Browse...** and selecting a folder.
5. Click **Create project** to confirm.

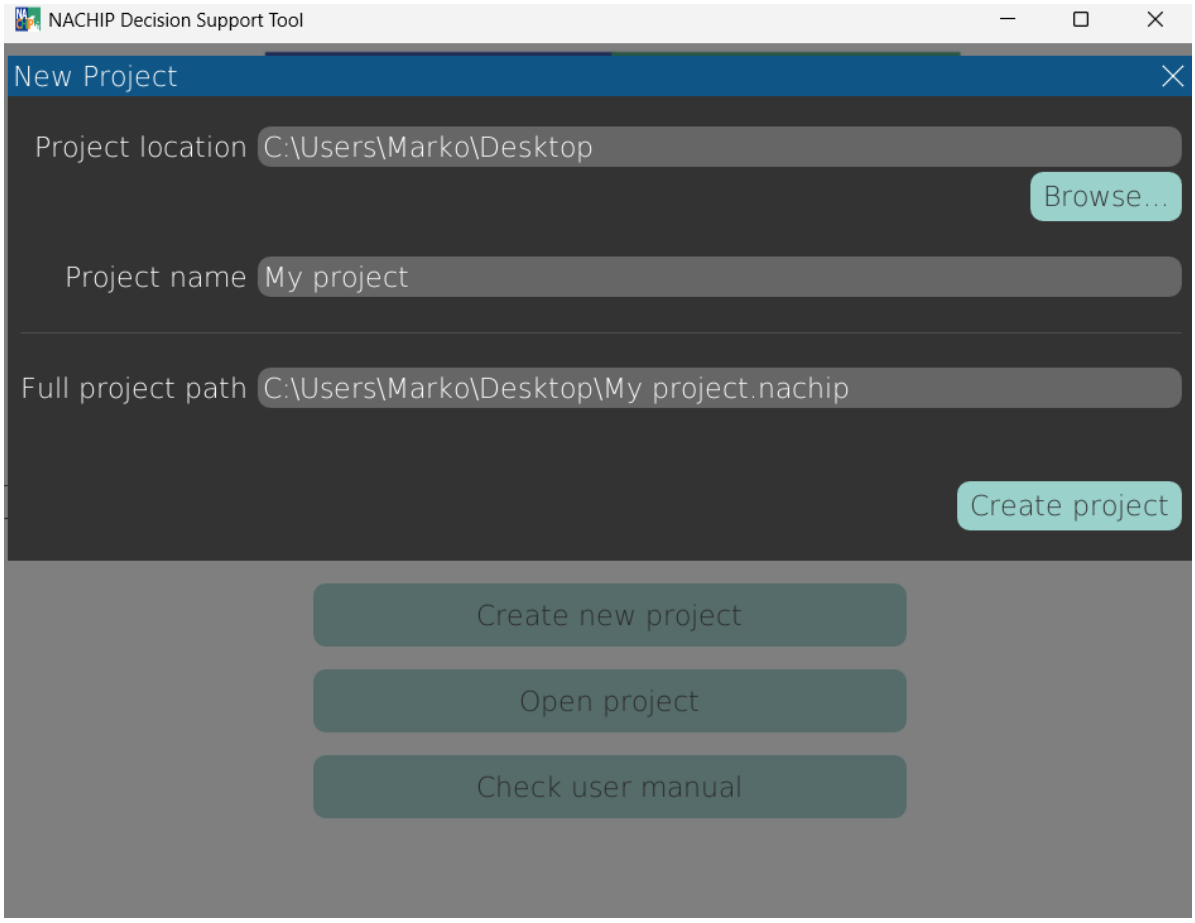


Figure 2.2: Creating a new project

A new project opens in the main window.

Step 3: Create a new scenario

1. Click **Add new scenario** in the sidebar (left panel).
2. A default scenario named `Scenario 1` will appear.
3. Expand the scenario and click **System structure** to view the system configuration interface.

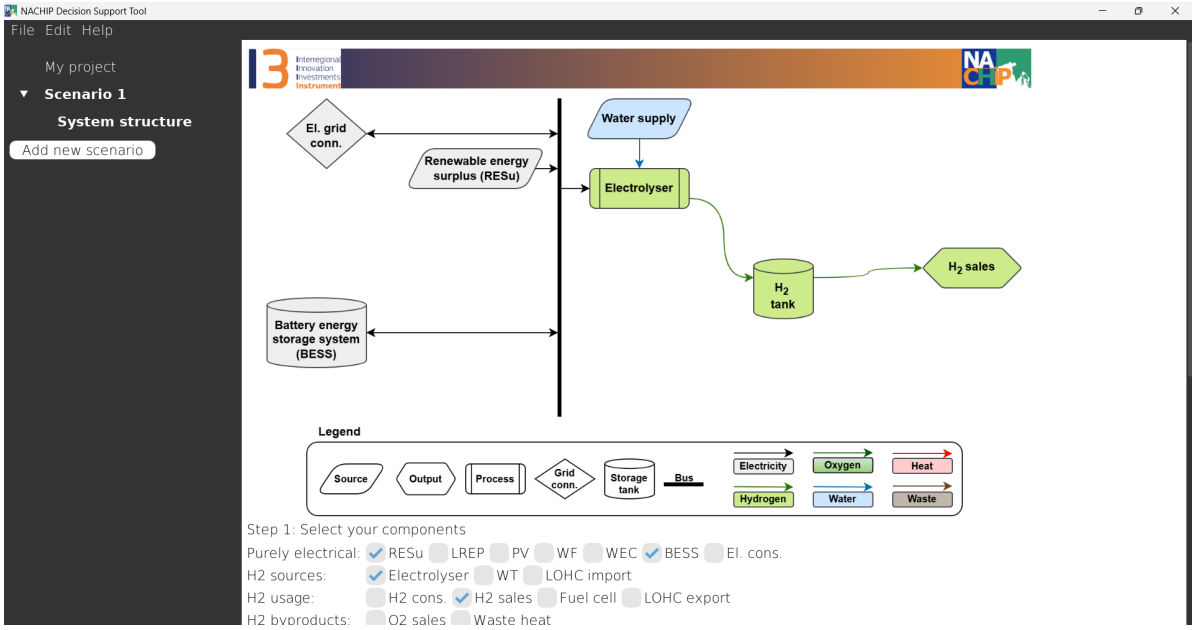


Figure 2.3: Project with one scenario

The *Main window* now shows the system configuration interface.

Step 4: Select components

In this example we will consider:

- PV system,
- BESS,
- fixed electricity consumption.

The scenario is initialized with some components already selected. To configure our desired configuration:

1. Deselect components that are not needed (e.g., Electrolyser, RESu if active).
2. Ensure the following components are selected: PV, BESS, El. cons.

The checkboxes are arranged in four rows: *Purely electrical* components (RESu, LREP, PV, WF, Leonitus, BESS, El. cons.), *H2 sources* (Electrolyser, WT, LOHC import), *H2 usage* (H2 cons., H2 sales, Fuel cell, LOHC export) and *H2 byproducts* (O2 sales, Waste heat). Unchecking a hydrogen source also unchecks and hides the hydrogen components that depend on it. See Chapter 4 for details.

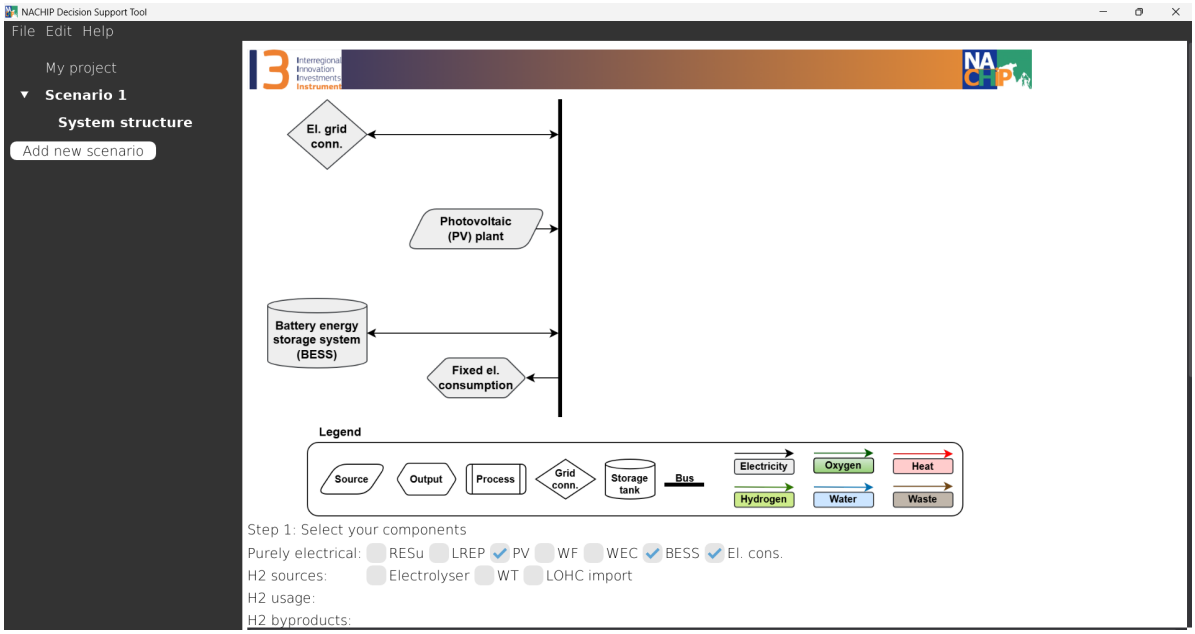


Figure 2.4: Selecting components for the configuration

Step 5: Configure location

The **Location** component defines where the energy hub is located. This affects solar irradiance data and other location-specific parameters.

1. Expand the **Location** section.
2. Search for *Zagreb*.
3. Select **Zagreb** in Municipality dropdown.
4. Verify that your coordintes are: 45.813, 15.977 and that country is Croatia.



Figure 2.5: Choosing location

Step 6: Configure electrical grid

The **Electrical grid connection** component defines electricity prices and connection conditions. Since the location is in Croatia, electricity tariff models from Croatia became available to use. To configure:

1. Expand the **Electrical grid connection** section.
2. Expand **Grid connection** tree node.
3. For **Use predefined connection type** select `SN - Bijeli - Zagreb`.
4. Check **Grid connection already exists**.
5. Set **Existing connection size** to 270 kW.
6. Expand **Electricity supply** tree node.
7. For **Use predefined supply models** select `HEP elektra - SN bijeli`.

Step 7: Configure PV system

The **PV** component represents solar energy generation capacity. In this example, we only define the available installation area.

1. Expand the **Photovoltaic (PV) system** section
2. Click **Add new array**, then click **Done** in the pop-up window.
3. You will see a default PV array. Configure the following:

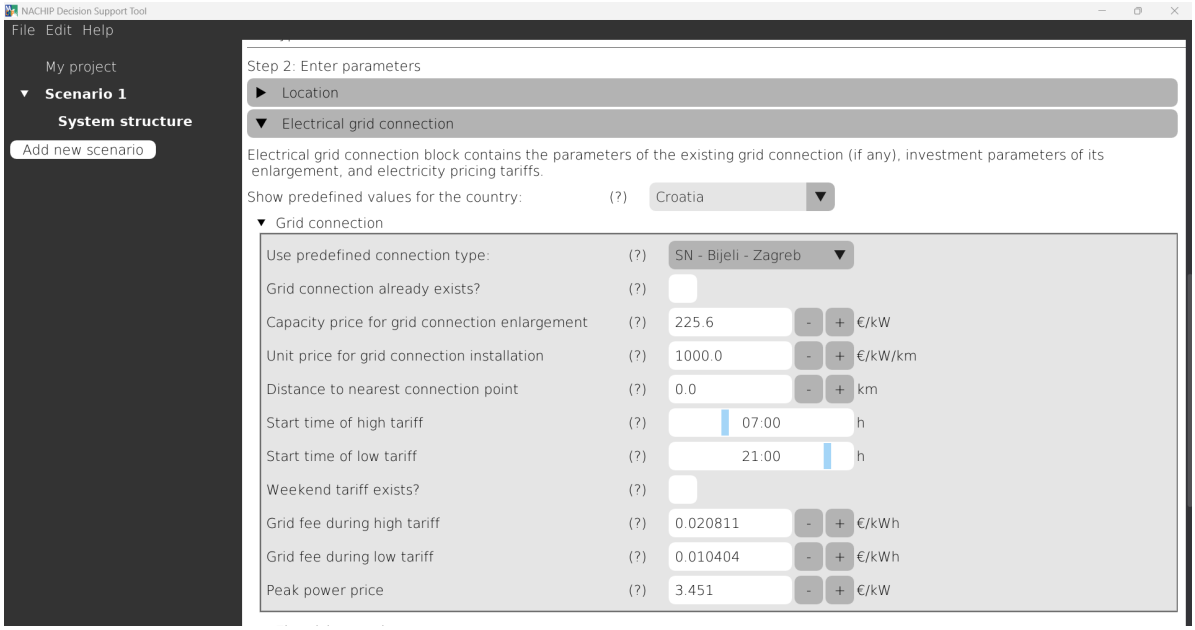


Figure 2.6: Configuring the Electrical grid connection

- **Available area**: Enter the area of 1000 m^2

Step 8: Configure BESS

The **Battery Energy Storage System (BESS)** stores excess solar energy for later use. Since the default parameters for BESS are already entered we can just skip BESS component.

Step 9: Configure electricity consumption

1. Expand the **Electricity consumption** section.
2. Click **Add new consumer**, then click **Done** in the pop-up window.
3. Select building type:
   - **Large hotel**
4. Enter:
   - **Annual electricity consumption**: 1000 MWh

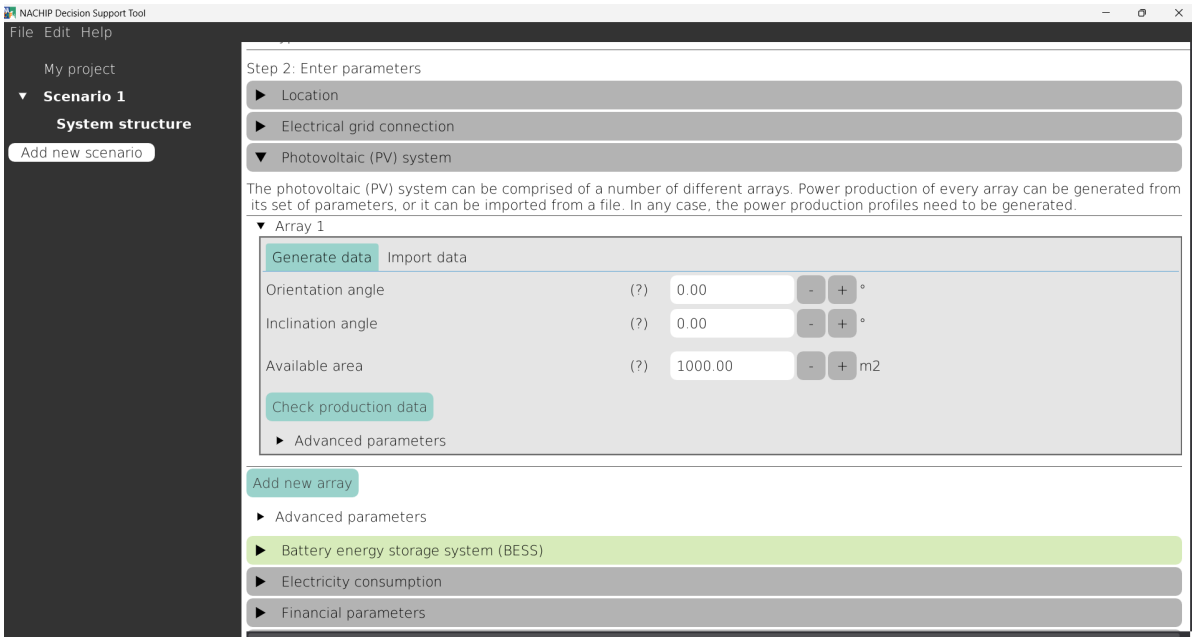


Figure 2.7: Configuring the PV system

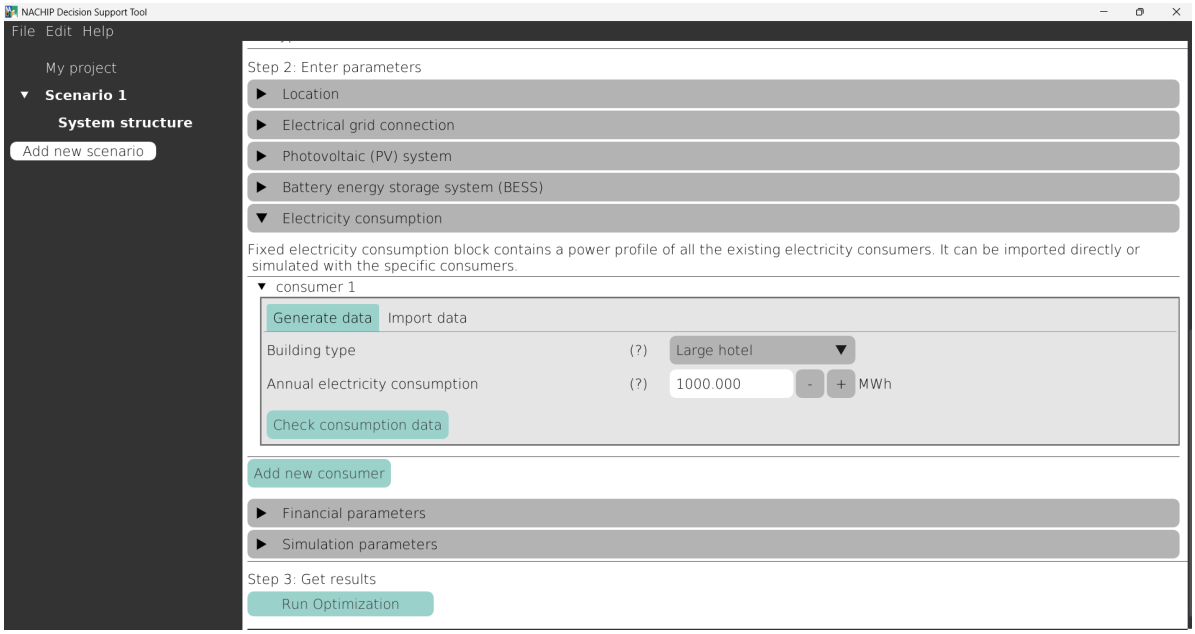


Figure 2.8: Configuring Electricity consumption

Step 10: Financial parameters

In this example we will set subsidy to 50%:

1. Click on **Apply subsidy to all components**
2. Enter 50% in **Investment subsidy** field

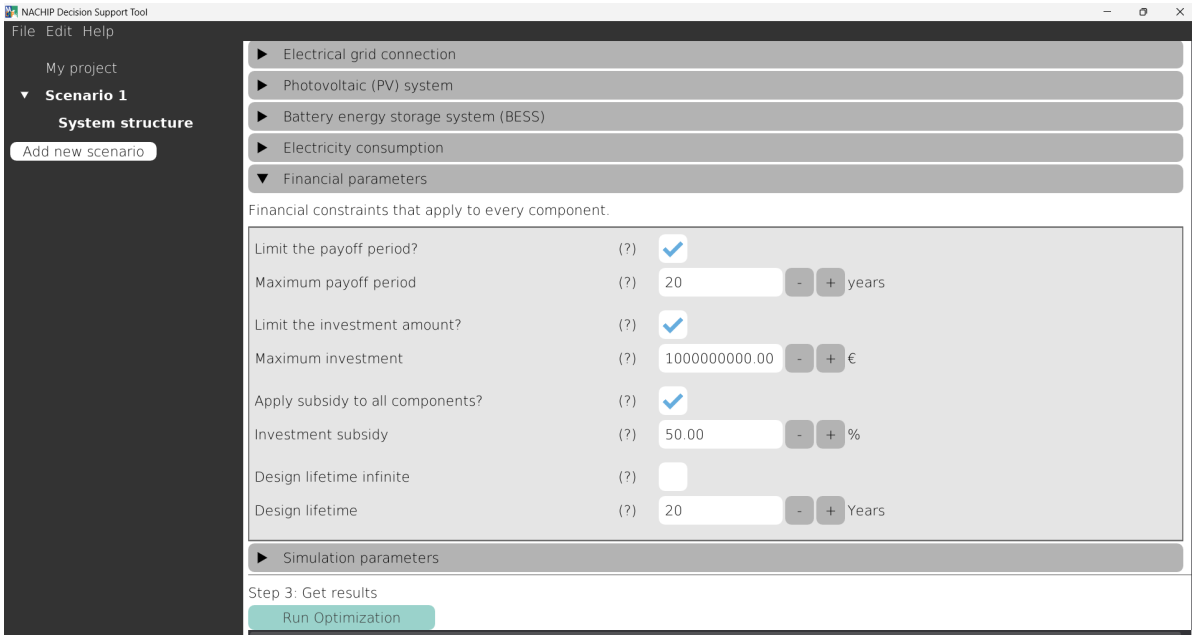


Figure 2.9: Adding subsidy to all components

Step 11: Configure simulation parameters

The **Simulation** component defines the optimization timeframe and time resolution. To reduce computation time, we shorten the simulation period:

1. Expand the **Simulation** section.
2. Configure:
    - **Start date**: June 1st
    - **End date**: June 30th

Step 12: Run optimization

Once all components are configured:

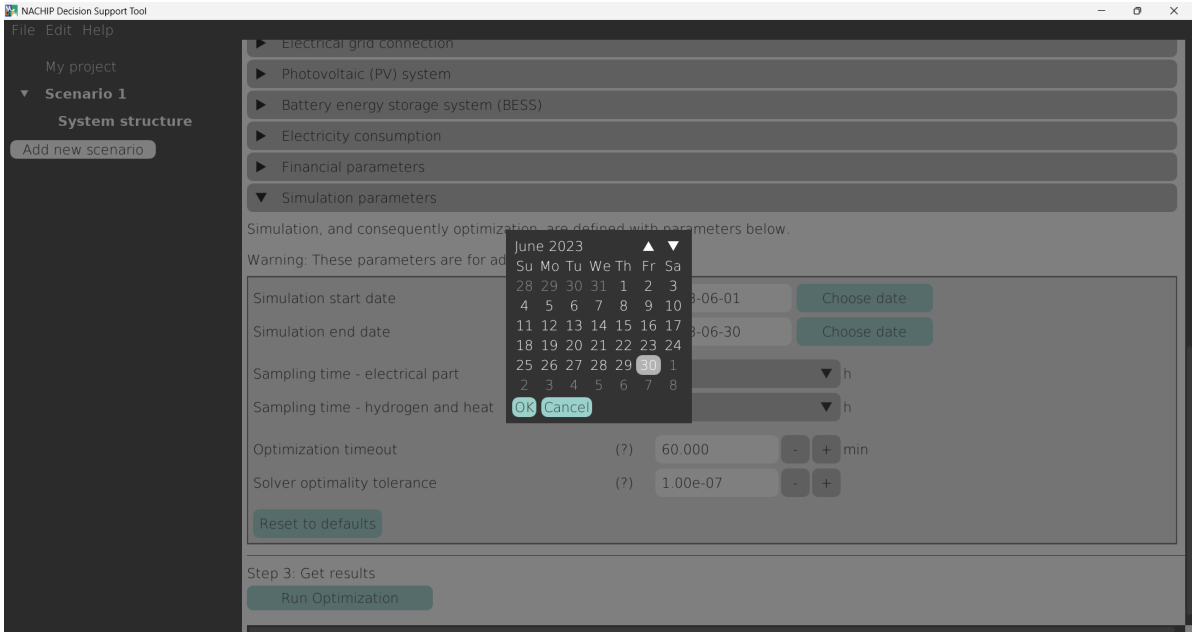


Figure 2.10: Configuring simulation period

1. Scroll to the bottom of the System structure view.
2. Click **Run Optimization** to start the optimization process.
3. A progress indicator will appear during computation.
4. After completion, the **Results** node will appear in the Sidebar under the scenario.

Step 13: View Results

1. In the Sidebar, click **Results** under the scenario.

The Main window now displays the calculated optimal system configuration, its optimal operation, and financial indicators.

2.3 Next steps

You have now created a project and optimized a scenario with a simple PV + BESS system. Based on your location's solar resources and your specified constraints, the optimization algorithm has determined:

- Optimal system sizes
- Operational schedule

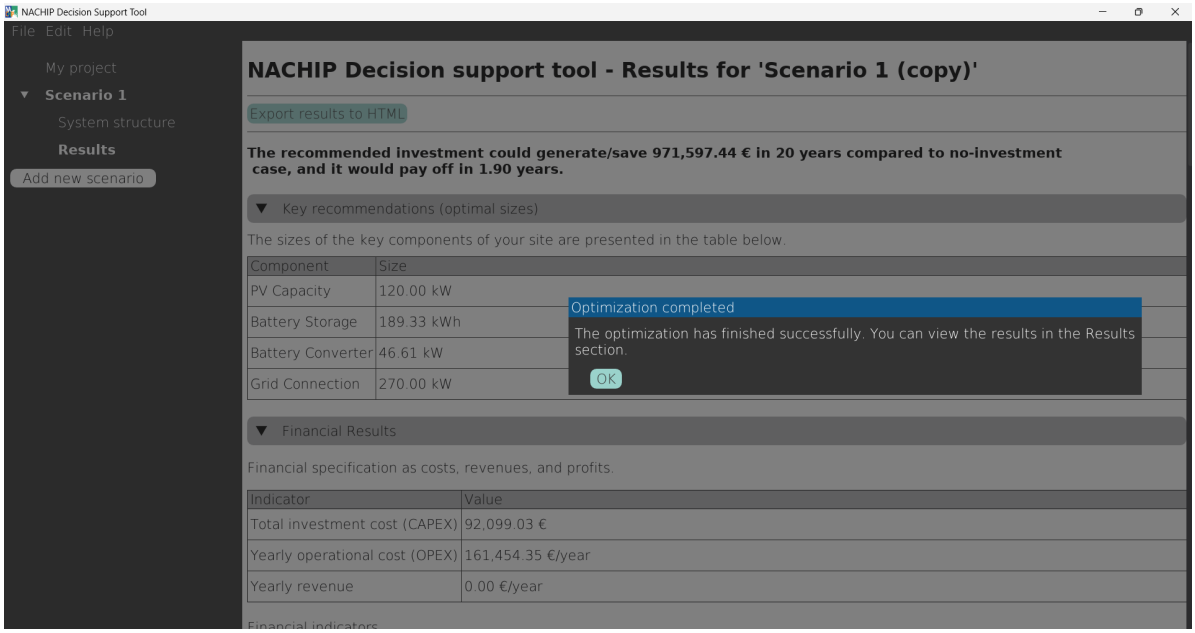


Figure 2.11: Successful optimization

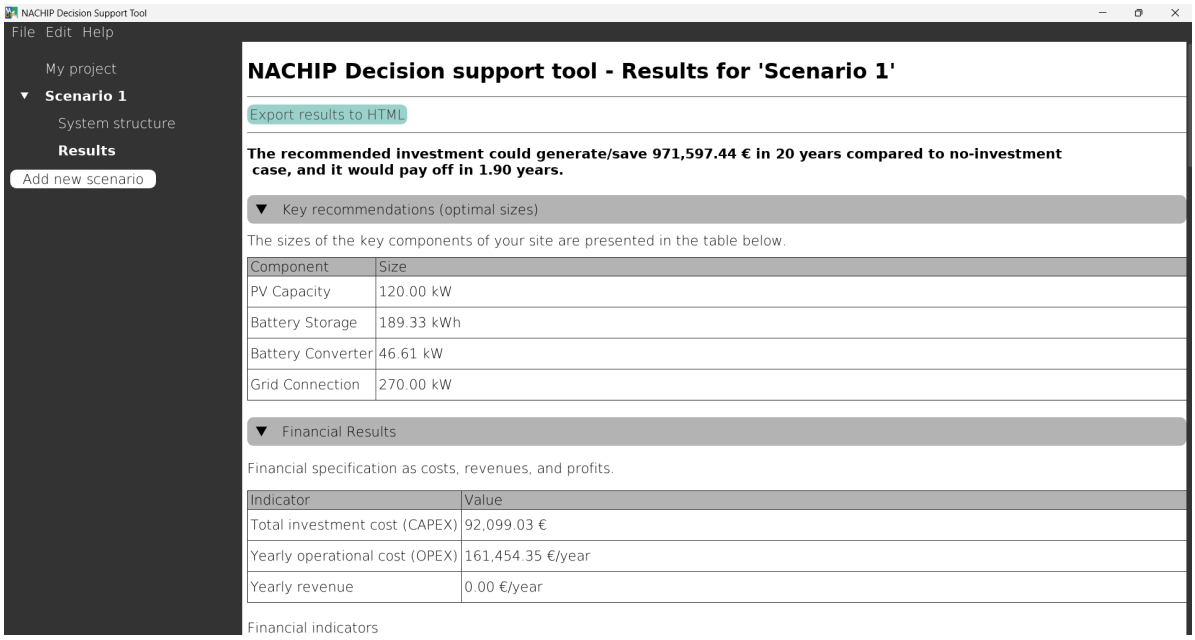


Figure 2.12: Showing results of the optimization

- Financial performance

After optimization, the original scenario becomes locked and inputs cannot be modified.

To test alternative assumptions:

1. Right-click the scenario in the Sidebar.
2. Select **Duplicate**.
3. Modify parameters in the duplicated scenario.
4. Run the optimization again.
5. Compare results.

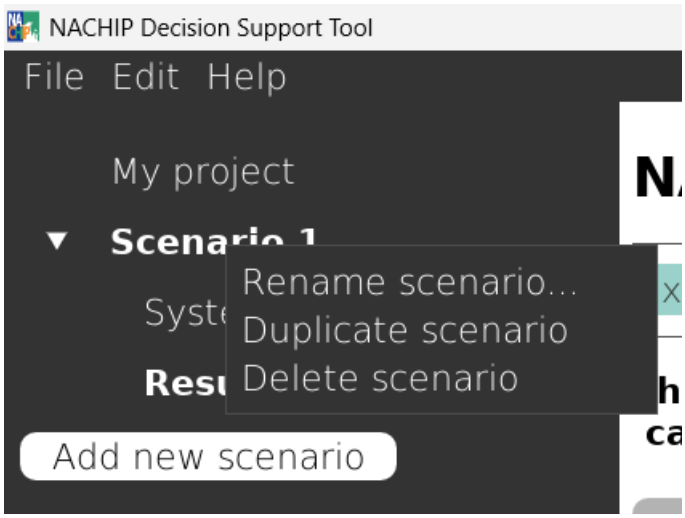


Figure 2.13: Right-clicking the scenario in the Sidebar

For example, try changing the subsidy from 50% to 0% and observe how this affects the investment decision.

For more detailed information about each component, refer to Section 4.2.

3 Interface overview

3.1 Keyword definitions

The DST is organized around **projects** and **scenarios**. Understanding these terms is essential before using the tool. The following definitions explain the most important concepts used throughout the DST and this user manual.

3.1.1 Project

A **project** represents one real-world investment case. It defines:

- The location of the hydrogen hub (geographical coordinates)
- The local conditions
- Any existing infrastructure already available at the site

You can think of a project as a **container** that holds all analyses related to one physical location. At any time, only one project can be opened in the tool.

Each project is saved as a separate `.NACHIP` file. A single project file can contain multiple scenarios.

Example: A municipality planning a hydrogen hub at an industrial zone would create one project for that location and then analyze different design options within it.

3.1.2 Scenario

A **scenario** represents one specific design and assumption set within a project. It includes:

- Selected technologies (e.g., PV, battery, electrolyser)
- Technical parameters (e.g., efficiencies, capacities)
- Economic parameters (e.g., electricity prices, investment costs)
- Policy or regulatory assumptions

The DST performs the optimization separately for each scenario.

After the optimization is completed, a scenario contains:

- **Input data** (what the user defined)
- **Optimization output** (optimal sizes, operation schedule, financial indicators)

This allows you to compare different scenarios within the same project.

As an example you may create:

- Scenario A: High electricity price assumption
- Scenario B: Without battery storage included
- Scenario C: With investment subsidy

Each scenario produces its own results.

3.1.3 Existing infrastructure

Existing infrastructure refers to technologies that are already installed at the project location. It encapsulates all parameters of that technologies.

Examples:

- An existing PV plant with its grid connection and power generation capability
- An industrial plant with its grid connection and electricity consumption
- A hydrogen refueling station with its hydrogen storage tank and hydrogen demand
- A drying facility with its heat demand and district heat connection

Instead of replacing them, the DST:

- Keeps their current capacity
- Optimizes possible expansions
- Optimizes the operation of existing and new components together

3.1.4 Parameters

In the DST, **parameters** (also referred to as *input parameters*) represent all user-defined data that describe a scenario. All parameters are configured in the **System structure** window of the selected scenario.

Certain parameters:

- Have default values (e.g., technical parameters of components),
- Are automatically populated based on selected location (e.g., rainfall amounts),
- Offer predefined values (e.g., electricity tariffs),
- Are mandatory to be entered by users.

Users can either keep predefined values or modify them according to the specific project assumptions. If required inputs are missing or inconsistent, the application will notify the user.

After optimization is executed, the parameters of that scenario become locked. To modify inputs, the scenario must be duplicated.

3.1.5 Optimization outputs

After running the optimization, the application generates **optimization outputs**, which represent the calculated results of the defined scenario. Optimization outputs include:

- Optimal installed capacities (e.g., PV size, BESS capacity)
- Optimal schedules of every components (e.g., grid import and export profiles, storage charging and discharging behavior)
- Financial indicators (e.g., NPV, ROI)

These outputs are available in the **Results** window of the scenario. They are presented in tables and in charts.

3.2 Welcome window

When you start the application, the welcome screen appears as shown in Figure 3.1.

With the welcome screen you have 4 options, i.e. buttons to click on:

- Create new project
- Open project
- Check user manual
- Go to Hydrogen Potential Model (HPM)

Each option opens a different workflow within the application.

3.2.1 Create new project

By clicking on **Create new project** a new window appears as in Figure 3.2.

The window contains three text fields:

- **Project location**: This is the folder on your computer where the project file (`.NACHIP`) will be saved. You can either type the folder path manually or click *Browse…* to select a folder. By default, the project location is your Desktop.

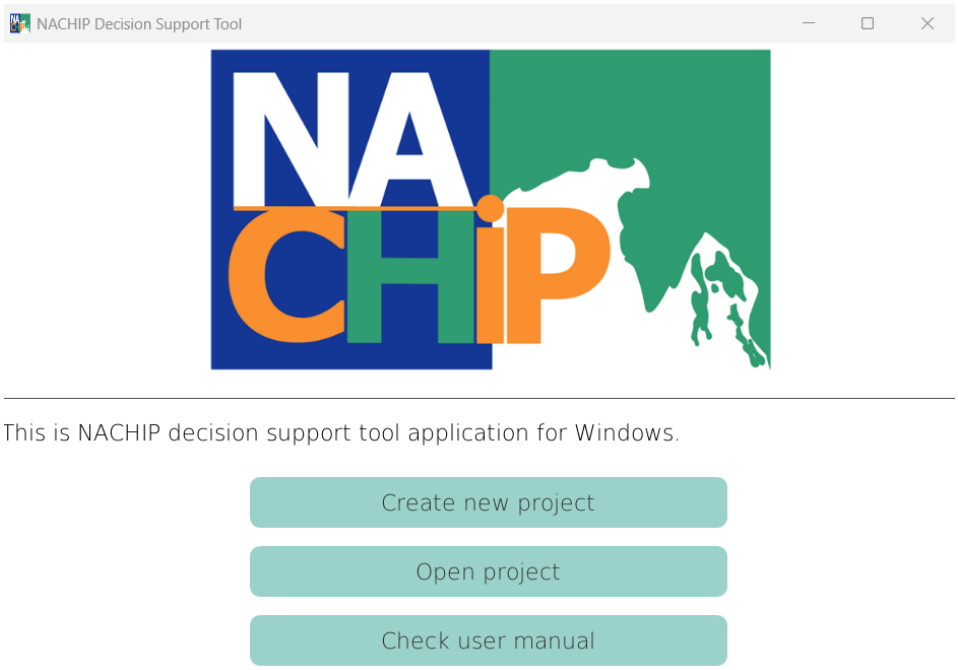


Figure 3.1: Welcome window

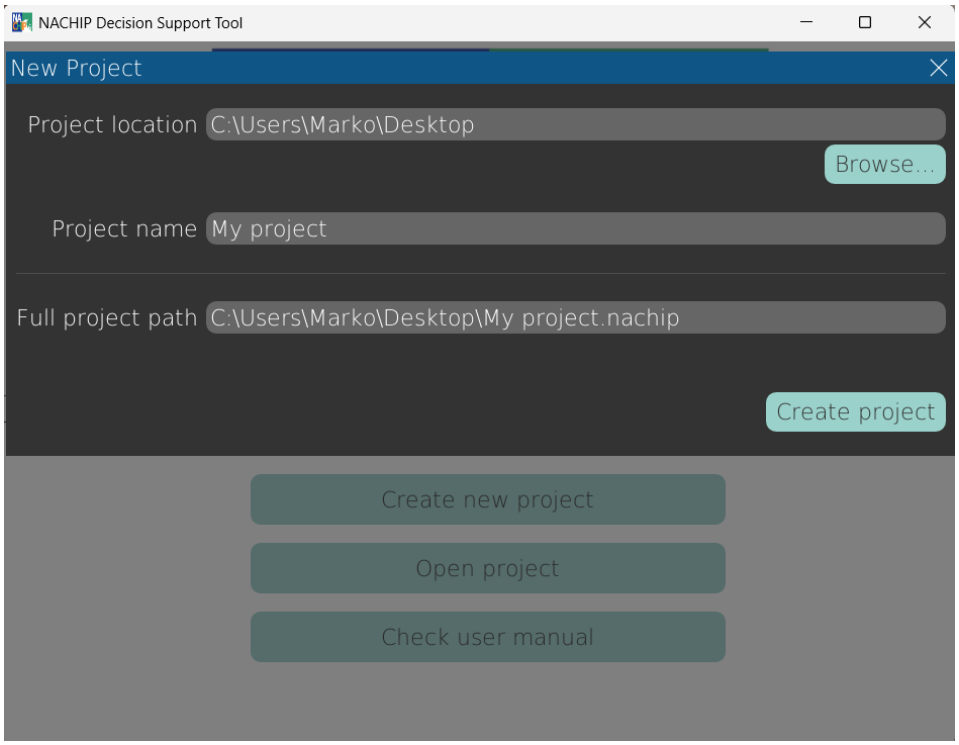


Figure 3.2: Creating a new project

- **Project name**: This is the name of your project. It will also be the name of the `.NACHIP` file. Two projects with the same name cannot exist in the same folder.
- **Full project path**: This field shows the complete file path that will be created. It is generated automatically and cannot be edited directly. The format is: `PROJECT_LOCATION\PROJECT_NAME.NACHIP`.

After entering the required information, click **Create project**. A new project file (`.NACHIP`) will be created in the selected location and project window will be opened. If a project with the same name already exists in that folder, the tool will display a notification.

3.2.2 Open project

By clicking on **Open project**, a file selection window appears. In this window, you can browse your computer and select an existing project file (`.NACHIP`) that you want to open.

Only folders and `.NACHIP` project files are displayed. Other file types are hidden to make selection easier and to prevent opening unsupported files.

After selecting the desired file and confirming, the project will be loaded into the application. You can then view, edit, or create scenarios within that project.

3.3 Project window

When a project is loaded (either newly created or opened from file), the **Project window** appears, as shown in Figure 3.3. This is the main working environment of the DST. All scenario creation, parameter input, optimization, and result analysis are performed here.

The Project window consists of three main parts:

- **Menu bar** – located at the top of the window
- **Sidebar** – located on the left side
- **Main window** – the central and largest area

3.3.1 Menu bar

The menu bar has three dropdown menus: **File**, **Edit**, and **Help**.

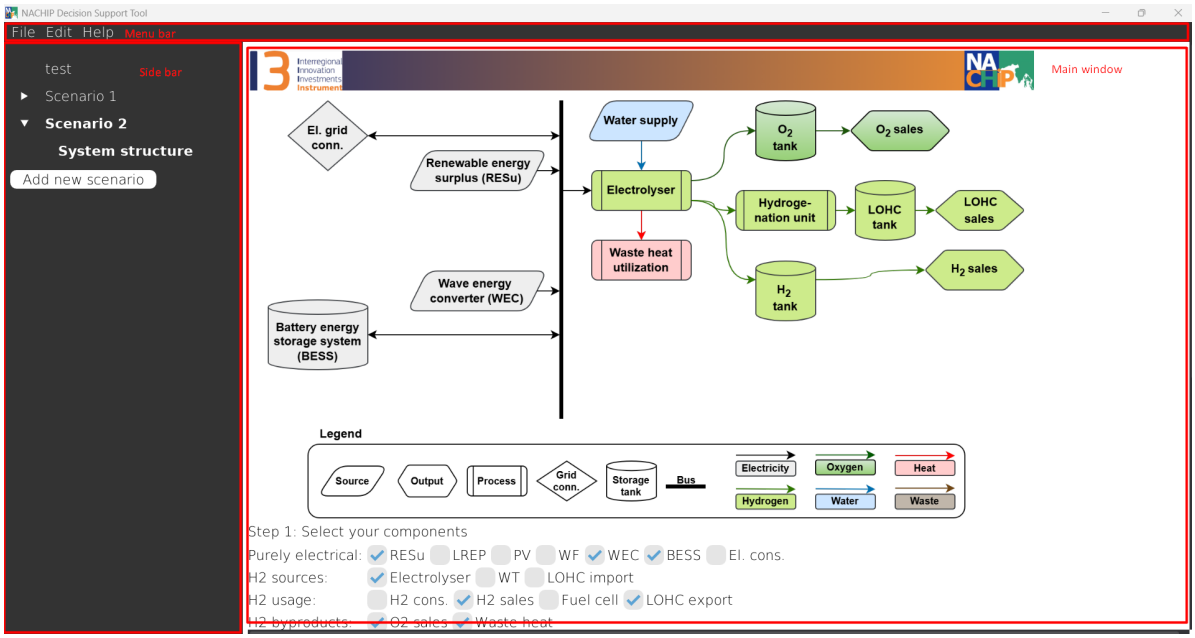


Figure 3.3: Project window

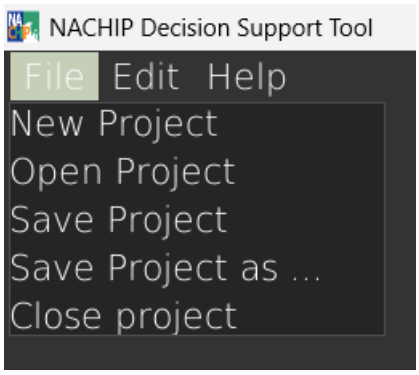


Figure 3.4: File menu

3.3.1.1 File

The **File** menu contains options related to project management.

- **New project**: creates a new project and opens it instead of the current project. The application will ask you whether you want to save the currect project before prceeding. The project creation procedure is described in Section 3.2.1.
- **Open project**: opens an existing `.NACHIP` project file. The application will ask you whether you want to save the currect project before prceeding. The procedure is described in Section 3.2.2.
- **Save project**: saves the current project to its existing `.NACHIP` file. No confirmation message is shown, the project is saved immediately.
- **Save project as ...**: opens a dialog window where you can choose a new location and name for the project file. This action effectively creates a copy of the current project. The copied project is then opened in the application.
- **Close project**: closes the current project and returns to the Welcome screen (Figure 3.1). The application will ask you whether you want to save the currect project before prceeding.

3.3.1.2 Edit

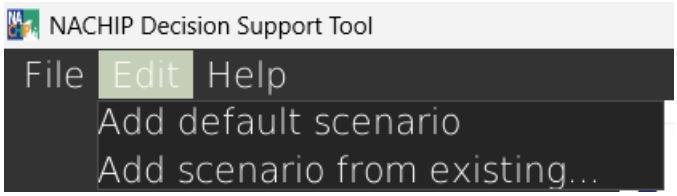


Figure 3.5: Edit menu

- **Add default scenario**: creates a new scenario in the sidebar. The new scenario contains default components with all parameters set to zero. It is automatically named `Scenario X`, where `X` is the next available number.
- **Add scenario from existing**: creates a new scenario as a copy of an existing one. A dialog window opens where you select the scenario to duplicate. The copied scenario is named `Scenario Y (copy)` where `Scenario Y` is the original scenario name.

3.3.1.3 Help

- **User manual**: opens this user manual in your default internet browser.
- **Contact**: opens a new email message in your default email application with the developers already added as recipients.
- **About**: opens a pop-up window with general information about the application, including: tool version, project information, developer information.

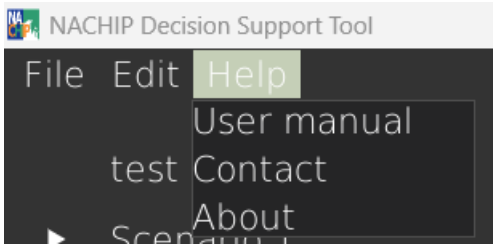


Figure 3.6: Help menu

3.3.2 Sidebar

The **Sidebar** is located on the left side of the Project window. It provides an overview of the project and all scenarios within it.



Figure 3.7: Side bar with 3 scenarios

At the top of the Sidebar, the **project name** is displayed. Below the project name, all scenarios belonging to the project are listed.

3.3.2.1 Scenario list

Each scenario appears as an expandable item in the Sidebar. When expanded, a scenario contains two sections:

- **System structure**: The input definition of the scenario.
- **Results**: The calculated outputs.

The **Results** section becomes visible only after the scenario has been successfully optimized.

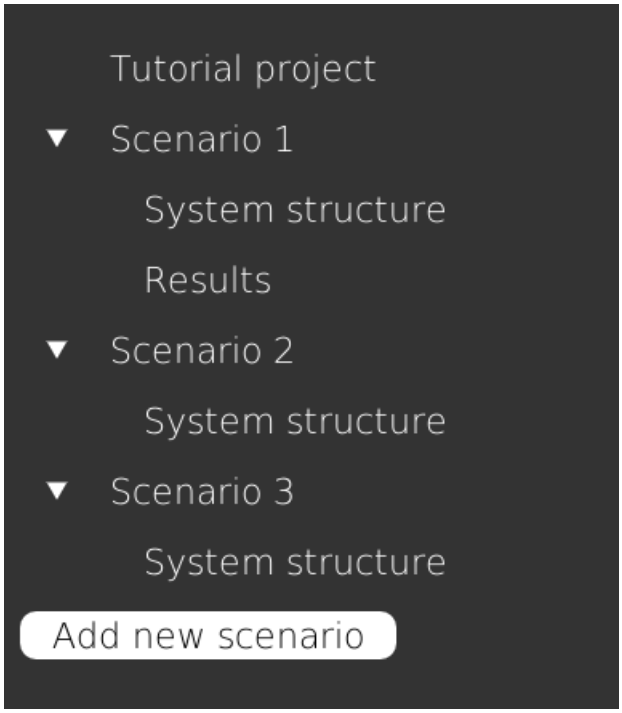


Figure 3.8: Side bar with fully expanded items

3.3.2.2 Active scenario and active window

The currently opened scenario is displayed in **bold font**. Within the scenario, the currently active section (either **System structure** or **Results**) is also shown in bold.

This makes it easy to see:

- Which scenario you are currently working on
- Whether you are editing inputs or reviewing results

3.3.2.3 Adding a new scenario

At the bottom of the Sidebar, there is a button: **Add new scenario**

Clicking this button creates a new scenario within the current project. The newly created scenario will appear in the list and can then be configured and optimized. It contains default components with all parameters set to zero. It is automatically named `Scenario X`, where `X` is the next available number.

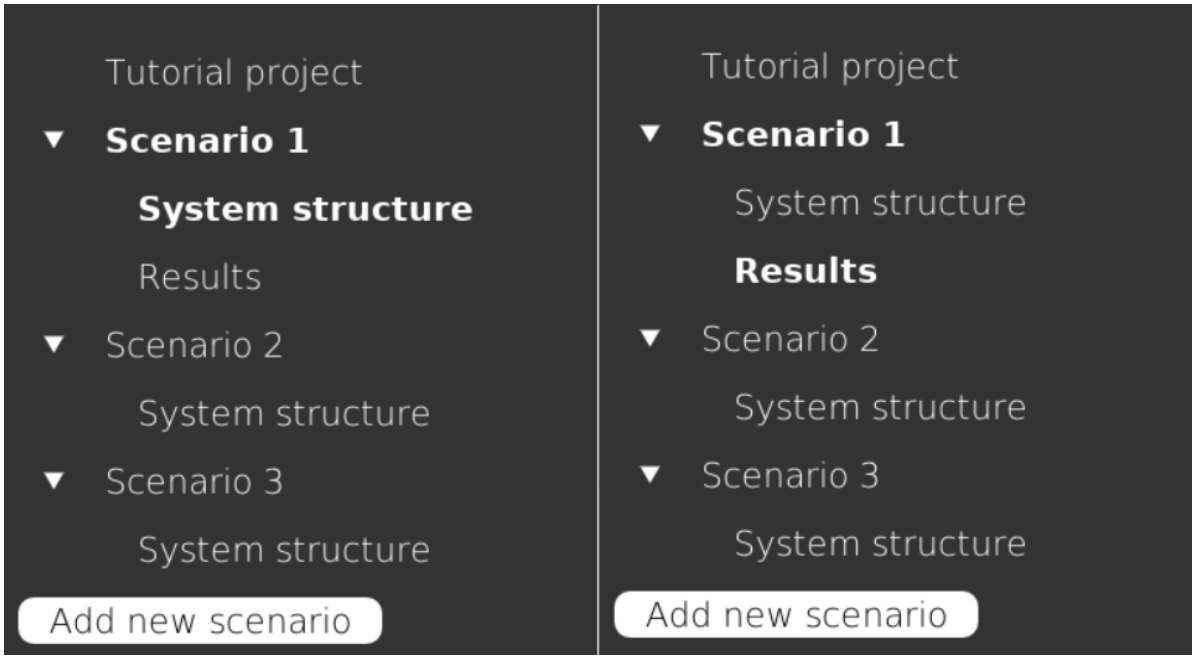


Figure 3.9: Side-by-side view of two possible scenario views: System structure, or Results

3.3.2.4 Scenario management (right-click menu)

You can manage individual scenarios by right-clicking on a scenario name in the Sidebar. A context menu appears with the following options:

- **Rename**: Allows you to change the name of the selected scenario. It opens a pop-up window where you type in a new name.
- **Duplicate**: Creates a copy of the selected scenario, including all defined parameters, but without any results. This is useful when testing small variations of an existing setup. The duplicated scenario is named `Scenario Y (copy)` where `Scenario Y` is the original scenario name.
- **Delete**: Permanently removes the selected scenario from the project. Results associated with that scenario are also deleted. The application will ask you to confirm deletion before proceeding.

3.3.2.5 Practical note

Using multiple scenarios within one project is recommended when comparing different technology configurations, cost assumptions, or policy conditions. This structure allows you to keep all variations organized under the same project location.

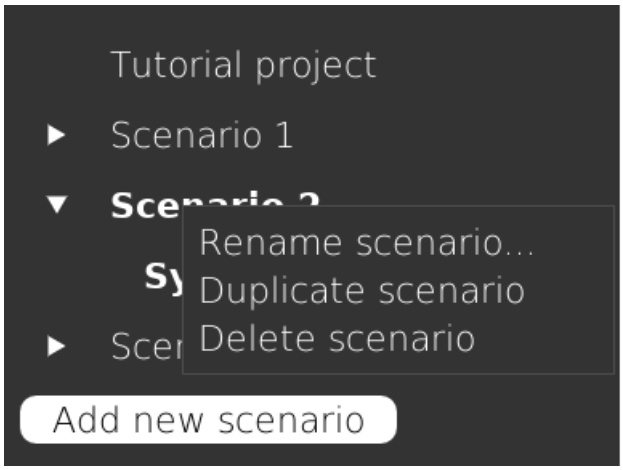


Figure 3.10: Right-click menu of a scenario

3.3.3 Main window

The **Main window** is the central and largest part of the **Project window**. It displays either the **System structure** view or the **Results** view of the currently selected scenario.

Which view is shown depends on what you select in the Sidebar.

- If you click **System structure**, the Main window displays all input parameters of the scenario.
- If you click **Results**, the Main window displays the optimization outputs (available only after a successful optimization).

This is the primary workspace of the application. All parameter definition, optimization control, and result analysis are performed here.

Examples of both views are shown in Figure 3.11 and Figure 3.12.

The detailed functionality of both views is explained in the following chapters.

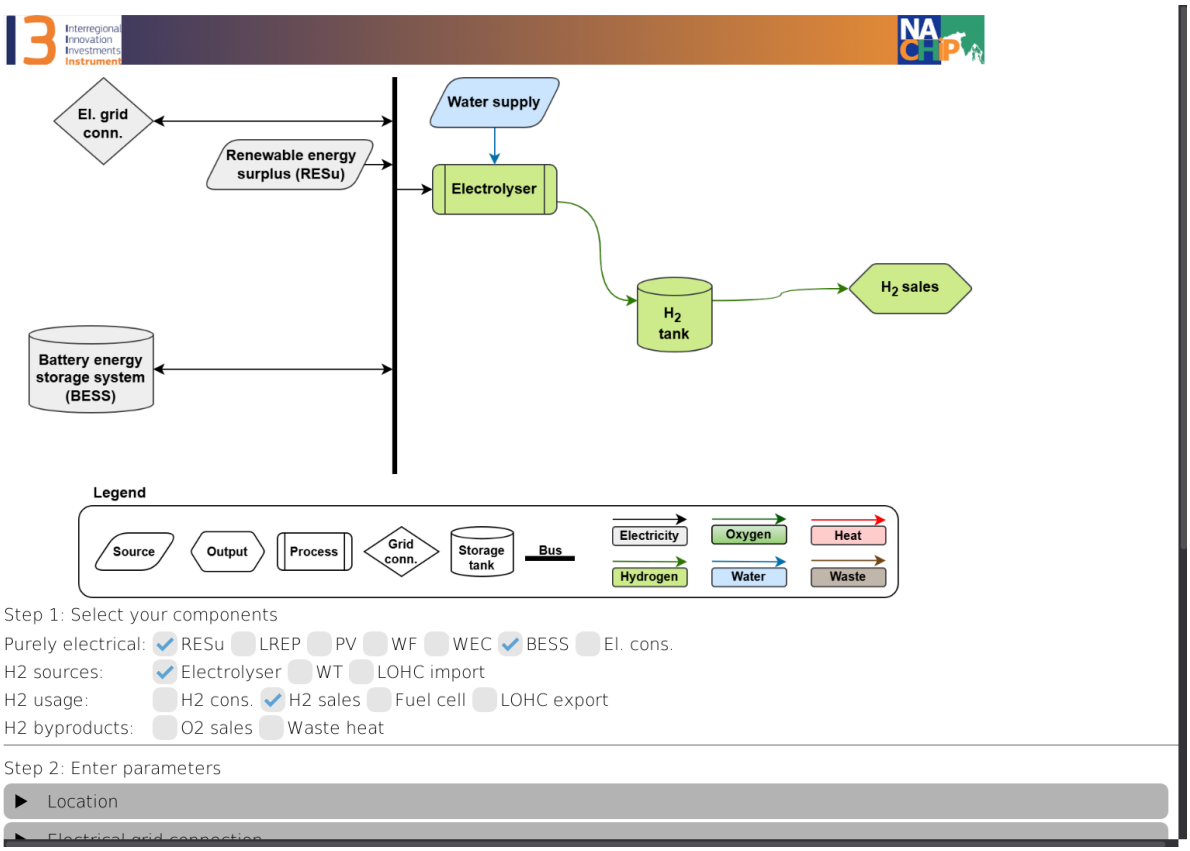


Figure 3.11: Example of System structure window

NACHIP Decision support tool - Results for 'Scenario 1'

Export results to HTML

The recommended investment could generate/save 11,456.16 € in 20 years compared to no-investment case, and it would pay off in 4.19 years.

▼ Key recommendations (optimal sizes)

The sizes of the key components of your site are presented in the table below.

Component	Size
PV Capacity	2.40 kW
Grid Connection	30.00 kW

▼ Financial Results

Financial specification as costs, revenues, and profits.

Indicator	Value
Total investment cost (CAPEX)	2,400.00 €
Yearly operational cost (OPEX)	21,909.10 €/year
Yearly revenue	0.00 €/year

Financial indicators

Indicator	Value
Payback period	4.19 years

Figure 3.12: Example of Results window

4 System structure

4.1 Available components

The HyEfRe Decision Support Tool allows you to configure your hydrogen infrastructure project by selecting different components through checkboxes in the **System structure** window. This section explains how to select components and which combinations are feasible.

4.1.1 Selecting components using checkboxes

When you open a scenario in the **System structure** view, you will see a panel with checkboxes that allow you to select which components to include in your analysis. The checkboxes are organized as follows:

1. **Navigate to System structure**: In the sidebar, expand your scenario and click on **System structure**.
2. **Locate the checkbox panel**: At the top of the main window, you will see "Step 1: Select your components" with the available components arranged in four labelled rows:

   - **Purely electrical**: RESu, LREP, PV, WF, Leonitus, BESS, El. cons.
   - **H2 sources**: Electrolyser, WT, LOHC import
   - **H2 usage**: H2 cons., H2 sales, Fuel cell, LOHC export
   - **H2 byproducts**: O2 sales, Waste heat

   The checkboxes in the *H2 usage* and *H2 byproducts* rows are shown only when the components they depend on are selected (see Component dependencies).
3. **Enable/disable components**: Click on a checkbox to include or exclude that component from your scenario:

   - **Checked** (enabled): The component will be included in the optimization
   - **Unchecked** (disabled): The component will not be considered in the analysis

The following components can be selected via checkboxes:

Component	Description
RESu	Renewable Energy Surplus - represents excess renewable energy that can be utilized
LREP	Local Renewable Energy Plant - dedicated renewable energy generation
PV	Photovoltaic (PV) system - solar electricity generation
WF	Wind Farm - wind-based electricity generation
Leonitus	Leonitus wave energy converter (WEC) - controllable wave-based electricity generation, sized by its power (see Section 4.2.7)
BESS	Battery Energy Storage System - electrical energy storage
El. cons.	Fixed Electricity Consumption - baseline electricity demand
Electrolyser	Electrolyser - hydrogen production through water electrolysis
WT	Waste treatment plant - reactors that treat waste and produce non-green hydrogen, electricity, heat and carbon black (waste source and storage is shown with it)
H2 cons.	Fixed Hydrogen Consumption - on-site hydrogen demand (see Section 4.2.13)
H2 sales	Hydrogen Sales - hydrogen product distribution
O2 sales	Oxygen Sales - oxygen by-product from electrolysis
Fuel cell	Fuel Cell - hydrogen-powered electricity generation
LOHC export	LOHC export stream - hydrogen is bound to a liquid organic hydrogen carrier and sold (LOHC sales including the hydrogenation unit, LOHC tank)
LOHC import	LOHC import stream - loaded LOHC is bought and its hydrogen is released on site (LOHC purchase including the dehydrogenation unit, LOHC tank)
Waste heat	Waste Heat Utilization - thermal energy recovery and use

> **Note**
>
> Some components are present in the system but do not have checkboxes:
>
> - **Location** - defines the geographic coordinates and local conditions
> - **Electrical grid connection** - connection to the grid is always present
> - **Water supply** - water input for the electrolyser (when Electrolyser is active)
> - **Hydrogen storage** - hydrogen compressor and tank (when a hydrogen source is active)
> - **Waste source and storage** - waste offered to the waste treatment plant and its storage (when WT is active)
> - **Financial parameters** - economic settings for the analysis
> - **Simulation parameters** - technical settings for optimization

4.1.2 Component dependencies

The DST enforces certain dependencies between components. Some components can only be selected if other components are already enabled. This ensures that the infrastructure configuration is technically coherent.

4.1.2.1 Hydrogen sources and hydrogen-dependent components

Hydrogen can be supplied on site by three components, the **hydrogen sources**:

- **Electrolyser**, which produces hydrogen by water electrolysis
- **WT** (waste treatment plant), whose reactors produce non-green hydrogen from waste
- **LOHC import**, which releases hydrogen from bought LOHC in the dehydrogenation unit of LOHC purchase

The following checkboxes are shown while at least one hydrogen source is selected:

- **H2 cons.** (Fixed Hydrogen Consumption)
- **H2 sales** (Hydrogen Sales)
- **Fuel cell** (Fuel Cell)

When the last selected hydrogen source is deselected, these components are deselected automatically and their checkboxes are hidden. If one source is deselected while another remains selected, they stay selected.

4.1.2.2 Waste heat

The **Waste heat** (Waste Heat Utilization) checkbox is shown only while the **Electrolyser**, the **Fuel cell** or **WT** is selected, because these are the components that produce waste heat. When the last of them is deselected, Waste heat is deselected automatically and its checkbox is hidden.

4.1.2.3 Components that need on-site hydrogen production

- **O2 sales** (Oxygen Sales) is shown only while the **Electrolyser** is selected, because oxygen is a by-product of electrolysis.
- **LOHC export** is shown only while the **Electrolyser** or **WT** is selected, because the hydrogenation unit of LOHC sales binds hydrogen produced on site. It is deselected when the last of them is deselected.

4.1.2.4 LOHC streams

The LOHC components (see Section 4.2.20) have no checkboxes of their own. They are shown according to the selected LOHC stream:

- **LOHC export**: LOHC sales (including the hydrogenation unit) and LOHC tank. Requires the Electrolyser or WT.
- **LOHC import**: LOHC purchase (including the dehydrogenation unit) and LOHC tank. The LOHC import checkbox is always visible.

The **LOHC tank** is shown for both streams. Only one LOHC stream can be active at a time: the two checkboxes behave like radio buttons, so checking one unchecks the other, and unchecking the active one leaves both unchecked.

4.1.2.5 Automatic selection

Some selections automatically select or deselect other components:

- Selecting **Electrolyser** automatically selects **H2 sales**, **O2 sales**, **Fuel cell** and **Waste heat**. H2 cons. and LOHC export are not selected automatically.
- Selecting **LOHC import** or **WT** automatically selects **H2 sales** if none of H2 cons., Fuel cell or H2 sales is selected yet.
- Deselecting **Electrolyser** deselects **O2 sales**, and **LOHC export** if WT is not selected.
- Deselecting the last hydrogen source (Electrolyser, WT or LOHC import) deselects **H2 cons.**, **H2 sales** and **Fuel cell**.

- **Waste heat** is deselected when none of the Electrolyser, Fuel cell and WT remains selected.
- Selecting **LOHC export** automatically deselects **LOHC import**, and vice versa.

4.1.3 Feasible component combinations

Below are examples of common and feasible infrastructure configurations. The DST validates component combinations based on the following rules:

4.1.3.1 Validation rules

The system validates your component selection against the following rules. A click that would break one of them is ignored.

1. **Electrolyser and WT need a hydrogen use**: If the **Electrolyser** or **WT** is selected, at least one of the following must also be selected:

   - **Fuel cell** (Fuel Cell)
   - **H2 sales** (Hydrogen Sales)
   - **H2 cons.** (Fixed Hydrogen Consumption)
   - **LOHC export** (LOHC export stream)

   Rationale: The electrolyser and the waste treatment reactors produce hydrogen, so there must be a way to utilize or sell the hydrogen.

2. **LOHC import needs a hydrogen use**: If **LOHC import** is selected, at least one of the following must also be selected:

   - **H2 cons.** (Fixed Hydrogen Consumption)
   - **Fuel cell** (Fuel Cell)
   - **H2 sales** (Hydrogen Sales)

   Rationale: The hydrogen released from the bought LOHC must be used or sold.

3. **Hydrogen users need a hydrogen source**: **H2 cons.**, **Fuel cell** and **H2 sales** can only be selected together with the **Electrolyser**, **WT** or **LOHC import**. **Waste heat** can only be selected together with the **Electrolyser**, the **Fuel cell** or **WT**.

4. **LOHC export requires on-site hydrogen production**: **LOHC export** can only be selected together with the **Electrolyser** or **WT**.

5. **One LOHC stream at a time**: **LOHC export** and **LOHC import** cannot both be selected.

6. **At least one electric source**: The configuration must include at least one of:

- **El. grid** (Electrical Grid)
- **LREP** (Local Renewable Energy Plant)
- **PV** (Photovoltaic system)
- **WF** (Wind Farm)
- **Leonitus** (Leonitus wave energy converter)

Rationale: The system needs at least one source of electricity.

7. **At least one electricity consumer**: The configuration must include at least one of:

   - **Electrolyser** (hydrogen production)
   - **El. cons.** (Fixed Electricity Consumption)
   - **BESS** (Battery Storage)
   - **LOHC import** (the dehydrogenation unit of LOHC purchase uses an electric heater)
   - **WT** (waste treatment reactors)

   Rationale: There must be a consumer for the electricity supplied by the sources.

4.1.3.2 Invalid combinations

If you try to select an invalid combination of components, the application will:

1. **Reject the change**: The component selection will not be updated
2. **Maintain valid state**: The previous valid configuration will remain active

For example, the following combinations are **invalid**:

- **Electrolyser or WT without Fuel cell, H2 sales, H2 cons. or LOHC export**: The produced hydrogen must be used somewhere
- **LOHC import without H2 cons., Fuel cell or H2 sales**: The released hydrogen must be used somewhere
- **Fuel cell, H2 sales or H2 cons. without Electrolyser, WT and LOHC import**: There is no hydrogen to supply them
- **Waste heat without Electrolyser, Fuel cell and WT**: There is no component that produces waste heat
- **LOHC export without Electrolyser and WT**: The hydrogenation unit needs hydrogen produced on site
- **LOHC export together with LOHC import**: Only one LOHC stream can be active
- **No electric source**: There must be at least one source of electricity (El. grid, LREP, PV, WF, or Leonitus)
- **No electricity consumer**: There must be at least one consumer (Electrolyser, El. cons., BESS, LOHC import or WT)

4.1.3.3 Examples of feasible combinations

- **Electricity only**: PV, BESS and El. cons. (no hydrogen components).
- **Hydrogen production and sale**: Electrolyser with H2 sales (O2 sales, Fuel cell and Waste heat are optional).
- **Hydrogen export as LOHC**: Electrolyser with LOHC export.
- **Hydrogen import as LOHC**: LOHC import with H2 sales and/or Fuel cell, without the Electrolyser.
- **Electrolyser supplemented by LOHC import**: Electrolyser and LOHC import with H2 sales and/or Fuel cell.
- **Hydrogen from waste**: WT with H2 sales, H2 cons. (non-green allowed), Fuel cell and/or LOHC export to non-green customers.

4.2 Component details

4.2.1 Location

Purpose
The backend uses the geographic coordinates for:

- Weather data which is used for rain collection in Water supply component
- Solar data which is used in PV component
- Wind data which is used in Wind farm component
- Detecting country which is used to set default parameters in:
  - Electricity grid connection
  - Electricity supply
  - Water connection

You can enter coordinates manually, choose one of predefined cities or choose location on a map.

4.2.1.1 Input parameters

Parameter	Description	How to input
Latitude/Longitude	Geographic coordinates	Click on map or enter manually
City	Predefined city selection	Select from dropdown list

Country	Automatic from coordinates	Auto-detected

Important Notes:

- Must be set before generating renewable production profiles

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4.2.2 Electrical grid connection

The **Electrical grid connection** component represents the interface between the renewable energy hub and the electrical grid. This component cannot be disabled.

It consists of 3 sections:

- **Grid connection**: the physical characteristics of the grid connection (e.g. grid fees, peak power pricing).
- **Electricity supply**: models and tariffs for purchasing electricity from the grid and feeding it into the grid.
- **Advanced features**: optional grid-related constraints.

Before the 3 sections, there is a parameter influencing both *Grid connection* and *Electricity supply*:

Show predefined values for the country (dropdown menu) The country is automatically selected according to the site location, but it can be changed manually. Available countries are those that participated in the NACHIP project: Croatia, Slovenia and Italy. Further available countries had participated in previous projects with the tool developers: Austria, Czechia, Germany, Hungary and Poland. If the selected location is in another country, the value of this parameter remains empty. With the selected country the tool offers grid connection and electricity supply types for that country, but any parameter can be entered manually.

4.2.2.1 Grid connection

The following parameters define technical and economic aspects of the physical grid connection. Depending on whether an existing connection is available, different cost calculation approaches are applied.

Parameter	Description
Use predefined connection type [dropdown menu]	Depending on the selected country, the tool offers predefined connection types. By selecting one of the connection types, all other parameters from this section (*Grid connection*) are populated. You can also enter data manually, even after selecting a connection type.
Grid connection already exists? [checkbox]	Whether the grid connection already exists at the location or not. It must be enabled if there is any other existing infrastructure.
Existing connection size [kW]	The size of the existing grid connection. The connection is assumed symmetrical, i.e. it has the same limit for both directions. The parameter is visible only if *Grid connection already exists?* is enabled.
Capacity price for grid connection enlargement [€/kW]	It represents the price for building and installation of the connection depending only on the capacity, i.e. size. It can be determined from existing projects: cost of the connection divided with the capacity of the connection.
Unit price for grid connection installation [€/(kW km)]	It represents the price for building and installation of the connection depending on the capacity and distance to the nearest *connection point*. A *connection point* represents a piece of infrastructure to which the renewable energy hub must be connected to, such as a substation. This parameter is used only for greenfield investments – it is visible only when *Grid connection already exists?* is disabled. It can be determined from existing projects: cost of the connection divided with the capacity and distance to the nearest connection point.
Distance to nearest connection point [km]	A *connection point* represents a piece of infrastructure to which the renewable energy hub must be connected to, such as a substation. This parameter is used only for greenfield investments – it is visible only when *Grid connection already exists?* is disabled. It is used to calculate the total price of the grid connection: *Unit price* * *Distance* + *Capacity price*
Start time of high tariff [0–24]	Hour of the day when high tariff for grid fees starts. If the tariff model has a single tariff, set it to 0, and enter your single tariff grid fee in *Grid fee during high tariff*.
Start time of low tariff [0–24]	Hour of the day when low tariff for grid fees starts. If the tariff model has a single tariff, set it to 24, and the model will use only *Grid fee during high tariff* as your single tariff grid fee.
Weekend tariff exists? [checkbox]	Whether there is weekend tariff for grid fees or not. If enabled, weekend tariff applies to every weekend, starting on Saturday at 00:00 and ending on Sunday at 23:00.

Parameter	Description
Grid fee during high tariff [€/kWh]	Grid fee, without taxes, during high tariff hours. Includes transmission and distribution fees, and any other fees paid to DSO and TSO, charged per amount of energy consumed.
Grid fee during low tariff [€/kWh]	Grid fee, without taxes, during low tariff hours. Includes transmission and distribution fees, and any other fees paid to DSO and TSO, charged per amount of energy consumed.
Grid fee during weekend tariff [€/kWh]	Grid fee, without taxes, during weekends (from Saturday at 00:00 to Sunday at 23:00). It is visible and applies only if *Weekend tariff exists?* is enabled. It includes both transmission and distribution fees, and any other fees paid to DSO and TSO, charged per amount of energy consumed.
Peak power price [€/kW/month]	Monthly charge based on the maximum recorded power demand. The fee is applied for each month included in the simulation period, regardless of the number of simulated days within that month. Even if the simulation lasts only 1 day in a month, it is assumed that the peak power happened that day.

4.2.2.2 Electricity supply

Electricity supply represents the only non-green electricity source for the hydrogen hub. The parameters in this section define the electricity purchase and feed-in prices, excluding grid fees, which are specified separately in the Grid connection section.

Electricity supply parameters can be defined in two ways, each available in a separate tab:

- **Standard tariffs**: defined through high, low, and optional weekend tariffs.
- **Custom price profile**: defined by importing external time-series data of electricity prices.

Parameters in *Standard tariffs* tab

Parameter	Description
Use predefined supply models [dropdown menu]	Select a predefined electricity supply model for the chosen country. When selected, all energy prices, feed-in prices, and additional fees in this section are automatically populated according to the selected model. Manual modification of individual parameters is possible after selecting a predefined model.
Start time of high tariff [0–24]	Hour of the day at which the high tariff period begins. If a single tariff model is used, set this parameter to 0 and define the energy price in *Energy price during high tariff*.

Parameter	Description
Start time of low tariff [0–24]	Hour of the day at which the low tariff period begins. If a single tariff model is used, set this parameter to 24. In that case, only the high tariff price is applied throughout the entire day.
Weekend tariff exists? [checkbox]	Indicates whether a separate tariff applies during weekends. If enabled, weekend tariff values apply from Saturday at 00:00 to Sunday at 23:00. If disabled, weekday high and low tariffs are applied during weekends as well.
Energy price during high tariff [€/kWh]	Energy price for electricity imported from the grid during high tariff hours. This price represents the supplier's energy component only and excludes grid fees and taxes.
Energy price during low tariff [€/kWh]	Energy price for electricity imported from the grid during low tariff hours. This price represents the supplier's energy component only and excludes grid fees and taxes.
Energy price during weekend tariff [€/kWh]	Energy price for electricity imported from the grid during weekend tariff hours. Visible and applied only if *Weekend tariff exists?* is enabled. This price excludes grid fees and taxes.
Feed-in price during high tariff [€/kWh]	Compensation received for electricity exported to the grid during high tariff hours. Defines the revenue per unit of exported electricity.
Feed-in price during low tariff [€/kWh]	Compensation received for electricity exported to the grid during low tariff hours. Defines the revenue per unit of exported electricity.
Feed-in price during weekend tariff [€/kWh]	Compensation received for electricity exported to the grid during weekend tariff hours. Visible and applied only if *Weekend tariff exists?* is enabled.
Additional fees during high tariff [€/kWh]	Additional supplier-related energy charges applied during high tariff hours. These may include supplier margins, balancing costs, or other contractual energy charges, excluding grid fees and taxes.
Additional fees during low tariff [€/kWh]	Additional supplier-related energy charges applied during low tariff hours. Excludes grid fees and taxes.
Additional fees during weekend tariff [€/kWh]	Additional supplier-related energy charges applied during weekend tariff hours. Visible and applied only if *Weekend tariff exists?* is enabled. Excludes grid fees and taxes.
Tax on electricity [%]	Percentage tax applied to the total electricity purchase price. The tax is calculated on the sum of: energy price, additional fees, and grid fees. Feed-in revenues are not subject to this tax.

GUI elements in *Custom price* tab

This option allows the user to define electricity purchase prices by importing a time-series

dataset from a CSV file. The imported values represent electricity prices including VAT, but excluding grid fees. The custom price profile replaces the standard tariff definition and is applied throughout the entire simulation period. Grid fees from Section 4.2.2.1 are applied.

GUI element	Description
Custom profile of prices [text field]	Defines the name of the CSV file containing the electricity price time series. It is not possible to write in this field – it is automatically populated after clicking on button *Browse…*.
Browse [button]	By clicking Browse, the user selects a CSV file from the local system. File format requirements: - File format: .csv - Single column - No header row Allowed number of rows: - 8,760 (hourly resolution) - 17,520 (30-minute resolution) - 35,040 (15-minute resolution) Each row must contain a numerical value representing the electricity purchase price in €/kWh, including VAT and excluding grid fees. The first value in the file corresponds to the beginning of the year (January 1st, 00:00), and the last value corresponds to the end of the year (December 31st, 23:00 for hourly resolution). If the dataset originates from a leap year (8,784 hourly values), one day of data must be removed to match one of the supported row counts. After selecting a file, a message is displayed indicating whether the data import was successful or if errors were detected.
Check imported data [button]	Opens a pop-up window displaying a chart of the imported electricity price profile. The chart is dynamic, meaning it can be zoomed in or out and moved around. This allows the user to visually verify data continuity, seasonal trends, extreme values, and possible data inconsistencies. The chart is for verification purposes only and does not modify the imported dataset. This button is visible only if data import was successful.
Percentage for feed-in remuneration [%]	Defines the percentage of the imported electricity purchase price used to calculate the feed-in price. The feed-in price is calculated as: *feed-in price = imported electricity price × defined percentage*. For example, if the percentage is set to 80%, the feed-in remuneration equals 80% of the corresponding purchase price at each time step.

4.2.2.3 Advanced features

This section provides additional constraints related to electricity import, export, and price behavior. Certain options are mutually exclusive and cannot be enabled simultaneously.

Parameter	Description
Electricity export is not possible [checkbox]	Enable this option if electricity can be imported from the grid but cannot be exported to it. When enabled, the model does not allow feed-in of surplus electricity to the grid. All locally generated electricity must either be consumed on-site or curtailed. This option is applicable in cases where grid injection is not permitted by the DSO or contractual arrangements. If enabled, the following options are automatically disabled: *Electricity import is not possible*, *Energy export limited to the imported amount*
Electricity import is not possible [checkbox]	Enable this option if electricity can be exported to the grid but cannot be imported from it. When enabled, the system cannot purchase electricity from the electricity supplier. This is typically relevant when optimizing a generation plant with storage, where charging from the grid is not permitted by the DSO. The only exception is the renewable energy surplus (RESu, Section 4.2.3): if RESu is active, its surplus energy can still be taken from the grid, while purchasing from the supplier remains blocked. If enabled, the following options are automatically disabled: *Electricity export is not possible*, *Energy export limited to the imported amount*.
Energy export limited to the imported amount [checkbox]	Limits total exported electricity to the total imported electricity over the simulation period. This reflects regulatory or contractual conditions where feed-in remuneration is restricted to the amount of previously imported energy. Such schemes are designed to ensure that the prosumer remains primarily a consumer rather than a net producer. If enabled, the following options are automatically disabled: *Electricity export is not possible*, *Electricity import is not possible*.
Price of electricity cannot be negative [checkbox]	Prevents the effective electricity purchase price from becoming negative. When using custom electricity price profiles (e.g., day-ahead market data), negative prices may occur. In practice, however, retail consumers purchasing electricity through a supplier are typically not remunerated for consumption during negative price periods. If this option is enabled, any negative energy price is set to zero. It is recommended to keep this option enabled when electricity is purchased or sold through a supplier rather than directly on wholesale energy markets.

4.2.2.4 Optimization output

After running the optimization, the tool calculates the optimal technical and economic values for the defined scenario. This subsection shows what variables are tied to the *Electrical grid connection* component, while the results are presented in Results (Chapter 5).

Variable	Description
Grid connection enlargement	If the existing grid connection capacity is insufficient to satisfy the optimized power flows, the model determines the required: - **Grid connection capacity enlargement** [kW] - **Investment cost of enlargement** [€] The investment cost is calculated according to the parameters defined in Section 4.2.2.1 – capacity price and/or distance-based cost components. If no enlargement is required, the investment cost is zero.
Electricity import from supplier	The optimization determines the time-dependent profile of electricity purchased from the supplier. Results include: - **Electricity import profile** as mean powers for each time step [kW] - **Operating cost of imported electricity** [€] The operating cost is calculated using: energy prices, additional fees, and taxes. If the total operating cost is negative, it indicates that revenue from electricity export exceeds the cost of electricity import during the simulation period. In that case, the hydrogen hub generates net income from electricity production.
Electricity flow through the grid connection	The tool also provides the time-dependent profile of electricity flowing through the physical grid connection. This profile represents the actual power exchange at the connection point and may differ from the electricity purchased from the supplier. The difference occurs because the RESu component (Section 4.2.3, if enabled) acts as a secondary supplier. Therefore, grid fees are calculated based on the total electricity flowing through the connection. Outputs include: - **Electricity flow profile** as mean powers for each time step [kW] - **Total cost of grid fees** [€]
Monthly peak power	For each month included in the simulation period, the model determines **peak import power** [kW] and **peak export power** [kW]. Peak values are used to calculate capacity-based charges according to the defined peak power price. The resulting output includes: - **Monthly peak values** [kW] - **Total peak power cost** over the simulation period [€] If the simulation covers only part of a month, the peak power charge is still applied for that month, according to the defined pricing rules.

i Time resolution and profile length

All resulting time-series outputs (electricity import, export, and grid flow profiles) are generated according to the Simulation parameters – sampling time, start date, end date. The length of the output profiles therefore directly corresponds to the simulation settings.

4.2.2.5 Related components

- Location, as the predefined models of *Grid connection* and *Electricity supply* depend on the selected country.
- RESu, as its electricity flow contributes to the total grid fee cost.

4.2.2.6 Troubleshooting

Common issues and solutions

⚠ Existing connection size and payoff period limit

If you are using any existing infrastructure, most commonly **Electricity consumption** (Section 4.2.9), make sure the **Existing connection size** (see Section 4.2.2.1) is large enough to cover current energy needs.
If the existing connection is too small, the tool will automatically enlarge it, regardless of whether the investment is economically viable. This enlargement incurs an investment cost, which means that it is **not possible to limit the payoff period** in the Financial parameters (Section 4.2.9) if the existing connection capacity is insufficient.
The tool calculates the minimum required connection size needed to enable the payoff period constraint. If this minimum is not reached, an error message is displayed.

ℹ Using RESu and limiting electricity import

Under the **Advanced features** of the **Electrical grid connection** (see Section 4.2.2.3), it is possible to forbid electricity import. This restriction applies to the electricity bought from the supplier. The **RESu component** (Section 4.2.3) can still be used together with it, so the system can take renewable surplus energy from the grid while buying no electricity from the supplier. This is the intended setup for e.g. a hydrogen plant that is allowed to run only on renewable surplus (HPM).
The RESu energy flows through the grid connection, so it is still counted in the peak power and connection size.
In time steps with available renewable surplus, electricity cannot be exported to the grid and the RESu energy cannot be sold back to the supplier. The RESu energy can only be consumed on site, or left unused.

4.2.3 Renewable energy surplus

The **Renewable Energy Surplus (RESu)** component represents excess energy available in the power distribution or transmission grid. This component allows the optimization to utilize renewable surplus energy that would otherwise be curtailed or wasted.

In the NACHIP DST, the RESu component serves as an alternative electricity source that can reduce grid electricity imports. When activated, the optimization algorithm can choose to use this surplus renewable energy (at the specified price) instead of purchasing electricity from the grid.

4.2.3.1 Input parameters

Basic parameters

Parameter	Description
HPM output file	The Hydrogen Potential Model (HPM) output file containing renewable surplus data. The file needs to be a CSV file, formatted as 2 columns and 8760 rows (no header). Currently, this option is not yet fully implemented.
File with RESu profile	A custom CSV file containing yearly renewable energy surplus profiles. The custom RESu needs to be in a CSV file, formatted in a single column and 8760, 17,520, or 35,040 rows (no header), where values represent renewable energy surplus in kWh. If the data in the file is for a leap year, please remove 1 day of data to match 8760, 17,520, or 35,040 rows. The first value in the file corresponds to the beginning of the year, while the last value corresponds to the end of the year.
Energy price [€/MWh]	Price at which the surplus of renewable energy is bought from the grid. Grid fees from *Electrical grid connection* are added to this value. Default value is 0 €/MWh.

4.2.3.2 Optimization output

Variable	Description
Used power [kW]	Power used from the renewable energy surplus. This represents the actual amount of surplus energy utilized in each time step, limited by the available surplus.

Variable	Description
Total cost of buying surplus energy [€]	Operational costs associated with using the renewable energy surplus. Calculated as the energy price multiplied by the power used.

The optimization model determines how much of the available renewable surplus to use in each time step.

4.2.3.3 Interface options

The RESu component provides the following options:

1. **From tabular tab**: Import a custom renewable surplus profile from a CSV file. Click *Browse...* to select the CSV file containing the RESu profile. Additionally, you can set the energy price at which the surplus renewable energy is bought from the grid.
2. **Check renewable energy surplus button**: After loading a RESu profile, click this button to visualize the renewable energy surplus data as a time series chart. This helps verify that the data has been correctly imported and shows the profile over time.

Data format requirements:

- Custom RESu profile: 1 column, 8760 rows (hourly), 17,520 rows (30-min), or 35,040 rows (15-min)

4.2.3.4 Related Components

- Electrical grid connection - grid fees are added to the RESu energy price

4.2.3.5 Troubleshooting

Common issues and solutions:

- **File with RESu profile is not defined**: When using the *From tabular* tab, ensure you have imported a valid CSV file with the RESu profile. Click *Browse...* to select a properly formatted CSV file.
- **Invalid file format**: The custom RESu file must be a single-column CSV with no header. Ensure your file meets this requirement.

- **Incorrect number of rows**: The RESu profile file must contain exactly 8760, 17,520, or 35,040 rows. If your data is from a leap year, remove one day of data to match the expected row counts. Common row counts: 8760 (hourly), 17520 (30-min), 35040 (15-min).
- **RESu and LREP combination error**: If you have both RESu and LREP active with the payoff period limit enabled, the optimization will fail. Disable the payoff period limit or remove one of these components from your configuration.
- **Using RESu without grid import**: RESu can be used even when *Electricity import is not possible* is enabled in the Electrical grid connection settings. In that case, no electricity is bought from the supplier and all imported energy comes from the renewable surplus. In time steps with available surplus, export to the grid and selling RESu energy back to the supplier are not allowed.

4.2.4 Local renewable energy plant

A **local renewable energy plant (LREP)** is considered as an existing plant that does not require any additional investments, i.e. this component is a source of renewable electricity that is **not sized by the optimization procedure**. The LREP can represent various renewable energy sources, such as wind, solar PV, hydro, biomass, biogas, and geothermal, but also a combination of those.

The user can specify multiple energy plants (REPs), each with its own production profile. A REP is added by clicking on **Add new REP** button.

By right-clicking on a REP's node, a pop-up menu appears with the following options:

- **Rename REP** – change the name of the REP.
- **Delete REP** – delete the REP.

Production profile of each REP can be parametrized in two ways:

- by selecting a **generic power production** profile available in the tool, or
- by importing a **custom power production** profile from a file.

The price for electricity export is set in Electrical grid connection.

> **Disclaimer**
>
> The electricity production of LREP is used for green hydrogen production and therefore must comply with Article 3 of EU regulation 2023/1184.

4.2.4.1 Input parameters

Generate data

Parameter	Description
REP type [dropdown menu]	A generic production profile of the selected type is used and scaled to match the *Annual energy production*.
Annual energy production [MWh]	Amount of electricity produced by the REP in a year. This value is used to scale the generic profile.

Import data

GUI element	Description
File with the production profile [text field]	Defines the name of the CSV file containing the electricity production time series. It is not possible to write in this field – it is automatically populated after clicking on button *Browse…*.
Browse [button]	By clicking Browse, the user selects a CSV file from the local system. File format requirements: - File format: .csv - Single column - No header row Allowed number of rows: - 8,760 (hourly resolution) - 17,520 (30-minute resolution) - 35,040 (15-minute resolution) Each row must contain a numerical value representing the electricity production power in kW. The first value in the file corresponds to the beginning of the year (January 1st, 00:00), and the last value corresponds to the end of the year (December 31st, 23:00 for hourly resolution). If the dataset originates from a leap year (8,784 hourly values), one day of data must be removed to match one of the supported row counts. After selecting a file, a message is displayed indicating whether the data import was successful or if errors were detected.

Check the production profile
A button that opens a pop-up window displaying a chart of the electricity production profile, based on the active tab. The chart is interactive and can be zoomed and panned. This allows the user to visually verify data continuity, seasonal trends, extreme values, and possible data inconsistencies. The chart is for verification purposes only and does not modify the data.

4.2.4.2 Optimization output

Variable	Description
Electricity production profile [kW]	Time-dependent profile of electricity production expressed as average power per time step, since every REP's output can be curtailed. A single profile is used for all REPs together. The electricity produced can be used within the renewable energy hub, or it can be exported to the grid. The exact power routing cannot be determined because all electricity is balanced through a common bus.

4.2.4.3 Interface options

Production profile of each REP can be parametrized in two ways, each with its own tab:

1. **Generate data**: the user selects the type of the REP (PV, Wind, or Hydro), and enters annual energy production. The tool scales the profile accordingly.
2. **Import data**: the user imports a CSV file with a possible power production profile.

4.2.4.4 Related components

- Electrical grid connection, where prices for electricity export are defined.

4.2.4.5 Troubleshooting

An error is displayed if the button *Check the production profile* is clicked on and:

- **Generate data** tab is active and a REP type is not selected, or
- **Generate data** tab is active and *Annual energy production* is zero, or
- **Import data** tab is active and no CSV file has been imported.

> ⚠ Using RESu and LREP together
>
> It is not possible to use both RESu and LREP in the same configuration while also limiting the payoff period.
> If such a configuration is optimized, the tool displays an error message and the optimization is not executed.
> In general, using both RESu and LREP in the same scenario is not recommended.

4.2.5 Photovoltaic system (PV)

The **Photovoltaic (PV)** component models solar power generation systems within the NACHIP DST. It enables the design and optimization of PV arrays by either generating production profiles from PVGIS (Photovoltaic Geographical Information System) or importing custom production data.

The PV system can consist of multiple arrays, each with different orientations, inclinations, and specifications. During optimization, the algorithm determines the optimal installed capacity for each array within the specified constraints, considering the available area, financial parameters, and the production profile.

4.2.5.1 Input parameters

Basic parameters

Parameter	Description
Orientation angle [°]	Orientation angle, or azimuth, is the angle of the PV modules relative to the direction due South. -90° is East, 0° is South and 90° is West.
Inclination angle [°]	Inclination angle, or slope, is the angle of the PV modules from the horizontal plane. 0° means horizontally placed PV modules, while 90° means vertically placed PV modules.
Available area [m²]	Total area available for placement of new PV modules. For example, if it is a roof, then enter the surface area of the roof.
Installed power for reference [kWp]	Installed peak power of the system for which the data is imported.

Advanced parameters

Technical parameters

Parameter	Description
PV size per m² [Wp/m²]	Amount of PV modules that is possible to install per unit of surface area, including obstacles. For example, if a 500 Wp panel covers area of 2.5 m², this parameter should be less than 200 Wp/m² because not all surface could be covered with PV panels.
Limit PV size directly [checkbox]	The limit of the PV array is posed by the available surface area and then calculated into the limit of installed power. With this option you can limit the maximum installed power directly.

Parameter	Description
Maximum PV size [kWp]	Upper limit on the size (installed power) of the PV array. It is calculated according to 'Available area' and 'Solar PV panel power per area', but it can be also entered manually.
Minimum PV size [kWp]	Lower limit on the size (installed power) of the PV array. If this value is greater than zero, 'Maximal payoff period' and 'Maximum investment' parameters are disabled.

Financial parameters

Parameter	Description
PV system [dropdown menu]	Choose a PV system from the application's database and its financial parameters will be filled automatically. Afterwards you can change any of its parameters.
Investment price [€/kWp]	Investment price, per unit of installed power, includes all necessary elements (PV modules, inverter, cables, etc.), materials, labor (design, mounting, commissioning, etc.), and financing until full operation.
Maintenance price [€/kWp]	Estimated annual operation and maintenance (O&M) price per installed power. O&M includes cleaning of PV modules, insurance, administration costs, and replacing broken components. O&M does not include degradation of the system.
Replacement price [€/kWp]	Same as investment price, but after the lifetime of the system, i.e. the future investment price. The main difference from the investment price is that the subsidy does not apply to the replacement price.
Lifetime [years]	Number of years after which the PV system needs to be replaced.
Subsidy [%]	Percentage that is deductible from the investment cost. Does not apply to the replacement price.

4.2.5.2 Optimization output

Variable	Description
Installed capacity [kWp]	Optimized installed power capacity for each PV array.
Total installed capacity [kWp]	Sum of installed capacities across all PV arrays.
Power production [kW]	Actual power production in each time step, optimized by the model.
Available power [kW]	Maximum potential power production based on installed capacity and irradiance profile.
Investment cost [€]	Total investment cost for all PV arrays.
Maintenance cost [€/year]	Annual operation and maintenance costs.

Variable	Description
Degradation cost [€/year]	Annual cost accounting for system degradation over its lifetime.

The optimization model determines the optimal installed capacity for each PV array within the specified minimum and maximum size constraints. The power production is constrained by the available irradiance profile and the installed capacity.

4.2.5.3 Interface options

The PV component provides a tabbed interface with two data input methods:

1. **Generate data tab**: Create PV production profiles using PVGIS (Photovoltaic Geographical Information System).
    - Set orientation and inclination angles for the PV array
    - Specify the available area for PV module placement
    - Use *Check generated data* to visualize the production profile
2. **Import data tab**: Import custom PV production profiles from a CSV file.
    - Click *Browse...* to select a CSV file with production data
    - Specify the peak power of the system for which the data was generated
    - Set available area and existing PV parameters
    - Click *Import power profile* to process and save the data
    - Use *Check imported data* to visualize the imported profile

Data format requirements for import:

- Single column CSV file with no header
- 8760 rows (hourly), 17,520 rows (30-min), or 35,040 rows (15-min)
- Values represent mean production power at the output of the PV inverter [kW]
- For leap year data, remove one day to match expected row counts

4.2.5.4 Related Components

- Location - Geographic coordinates are used for PVGIS data generation

4.2.5.5 Troubleshooting

Common issues and solutions:

- **PV production profile is not yet generated**: Before running optimization, you must either generate a production profile from PVGIS or import one from a file. Use the "Generate power profile" or "Import power profile" button.
- **PVGIS API connection failed**: Check your internet connection. The PVGIS service may be temporarily unavailable.
- **Invalid file format for import**: The CSV file must have exactly one column with 8760, 17,520, or 35,040 numeric values and no header row.
- **Available area must be positive**: If "Limit PV size directly" is not checked, the available area must be greater than zero. Enter a valid surface area for PV placement.
- **Minimum size too large for available area**: The minimum PV size cannot exceed the maximum possible size calculated from the available area and power per area ratio. Either reduce the minimum size or increase the available area.
- **Reference power must be larger than 0**: When importing production data, you must specify the peak power of the system for which the data was generated.
- **Coordinates not set**: PVGIS data generation requires location coordinates. Ensure the location is properly set in the Location component.

4.2.6 Wind farm

The **Wind farm** (WF) component represents a wind power installation consisting of one type of wind turbine. Once the location of the wind farm is defined, the power production profile for a specific installed capacity can be generated. The tool then scales this profile to determine the optimal size of the wind farm. The scaling factor of the production profile is multiplied by the reference capacity to calculate the optimal capacity for installation.

4.2.6.1 Input parameters

Basic parameters

The basic parameters are organized in two tabs, allowing users to either generate data from an API or import their own reference power profiles.

Parameters in *Generate data* tab

Parameter	Description
Hub height [m]	The height of the turbine's tower, that is, how far the blades are centred above the ground. Hub heights are limited to between 10 and 150 m.

Parameters in *Import data* tab

This option allows the user to define the wind power production profile by importing a time-series dataset from a CSV file.

Parameter	Description
File with the production profile [text field]	The custom profile of production power needs to be in a CSV file, formatted in a single column and 8760, 17,520, or 35,040 rows (no header), where values represent hourly, half-hourly, or 15-min mean power in kW, respectively. If the data in the file is for a leap year, please remove 1 day of data to match 8760, 17,520, or 35,040 rows. The first value in the file corresponds to the beginning of the year, while the last value corresponds to the end of the year.
Reference power [kW]	The installed capacity, i.e. power, of the wind power plant from which the data was collected.

Advanced parameters

Parameter	Description
Minimum capacity [kW]	Lower limit on the installed capacity. If this value is greater than zero, "Maximal payoff period" and "Maximum investment" parameters are disabled.
Maximum capacity [kW]	Upper limit on the installed capacity.
Type of the wind turbine [dropdown menu]	Choose the type of the wind turbines used in the wind farm. The parameters will be filled with predefined values. After selection, you can change any of the parameters.
Investment price [€/kW]	Investment price of the wind farm, per unit of power, includes wind turbine(s), all auxiliary materials, labour (mounting, commissioning, etc.), and financing until full operation.
Maintenance price [€/kW]	Estimated annual maintenance price per unit of power. It does not include degradation of the wind turbine(s).

Parameter	Description
Replacement price [€/kW]	Same as investment price, but after the lifetime of the wind turbine(s), i.e. the future investment price. Also, it is not influenced by the subsidy.
Lifetime [years]	Number of years after which the wind turbine(s) needs to be replaced.
Subsidy [%]	Percentage that is deductible from the investment cost. It does not apply to the replacement cost.

4.2.6.2 Optimization output

After running the optimization, the tool calculates the optimal technical and economic values for the defined scenario. This subsection shows what variables are tied to the *Wind farm* component.

Variable	Description
Optimal installed capacity	The optimization determines the optimal installed capacity of the wind farm [kW]. This value is calculated by scaling the reference production profile to minimize the total cost while meeting the energy demands of the hydrogen hub.
Power production profile	The time-dependent profile of electricity generated by the wind farm. Results include: - **Power production profile** as mean powers for each time step [kW] - **Total energy produced** over the simulation period [kWh]
Investment cost	The total investment cost for the optimal wind farm capacity [€]. This includes the wind turbine(s), all auxiliary materials, labour, and financing. If a subsidy is defined, it is deducted from this cost.
Annual maintenance cost	The estimated annual maintenance cost based on the optimal installed capacity [€/year].
Replacement cost	The future investment cost for replacing the wind turbine(s) after their lifetime [€].

4.2.6.3 Interface options

The Wind farm component provides a tabbed interface with two data input methods:

1. **Generate data tab**: Automatically generate a wind power production profile based on the location and hub height. The tool uses the renewables.ninja API to fetch typical wind power production data for the specified location and turbine configuration. After setting the hub height, click “Check power production” to generate and visualize the profile.

2. **Import data tab**: Import a custom wind power production profile from a CSV file. This is useful when you have site-specific measurements or want to use data from a particular wind turbine model. After importing the file and setting the reference power, click "Check power production" to visualize the profile.

Data format requirements for custom profiles:

- File format: .csv
- Single column (no header row)
- Allowed number of rows:
  - 8,760 (hourly resolution)
  - 17,520 (30-minute resolution)
  - 35,040 (15-minute resolution)
- Each row must contain a numerical value representing the mean power output in kW
- The first value corresponds to the beginning of the year (January 1st, 00:00)
- The last value corresponds to the end of the year (December 31st, 23:00 for hourly resolution)

4.2.6.4 Related components

- Location, as the generated wind power profile depends on the geographic coordinates..

4.2.6.5 Troubleshooting

Common issues and solutions:

- **Hub height out of range**: Ensure the hub height is between 10 and 150 m. Values outside this range will cause an error.
- **Production profile is not defined**: When using the "Import data" tab, ensure you have imported a valid CSV file and set the reference power. Click "**Browse...**" to select a properly formatted CSV file.
- **Reference power must be greater than 0**: When importing custom data, the reference power must be a positive value representing the installed capacity of the wind power plant from which the data was collected.
- **Invalid file format**: The custom production profile file must be a single-column CSV with no header. Ensure your file meets this requirement.
- **Incorrect number of rows**: The production profile file must contain exactly 8,760, 17,520, or 35,040 rows. If your data is from a leap year, remove one day of data to match the expected row counts.

- **API connection failed**: If generating data fails, check your internet connection. The tool requires access to the renewables.ninja API to fetch wind power production data.
- **Minimum capacity greater than maximum capacity**: Ensure the minimum capacity is not greater than the maximum capacity. This will cause a validation error.

4.2.7 Leonitus wave energy converter

The **Leonitus wave energy converter** (WEC) component represents the Leonitus wave power plant that produces electricity from waves. The plant is sized by its installed power S_{LWEC} [kW], which the optimization determines. The plant is controllable, so in every time step it can produce any power between zero and its installed power:

$$0 \leq P_{\mathrm{LWEC}}(k) \leq S_{\mathrm{LWEC}} \leq S^{\max}_{\mathrm{LWEC}}.$$

Because the production can always be lowered, the plant never makes the problem infeasible when its electricity cannot be used.

The wave potential of the site is expressed by the **unit price** [€/kW]. One Leonitus plant (19 m x 16 m x 8 m) costs about 1.5 M€ and gives a constant 1 MW at the location Žigljen, i.e. 1500 €/kW. The same plant gives a lower constant power at a site with a weaker wave potential (e.g. 900 kW, i.e. 1667 €/kW) and a higher one at a site with a stronger potential (e.g. 1100 kW, i.e. 1364 €/kW).

The classic wave energy converter, sized by its length and a wave power profile, is no longer available. Projects saved with it are opened without it, and the Leonitus WEC is not selected in them.

4.2.7.1 Input parameters

Basic parameters

Parameter	Description
Unit price [€/kW]	Investment price per kW of installed power, which expresses the wave potential of the site (1500 €/kW at Žigljen). It includes the device, all auxiliary materials, labour (mounting, commissioning, etc.), and financing until full operation.

Advanced parameters

Parameter	Description
Maximum installed power [kW]	Upper limit on the installed power of the Leonitus WEC.
Maintenance price [€/kW]	Estimated annual maintenance price per kW of installed power. It does not include degradation of the plant.
Replacement price [€/kW]	Same as unit price, but after the lifetime of the plant, i.e. the future investment price. Also, it is not influenced by the subsidy.
Lifetime [years]	Number of years after which the plant needs to be replaced.
Subsidy [%]	Percentage that is deductible from the investment cost. It does not apply to the replacement cost. If a subsidy is set in the Financial parameters, that value is used instead.

4.2.7.2 Optimization output

After running the optimization, the tool calculates the optimal technical and economic values for the defined scenario. This subsection shows what variables are tied to the *Leonitus WEC* component.

Variable	Description
Optimal installed power	The optimal installed power of the Leonitus WEC [kW].
Power production profile	Electricity produced by the plant as mean powers for each time step [kW] and the total energy produced over the simulation period.
Capacity factor	Ratio of the produced energy and the energy the plant would produce at its installed power during the whole simulation period [%].
Investment cost	Unit price multiplied by the installed power [€]. If a subsidy is defined, it is deducted from this cost.
Annual maintenance cost	Maintenance price multiplied by the installed power [€/year].
Replacement cost	The future investment cost for replacing the plant after its lifetime, if it is shorter than the design lifetime of the project [€/year].

4.2.8 Battery energy storage system

The **battery energy storage system (BESS)** consists of:

- a **battery pack**, which determines the energy capacity of the BESS [kWh], and
- a **power converter**, which determines the charge/discharge power of the BESS [kW].

4.2.8.1 Input parameters

Basic parameters

Parameter	Description
BESS already exists? [checkbox]	Whether there is an existing battery energy storage system at the site or not. If enabled, existing capacity and power parameters are shown.
Existing capacity [kWh]	Rated capacity of the existing battery pack. It is visible only if *BESS already exists?* is activated.
Existing power [kW]	Rated power of the existing power converter. The rated power is assumed to be bidirectional, i.e. the same power is considered for both charging (rectifier mode) and discharging (inverter mode) of the battery. It is visible only if *BESS already exists?* is activated.

Advanced parameters

Parameter	Description
Minimum battery pack capacity [kWh]	Lower limit on the battery pack capacity. If this value is greater than zero, the options *Maximal payoff period* and *Maximum investment* under *Financial parameters* must be disabled.
Maximum battery pack capacity [kWh]	Upper limit on the battery pack capacity.
Minimum power rating [kW]	Lower limit on the power converter's nominal power. If this value is greater than zero, the options *Maximal payoff period* and *Maximum investment* under *Financial parameters* must be disabled.
Maximum power rating [kW]	Upper limit on the power converter's nominal power.
Battery system type [dropdown menu]	A battery system from the application's database. Once selected, all parameters below are automatically populated. The user may modify any parameter afterward.
Charging efficiency (overall) [%]	Overall efficiency of the BESS during battery charging. It includes both the power converter efficiency and the battery efficiency.
Discharging efficiency (overall) [%]	Overall efficiency of the BESS during battery discharging. Includes both the power converter efficiency and the battery efficiency.

Parameter	Description
Number of cycles [-]	Number of charging-discharging cycles that the battery is rated for without significant drop in capacity. Usually, the *number of cycles* is presented together with *depth of discharge* in datasheets of battery packs.
Depth of discharge [%]	Percentage representing the portion of the battery capacity that can be repeatedly discharged without significantly affecting battery health. Usually, the *number of cycles* is presented together with *depth of discharge* in datasheets of battery packs.
Maximum charging rate [1/h]	The maximum charging rate defines the upper limit of charging power relative to the battery energy capacity. It represents the C-rate at which the battery can be charged. A charging rate of 1 means the battery can be fully charged in one hour, 0.5 means it takes two hours, and 2 means it can be charged in half an hour.
Maximum discharging rate [1/h]	The maximum discharging rate defines the upper limit of charging power relative to the battery energy capacity. It represents the C-rate at which the battery can be discharged. A discharging rate of 1 means the battery can be fully discharged in one hour, 0.5 means it takes two hours, and 2 means it can be discharged in half an hour.
Investment price (battery) [€/kWh]	Investment price of the battery pack, per unit of energy capacity. It includes the battery pack itself, all auxiliary materials (cables, mounting brackets, etc.) and labour (mounting, commissioning, etc.) required to bring the system into full operation.
Maintenance price (battery) [€/kWh]	Estimated annual maintenance cost per unit of energy capacity. Battery degradation is not included.
Replacement price (battery) [€/kWh]	Same as investment price, but after the lifetime of the battery pack, i.e. the future investment price. After the battery has degraded, it will be replaced with this price. Degradation depends on battery usage, number of cycles, and depth of discharge. The replacement cost is not affected by the subsidy.
Investment price (converter) [€/kW]	Investment price of the power converter, per unit of power. It includes the power converter itself, all auxiliary materials (cables, mounting brackets, etc.) and labour (mounting, commissioning, etc.) required to bring the system into full operation.
Maintenance price (converter) [€/kW]	Estimated annual maintenance price per unit of power. Power converter degradation is not included.
Replacement price (converter) [€/kW]	Same as the investment price, but applied after the converter lifetime has expired, i.e. the future investment price. Also, it is not influenced by the subsidy.
Lifetime (converter) [years]	Number of years after which the power converter needs to be replaced.

Parameter	Description
Subsidy [%]	Percentage deducted from the investment cost. The same subsidy applies to both the battery pack and the power converter. It does not apply to replacement costs.

4.2.8.2 Optimization output

Variable	Description
Battery storage capacity [kWh]	Optimized nominal energy capacity of the battery pack. It represents the required usable storage size that enables the selected operational strategy and economic constraints.
Battery power converter [kW]	Optimized nominal bidirectional power of the battery's power converter. It defines the maximum charging and discharging power available for battery operation.
Battery charging profile [kW]	Time series of battery charging power over the simulation period. Positive values indicate charging from the electrical system, subject to converter limits, constraints, and battery availability.
Battery discharging profile [kW]	Time series of battery discharging power over the simulation period. Positive values indicate energy supplied from the battery to the electrical system, respecting converter limits and state-of-charge constraints.
Battery state-of-charge (SoC) [kWh]	Time series of the battery's state of charge during the simulation. It represents the available stored energy.

4.2.8.3 Interface options

The BESS parameters can be predefined by selecting one of the available BESS types from the database. Any parameter can still be manually modified after selection. By default, the first BESS type in the list is used.

4.2.8.4 Troubleshooting

- **Existing sizes**: The existing capacity and power must respect the ratio defined by *Maximum charging rate* and *Maximum discharging rate*. Otherwise, the optimization may fail.

4.2.9 Fixed electrical consumption

The **Fixed electricity consumption** component represents all existing electricity consumers at the location. It contains a power profile of all the existing electricity consumers, which can be imported directly or simulated with specific consumer types from the application's database.

This component defines the baseline electricity demand that must be supplied by the electrical grid connection or other electricity sources in the system.

4.2.9.1 Input parameters

The component offers two methods for defining the electricity consumption profile:

- **Generate profile**: Create a profile by selecting consumer types from the database.
- **Import data**: Import an existing consumption profile from a CSV file.

Parameters in *Generate profile* tab

Parameter	Description
Consumer type [dropdown menu]	Choose a consumer type from the application's database.
Annual electrical energy consumption [MWh]	Annual energy consumption of selected consumer. The profile selected by *Consumer type* will be scaled so that total energy is equal to this number.

GUI elements in *Import data* tab

This option allows the user to import an existing electricity consumption profile from a CSV file.

GUI element	Description
Browse [button]	The power consumption profile needs to be in a CSV file, formatted in a single column with 8,760, 17,520, or 35,040 rows (no header), where values represent mean consumption power at the grid connection point. If the data in the file is for a leap year, please remove 1 day of data to match 8,760, 17,520, or 35,040 rows. The first value in the file corresponds to the beginning of the year, while the last value corresponds to the end of the year. After selecting a file, a message is displayed indicating whether the data import was successful or if errors were detected.

GUI element	Description
Check imported data [button]	Opens a pop-up window with an interactive chart showing the imported fixed electricity consumption data.

Additional GUI elements

GUI element	Description
Delete consumer [button]	Click if you want to remove this consumer from your fixed electricity consumption system. All data will be lost.
Add new consumer [button]	Click if you want to add a new electricity consumer to the overall fixed electricity consumption.
Fixed electricity consumption profile [button]	Opens a pop-up window displaying the fixed electricity consumption profile chart.

4.2.9.2 Optimization output

This component does not introduce any decision variables in the optimization model.

4.2.9.3 Interface options

The Fixed electricity consumption component provides the following interface options:

- **Generate profile**: Select consumer types from the database and specify their installed power and annual energy consumption to automatically generate a consumption profile.
- **Import data**: Import an existing consumption profile from a CSV file with 8,760 (hourly), 17,520 (30-minute), or 35,040 (15-minute) rows.
- **Multiple consumers**: Add multiple consumers to create a combined consumption profile representing all existing electricity loads at the location.
- **Data verification**: Use the *Check imported data* button to visually verify the consumption profiles.

4.2.9.4 Related components

- Electrical grid connection, as the fixed electricity consumption determines the minimum grid connection capacity required.

4.2.9.5 Troubleshooting

Common issues and solutions:

- **Issue 1**: CSV file import fails with row count error. Ensure your file has exactly 8,760, 17,520, or 35,040 rows with no header. If using leap year data, remove one day (24, 48, or 96 rows depending on resolution).

4.2.10 Electrolyser

The **electrolyser** is the key component in hydrogen production. It uses electricity to split water molecules into hydrogen and oxygen, while producing heat due to inefficiencies in the conversion process. Therefore, it has 2 inputs (electricity and water) and 3 outputs:

- **Hydrogen (H_2)** – the main output of the electrolyser.
- **Oxygen (O_2)** – a byproduct of electrolysis that can be utilized or released into the atmosphere.
- **Heat** – thermal energy resulting from conversion losses. The tool distinguishes between low-temperature (LT) and high-temperature (HT) heat. An electrolyser may produce either of these, both, or none, depending on the selected configuration.

The user can specify multiple electrolysers, each with its own set of parameters. An electrolyser is added by clicking on the **Add new electrolyser** button.

By right-clicking on an electrolyser node, a pop-up menu appears with the following options:

- **Rename electrolyser** – change the name of the electrolyser.
- **Delete electrolyser** – removes the electrolyser from the overall setup.

4.2.10.1 Input parameters

Basic parameters – *Electrolyser*

Parameter	Description
Electrolyser type [dropdown menu]	An electrolyser from the tool catalog. Once selected, all required parameters are automatically populated according to the chosen type. Any parameter can be modified afterward.

Parameter	Description
Electrical efficiency [%]	Efficiency of converting electrical energy into hydrogen on a higher heating value (HHV) basis. A value of 100% corresponds to the thermodynamic minimum energy requirement: - 285.8 kJ/mole - 39.41 kWh/kg - 3.54 kWh/Sm3 - 3.36 kWh/Nm3 Sm3 = Standard cubic meter means temperature of 0 °C and pressure of 1.01325 bar (standard DIN 1343). Nm3 = Normal cubic meter means temperature of 15 °C and pressure of 1.01325 bar (standard ISO 2533). This parameter is tied to *Power consumption (by mass)* and *Power consumption (by volume).*
Power consumption (by mass) [kWh/kg]	Electrical energy required to produce 1 kg of hydrogen. The minimum value is 39.41 kWh/kg, corresponding to 100% efficiency (HHV limit). This parameter is tied to *Electrical efficiency* and *Power consumption (by volume).*
Power consumption (by volume) [kWh/Nm3]	Electrical energy required to produce 1 Nm3 of hydrogen. Nm3 (normal cubic meter) denotes hydrogen at 15 °C and 1.01325 bar (11.128 Nm^3/kg). The minimum value is 3.36 kWh/Nm3, corresponding to 100% efficiency. This parameter is tied to *Electrical efficiency* and *Power consumption (by mass).*
Investment price [€/kW]	Specific investment cost per unit of input electrical power. It includes all required equipment, materials, and labor (design, installation, commissioning, etc.) until full operation.
Electrolyser already exists? [checkbox]	Indicates whether an electrolyser is already installed at the site.
Existing electrolyser size [kW]	Nominal electrical power of the existing electrolyser. Visible only if *Electrolyser already exists?* is enabled. The parameters above apply to both the existing unit and any optimized expansion.

Advanced parameters – *Electrolyser*

Parameter	Description
Recoverable high temperature heat? [checkbox]	Indicates whether heat from the electrolyser cooling circuit can be recovered for high-temperature heating applications.
Heating efficiency (high temperature) [%]	The *Electrical efficiency* determines hydrogen production per unit of electricity, while the remaining share represents total heat losses. *Heating efficiency* defines the fraction of this recoverable heat available at high temperature. The sum of high- and low-temperature heating efficiencies must not exceed 100%. This parameter is tied to *Usable heat output.*

Parameter	Description
Usable heat output (high temperature) [kWh/kg]	The amount of usable heat (at high temperature) produced by the electrolyser for every 1 kg of produced hydrogen. This parameter is calculated as: *Power consumption (by mass)* × (100% - *Electrical efficiency*) × *Heating efficiency (high temperature)*
Operating temperature (high temperature) [°C]	Temperature at which the high-temperature heat is released from the electrolyser at nominal conditions, i.e. nominal power.
Recoverable low temperature heat? [checkbox]	Indicates whether heat from the electrolyser cooling circuit can be recovered for low-temperature heating applications.
Heating efficiency (low temperature) [%]	The *Electrical efficiency* determines hydrogen production per unit of electricity, while the remaining share represents total heat losses. *Heating efficiency* defines the fraction of this recoverable heat available at low temperature. The sum of high- and low-temperature heating efficiencies must not exceed 100%. This parameter is tied to *Usable heat output*.
Usable heat output (low temperature) [kWh/kg]	The amount of usable heat (at low temperature) produced by the electrolyser for every 1 kg of produced hydrogen. This parameter is calculated as: *Power consumption (by mass)* × (100% - *Electrical efficiency*) × *Heating efficiency (low temperature)*
Operating temperature (low temperature) [°C]	Temperature at which the low-temperature heat is released from the electrolyser at nominal conditions, i.e. nominal power.
Hydrogen outlet pressure [bar]	Gauge pressure of hydrogen at the electrolyser outlet under nominal conditions, i.e. at nominal power, expressed as difference from the atmospheric pressure (sometimes denoted with "barg"). The lowest outlet pressure of all hydrogen sources is the pressure of the hydrogen bus, i.e. the inlet pressure of the hydrogen compressor in Hydrogen storage.
Oxygen outlet pressure [bar]	Gauge pressure of oxygen at the electrolyser outlet under nominal conditions, expressed as difference from the atmospheric pressure (barg). In many electrolysers the oxygen side is at a lower pressure than the hydrogen side. The oxygen outputs of all electrolysers are joined, so the lowest oxygen outlet pressure is the inlet pressure of the oxygen compressor in O2 sales and is used to calculate its *Power consumption*.
Water consumption [l/kg]	Water required for producing 1 kg of hydrogen. The theoretical minimum is approximately 9 kg (9 l) of water per kg of hydrogen.
Minimum electrolyser size [kW]	Lower limit on the size (installed power in [kW]) of the electrolyser. If greater than zero, *Maximal payoff period* and *Maximum investment* in *Financial parameters* must be disabled.

Parameter	Description
Maximum electrolyser size [kW]	Upper limit on the size (installed power in [kW]) of the electrolyser.
Maintenance price [€/kW]	Estimated annual operation and maintenance (O&M) price per installed power. O&M does not include degradation of the system.
Replacement price [€/kW]	Future specific investment cost applied when the electrolyser reaches end of life. The subsidy does not apply to replacement costs.
Lifetime [years]	Number of years after which the whole electrolyser must be replaced.
Subsidy [%]	Percentage that is **deductible** from the investment cost. Does not apply to the replacement price.

4.2.10.2 Optimization output

Electrolyser

Variable	Description
Installed capacity [kW]	Optimized installed electrical power for each electrolyser and in total.
Power consumption [kW]	Electrical power consumption of each electrolyser in every time step, optimized by the model. Based on the power consumption, the following time series are derived: - **Hydrogen production** [kg/h] - **Oxygen production** [kg/h] - **Low-temperature heat generation** [kW] - **High-temperature heat generation** [kW] - **Water consumption** [kg/h]
Investment cost [€]	Investment costs for each electrolyser and in total.
Maintenance cost [€]	Annual operation and maintenance costs for each electrolyser and in total.
Degradation cost [€]	Annual cost associated with system degradation over its lifetime, reported for each electrolyser and in total.

4.2.10.3 Interface options

Parameters of electrolysers can be predefined by selecting one of the available types from the database. Any parameter can still be manually modified after selection. By default, the first type in the list is used.

4.2.10.4 Related Components

- Water supply – defines water supply used by the electrolysers
- Hydrogen storage – hydrogen compressor and hydrogen storage tank, which take hydrogen from the electrolysers through the hydrogen bus
- H2 sales – hydrogen sales, where another hydrogen compressor is defined
- O2 sales – oxygen sales, where oxgen compressor and oxygen storage tank are defined
- Waste heat – waste heat utilization, where heat consumption is defined

4.2.10.5 Troubleshooting

- **No electrolyser defined**: At least one electrolyser must be defined in the tool. Otherwise, the optimization will return an error.
- **Deleting electrolyser**: If an electrolyser is removed, all associated data are deleted as well. Once an electrolyser is deleted, it cannot be restored.
- **Efficiency and power consumption do not match**: When reviewing a product datasheet, the manufacturer may provide both the efficiency and the specific power consumption of the electrolyser. If these values do not match those in the tool, the efficiency may be reported with respect to the lower heating value (LHV) instead of HHV, or it may refer only to the electrolyser stack while excluding auxiliary equipment (e.g. power electronics).

4.2.11 Water supply

Water is a feedstock for hydrogen production via electrolysis. The primary source is the **water grid**, while **rainwater** and a **body of water** can serve as additional renewable sources. All water used must be demineralized before entering the electrolyser. This section describes the parameters for the following components:

- **Water grid connection**: the primary source of water. It is assumed that it has no limit on usage. It is used if any hydrogen is being produced.
- **Rain collection**: a secondary water source limited by the local rainfall and the collection area. If enabled, it requires a *water pump* and a *water tank*. Monthly precipitation data is obtained from Open-Meteo API through internet.
- **Body of water**: a secondary water source such as a river, lake, stream, or estuary. Daily availability is limited. If enabled, it requires a water pump.

- **Demineralizer**: prepares water for the electrolyser by removing minerals and impurities. It is used for all water sources.
- **Water pump**: pumps water from the water tank and from a body of water. Its electricity consumption comprises overcoming the height difference and any (ultra-)fine filtration. It is used if at least one of *Rain collection* and *Body of water* is enabled.
- **Water tank**: stores collected rainwater. It is used only when *Rain collection* is enabled.

4.2.11.1 Input parameters

Water grid connection parameters

Parameter	Description
Show predefined values for the country [dropdown menu]	The country is automatically selected according to the site location, but it can be changed manually. Available countries are those that participated in the NACHIP project: Croatia, Slovenia and Italy. Further available countries had participated in previous projects with the tool developers: Austria, Czechia, Germany, Hungary and Poland. If the selected location is in another country, the value of this parameter remains empty. By selecting one of the countries, all other parameters are populated. Any parameter can be modified afterward.
Water price [€/m3]	The price of water from the water grid, including all the fees and taxes.
Existing water connection point? [checkbox]	Whether the grid connection already exists at the location or not.
Existing water connection capacity [m3/h]	The size of the existing grid connection in terms of water mass flow. The parameter is visible only if *Existing water connection point?* is enabled.
Capacity price for water grid connection [€/(m3/h)]	It represents the price for building and installation of the connection depending only on the capacity, i.e. size. It can be determined from existing projects: cost of the connection divided with the capacity of the connection.
Unit price for water grid connection [€/(m3/h km)]	It represents the price for building and installation of the connection depending on the capacity and *Distance to the nearest connection point*. This parameter is used only for greenfield investments – it is visible only when *Existing water connection point?* is disabled. It can be determined from existing projects: cost of the connection divided with the capacity and distance to the nearest connection point.

Parameter	Description
Distance to the nearest connection point [km]	A *connection point* represents a piece of infrastructure to which the renewable energy hub must be connected to. This parameter is used only for greenfield investments – it is visible only when *Existing water connection point?* is disabled. It is used to calculate the total price of the grid connection: *Unit price* * *Distance* + *Capacity price*

Rain collection parameters

Parameter	Description
Available area [m2]	Area that is available for rain collection. For example, if rain can be collected from a roof, the available area is the area of the roof.
Expected rainfall in *Month* [mm]	The depth of precipitation that occurs over a unit area, i.e. measure of rainfall amount. There is *Expected rainfall* for every month of the year. The values for each month are automatically populated based on the selected location, but can be modified afterward. Source of rainfall data is Open-Meteo and it is based on the entered Location.
Existing rain collection area [m2]	Rainfall catchment area that is already present at the site.
Minimum rain collection area [m2]	Lower limit on the size of the rainfall catchment area. If this value is greater than zero, *Maximal payoff period* and *Maximum investment* in *Financial parameters* must be disabled.
Investment price [€/m2]	Specific investment cost per unit of area. It includes all required equipment, materials, and labor until full operation.
Maintenance price [€/m2]	Estimated annual operation and maintenance (O&M) price per installed area. O&M does not include degradation of the system.
Replacement price [€/m2]	Future specific investment cost applied when the rain collection system reaches end of life. The subsidy does not apply to replacement costs.
Lifetime [years]	Number of years after which the rain collection system must be replaced.
Subsidy [%]	Percentage that is deductible from the investment cost. Does not apply to the replacement price.

Body of water parameters

Parameter	Description
Price of extracted water [€/m3]	Price of water extracted from the body of water, including all the fees and taxes.

Parameter	Description
Maximum amount of water for extraction (daily) [m3]	Maximum amount of water for extraction permitted by, e.g. local authorities. Same value applies for every day of the year.
Investment price [€/(m3/h)]	Specific investment cost of the water extraction system, per unit of water flow. It includes all required equipment, materials, and labor until full operation.
Maintenance price [€/(m3/h)]	Estimated annual operation and maintenance (O&M) price of the water extraction system, per unit of water flow. O&M does not include degradation of the system.
Replacement price [€/(m3/h)]	Future specific investment cost applied when the water extraction system reaches end of life. The subsidy does not apply to replacement costs.
Lifetime [years]	Number of years after which the water extraction system must be replaced.
Subsidy [%]	Percentage that is deductible from the investment cost. Does not apply to the replacement price.

Demineralizer parameters

Parameter	Description
Demineralizer already exists? [checkbox]	Indicates whether a demineralizer is already installed at the site.
Existing demineralizer size [m3/h]	Nominal output water flow of the existing demineralizer. Visible only if *Demineralizer already exists?* is enabled. The parameters below apply to both the existing unit and any optimized expansion.
Demineralizer type [dropdown menu]	A demineralizer from the tool catalog. Once selected, all required parameters are automatically populated according to the chosen type. Any parameter can be modified afterward.
Power consumption [kWh/m3]	Amount of electrical energy required to produce 1 m3 of demineralized water.
Water losses [%]	Percentage of water that is lost at the demineralizer due to its way of operation.
Minimum demineralizer size [m3/h]	Lower limit on the size (installed capacity in [m3/h]) of the demineralizer. If greater than zero, *Maximal payoff period* and *Maximum investment* in *Financial parameters* must be disabled.
Maximum demineralizer size [m3/h]	Upper limit on the size (installed capacity in [m3/h]) of the demineralizer.

Parameter	Description
Investment price [€/(m3/h)]	Specific investment cost of the demineralizer system, per unit of water flow. It includes all required equipment, materials, and labor until full operation.
Maintenance price [€/(m3/h)]	Estimated annual operation and maintenance (O&M) price of the demineralizer system, per unit of water flow. O&M does not include degradation of the system.
Replacement price [€/(m3/h)]	Future specific investment cost applied when the demineralizer system reaches end of life. The subsidy does not apply to replacement costs.
Lifetime [years]	Number of years after which the demineralizer system must be replaced.
Subsidy [%]	Percentage that is deductible from the investment cost. Does not apply to the replacement price.

Water pump parameters

Parameter	Description
Water pump already exists? [checkbox]	Indicates whether a water pump unit is already installed at the site.
Existing water pump size [kW]	Nominal electrical power of the existing water pump unit. Visible only if *Water pump already exists?* is enabled. The parameters below apply to both the existing unit and any optimized expansion.
Water pump type [dropdown menu]	A water pump unit from the tool catalog. Once selected, all required parameters are automatically populated according to the chosen type. Any parameter can be modified afterward.
Water pump power consumption [kWh/m3]	Specific power consumption for pumping water. The power is used for overcoming the height difference and (ultra-)fine filtration.
Minimum water pump size [kW]	Lower limit on the size (installed power) of the water pump unit. If greater than zero, *Maximal payoff period* and *Maximum investment* in *Financial parameters* must be disabled.
Maximum water pump size [kW]	Upper limit on the size (installed power) of the water pump unit.
Investment price [€/kW]	Specific investment cost of the water pump unit, per unit of electrical power. It includes all required equipment, materials, and labor until full operation.
Maintenance price [€/kW]	Estimated annual operation and maintenance (O&M) price of the water pump unit, per unit of electrical power. O&M does not include degradation of the system.

Parameter	Description
Replacement price [€/kW]	Future specific investment cost applied when the water pump unit system reaches end of life. The subsidy does not apply to replacement costs.
Lifetime [years]	Number of years after which the water pump unit must be replaced.
Subsidy [%]	Percentage that is deductible from the investment cost. Does not apply to the replacement price.

Water tank parameters

Parameter	Description
Water tank already exists? [checkbox]	Indicates whether a water tank is already installed at the site.
Existing water tank size [m3]	Capacity of the existing water tank. Visible only if *Water tank already exists?* is enabled. The parameters below apply to both the existing unit and any optimized expansion.
Water tank type [dropdown menu]	A water tank from the tool catalog. Once selected, all required parameters are automatically populated according to the chosen type. Any parameter can be modified afterward.
Minimum water tank size [m3]	Lower limit on the size (available capacity) of the water tank. If greater than zero, *Maximal payoff period* and *Maximum investment* in *Financial parameters* must be disabled.
Maximum water tank size [m3]	Upper limit on the size (available capacity) of the water tank.
Investment price [€/m3]	Specific investment cost of the water tank, per unit of storage capacity. It includes all required equipment, materials, and labor until full operation.
Maintenance price [€/m3]	Estimated annual operation and maintenance (O&M) price of the water tank, per unit of storage capacity. O&M does not include degradation of the system.
Replacement price [€/m3]	Future specific investment cost applied when the water tank reaches end of life. The subsidy does not apply to replacement costs.
Lifetime [years]	Number of years after which the water tank must be replaced.
Subsidy [%]	Percentage that is deductible from the investment cost. Does not apply to the replacement price.

4.2.11.2 Optimization output

Variable	Description
Installed capacities	Optimized installed sizes: - **Capacity of the water grid connection** [m3/h] - **Rain collection area** [m2] - **Capacity of the water extraction system** for the body of water [m3/h] - **Capacity of the demineralizer** [m3/h] - **Water pump installed power** [kW] - **Capacity of the water tank** [m3]
Water flows	Profiles of water sources as flows in every time step, optimized by the model: - **From the water grid connection** [m3/h] - **From the rain collection system** [m3/h] - **From the body of water** [m3/h]
Power consumptions	Electrical power consumption of water preparation components in each time step, optimized by the model: - **Power consumption of the demineralizer** [kW] - **Power consumption of the water pump** [kW]
Tank state of charge (SoC)	Amount of water stored in the tank at each time step optimized by the model [m3].
Investment costs [€]	Total investment cost of each component: water grid connection, rain collection system, water extraction system, demineralizer, water pump, and water tank.
Maintenance cost [€]	Annual operation and maintenance cost of each component: water grid connection, rain collection system, water extraction system, demineralizer, water pump, and water tank.
Degradation cost [€]	Annual cost accounting for system degradation over its lifetime, of each component: water grid connection, rain collection system, water extraction system, demineralizer, water pump, and water tank.

4.2.11.3 Interface options

Water supply components are visible depending on the system configuration:

- **Water grid connection**: It is always visible, because it is the primary water source of any electrolyser.
- **Rain collection system**: It is visible if checkbox *Collect rain?* is enabled.
- **Body of water**: It is visible if checkbox *Use a body of water?* is enabled.
- **Demineralizer**: It is the main water processing component, so it is always visible.
- **Water pump**: It is visible if *Rain collection system* or *Body of water* are enabled.
- **Water tank**: It is visible only if *Rain collection system* is enabled.

Parameters of all water supply components can be predefined by selecting one of the available types from the database. Any parameter can still be manually modified after selection. By default, the first type in the list is used.

4.2.11.4 Related Components

- Electrolyser – water supply is necessary for the electrolysis.
- Location – monthly rainfall is dependent on the geographic location.

4.2.11.5 Troubleshooting

Common issues and solutions:

- **Monthly rainfall is not updated with location change**: Check your internet connection. The Open-Meteo service may be temporarily unavailable.
- **Coordinates not set**: Open-Meteo data generation requires location coordinates. Ensure the location is properly set in the Location component.

4.2.12 Hydrogen storage

The **Hydrogen storage** contains the hydrogen compressor (HCOMP1) and the hydrogen storage tank (HTANK). They are needed whenever there is a hydrogen source, i.e. the Electrolyser, the waste treatment plant (WT) or the LOHC import stream, so the header is shown when at least one of them is checked. It has no checkbox of its own.

Low-pressure hydrogen bus. Hydrogen producers (electrolysers, the waste treatment reactors of WT and the dehydrogenation unit of LOHC purchase) deliver hydrogen to a low-pressure hydrogen bus. Hydrogen from waste treatment is always non-green. The bus has no storage: in every time step, all hydrogen that is produced must be used by the consumers connected to it:

- **Hydrogen compressor** – compresses hydrogen from the bus into the hydrogen storage tank,
- **Hydrogenation unit** of LOHC sales – loads hydrogen into the LOHC,
- Fuel cell stacks – if they take hydrogen from the hydrogen bus,
- Fixed hydrogen consumption – if it uses low pressure hydrogen.

Green and non-green hydrogen are kept apart on the bus and in the tank. The pressure of the bus is the lowest outlet pressure of the electrolysers, of the waste treatment reactors that produce hydrogen and of the dehydrogenation unit, or 30 bar if there are none. It is shown as *Hydrogen bus pressure* in the hydrogen compressor parameters. With the predefined reactors and dehydrogenation unit (2 bar), the bus is at 2 bar while WT or LOHC import is selected, which raises the power consumption of the hydrogen compressor and limits the fixed hydrogen consumption at low pressure.

- **Hydrogen compressor**: compresses hydrogen from the hydrogen bus to the hydrogen storage tank using electricity. Its size is determined by the hydrogen mass flow rate, while its power consumption is calculated from the inlet pressure (pressure of the hydrogen bus) and outlet pressure (rated pressure of the hydrogen tank).
- **Hydrogen storage tank**: stores hydrogen between production and consumption or sale. It supplies H2 sales (including gas grid injection), and the fuel cells and the fixed hydrogen consumption when they take hydrogen from the tank. The tank must be sized to store at least one full day of hydrogen sales and of the fixed hydrogen consumption taken from the tank.

Projects saved with earlier versions had these parameters in the Electrolyser header; they are moved here automatically when the project is opened.

4.2.12.1 Input parameters

Hydrogen compressor (advanced)

Parameter	Description
Hydrogen compressor already exists [checkbox]	Indicates whether a hydrogen compressor is already installed at the site.
Existing compressor size [kg/h]	Nominal mass flow of the existing hydrogen compressor. Visible only if *Hydrogen compressor already exists* is enabled. The parameters below apply to both the existing unit and any optimized expansion.
Hydrogen compressor unit [dropdown menu]	A hydrogen compressor unit from the tool catalogue. Once selected, all required parameters are automatically populated according to the chosen type. Any parameter can be modified afterward.
Efficiency [%]	The tool first calculates the theoretical energy required to compress hydrogen from the pressure of the hydrogen bus to the rated pressure of the hydrogen storage tank, using the adiabatic compression model. This energy is then divided by the compressor efficiency to get the electrical *Power consumption*. When entering *Efficiency*, *Power consumption* is recalculated.
Hydrogen bus pressure [bar]	Pressure of the hydrogen bus, i.e. the inlet pressure of the compressor, expressed as difference from the atmospheric pressure (barg). This value is calculated by the tool and cannot be entered. It is the lowest *Hydrogen outlet pressure* of the active electrolysers, of the waste treatment reactors that produce hydrogen and of the dehydrogenation unit of LOHC purchase, or 30 bar if there are none.

Parameter	Description
Power consumption [kWh/kg]	Electrical energy required to compress 1 kg of hydrogen from the hydrogen bus to the hydrogen storage tank. This value is calculated by the tool and cannot be entered (see the formula below the table). It is recalculated whenever *Efficiency*, *Hydrogen bus pressure* or the *Rated pressure* of the hydrogen storage tank changes.
Minimum inlet pressure [bar]	Minimum allowable hydrogen pressure at the compressor inlet. This ensures proper lubrication and cooling (especially for oil-lubricated compressors). This parameter is used only to validate compatibility between the electrolyser and the hydrogen compressor.
Maximum discharge pressure [bar]	Maximum allowable compressor discharge pressure. Exceeding this limit may lead to system overpressurization and potential mechanical failure. This parameter is used only to validate compatibility between the hydrogen compressor and the storage tank.
Investment price [€/(kg/h)]	Specific investment cost per unit of mass flow. It includes the compressor itself, additional equipment (e.g. metering devices), installation, commissioning, and all other necessities until full operation.
Maintenance price [€/(kg/h)]	Estimated annual operation and maintenance (O&M) price per mass flow. O&M does not include degradation of the system.
Replacement price [€/(kg/h)]	Future specific investment cost applied when the compressor reaches end of life. The subsidy does not apply to replacement costs.
Lifetime [years]	Number of years after which the whole compressor unit must be replaced.
Subsidy [%]	Percentage that is deductible from the investment cost. Does not apply to the replacement price.

Calculation of the compressor power consumption

The inlet pressure of the compressor is the *Hydrogen bus pressure* p^{hbus} and its outlet pressure p^{htank} is the *Rated pressure* of the hydrogen storage tank. Both are gauge pressures, so the atmospheric pressure $p_{atm} = 1.01325$ bar is added to each of them to get absolute pressures. Assuming adiabatic compression of a real gas, the power consumption e_{el} in [kWh/kg] is:

$$e_{el} = \frac{1}{\eta_{el}} \frac{\kappa}{\kappa - 1} Z\, R\, T \left[\left(\frac{p^{htank} + p_{atm}}{p^{hbus} + p_{atm}} \right)^{\frac{\kappa-1}{\kappa}} - 1 \right]$$

where η_{el} is the compressor *Efficiency* and the physical constants of hydrogen are:

- $\kappa = 1.41$ – ratio of specific heat at constant pressure to that at constant volume,

- $Z = 1.0059$ – compressibility factor, which accounts for the deviation of hydrogen from an ideal gas (taken as constant, which has negligible effect on accuracy),
- $R = 0.00114563$ kWh/(kg K) – individual gas constant,
- $T = 298$ K – temperature of hydrogen during compression.

If the tank pressure is not higher than the bus pressure, no compression is needed and the power consumption is 0. The electrical power of the compressor in each time step is e_{el} multiplied by the mass flow of hydrogen compressed into the tank.

Hydrogen storage tank (advanced)

Parameter	Description
Hydrogen tank already exists? [checkbox]	Indicates whether a hydrogen storage tank is already installed at the site.
Existing storage tank size [kg]	Nominal hydrogen mass that can be stored in the existing tank at rated pressure. Visible only if *Hydrogen tank already exists?* is enabled. The parameters below apply to both the existing unit and any optimized expansion.
Hydrogen storage tank type [dropdown menu]	A hydrogen storage tank from the tool catalogue. Once selected, all required parameters are automatically populated according to the chosen type. Any parameter can be modified afterward.
Rated pressure [bar]	Maximum allowable operating pressure of the storage tank, expressed as difference from the atmospheric pressure (barg). It is also the outlet pressure of the hydrogen compressor and is used to calculate its *Power consumption*, and the inlet pressure of the hydrogen compressor for H2 sales.
Minimum capacity [kg]	Lower limit on the size (mass capacity) of the tank. If this value is greater than zero, *Maximal payoff period* and *Maximum investment* in *Financial parameters* must be disabled.
Maximum capacity [kg]	Upper limit on the size (mass capacity) of the tank.
Investment price [€/kg]	Specific investment cost per unit of storage capacity which includes the storage tank itself, additional equipment (e.g. metering devices), installation and commissioning, and all other necessities until full operation.
Maintenance price [€/kg]	Estimated annual operation and maintenance (O&M) price per unit of storage capacity. O&M does not include degradation of the system.
Replacement price [€/kg]	Future specific investment cost applied when the tank reaches end of life. The subsidy does not apply to replacement costs.
Lifetime [years]	Number of years after which the whole storage tank unit must be replaced.

Parameter	Description
Subsidy [%]	Percentage that is deductible from the investment cost. Does not apply to the replacement price.

4.2.12.2 Optimization output

Hydrogen compressor

Variable	Description
Installed capacity [kg/h]	Optimized installed mass flow capacity of the compressor.
Power consumption [kW]	Electrical power consumption of the compressor in each time step, optimized by the model.
Investment cost [€]	Total investment cost of the hydrogen compressor.
Maintenance cost [€]	Annual operation and maintenance costs.
Degradation cost [€]	Annual cost accounting for system degradation over its lifetime.

Hydrogen storage tank

Variable	Description
Installed capacity [kg]	Optimized installed hydrogen mass capacity.
State of charge (SoC) [kg]	Amount of hydrogen stored in the tank at each time step, reported separately for green and non-green hydrogen.
Investment cost [€]	Total investment cost of the hydrogen tank.
Maintenance cost [€]	Annual operation and maintenance costs.
Degradation cost [€]	Annual cost accounting for system degradation over its lifetime.

Variable	Description
Hydrogen bus	Hydrogen compressed into the tank in each time step equals the hydrogen produced minus the hydrogen used directly from the bus by the hydrogenation unit, the fuel cells and the fixed hydrogen consumption.

4.2.12.3 Interface options

Parameters of the hydrogen compressor and the hydrogen storage tank can be predefined by selecting one of the available types from the database. Any parameter can still be manually modified after selection. By default, the first type in the list is used.

4.2.12.4 Related components

- Electrolyser, waste treatment reactors and the dehydrogenation unit of LOHC purchase – hydrogen producers connected to the hydrogen bus, which set its pressure.
- H2 sales, Fuel cell stack, Fixed hydrogen consumption – hydrogen consumers supplied from the tank.

4.2.12.5 Troubleshooting

- **Hydrogen storage header not visible**: Check the *Electrolyser* or *LOHC import* checkbox first.

4.2.13 Fixed hydrogen consumption

The **Fixed hydrogen consumption** represents hydrogen consumed within the hydrogen hub, for example by an industrial process, a refuelling station for the hub's own vehicles, or a laboratory. The demand is fixed: it must be covered every day, either by hydrogen produced in the hub or by hydrogen bought from outside at the given price. The consumer can require purely green hydrogen. The demand can be generated from a simple weekly pattern or imported as a custom profile.

The hydrogen is supplied by the Electrolyser or by the LOHC import stream (dehydrogenation unit of LOHC purchase). If they do not cover the demand, the rest is bought from outside the hub.

The component is selected with the **H2 cons.** checkbox in the *H2 usage* row. The checkbox is shown only while at least one hydrogen source (Electrolyser or LOHC import) is selected, and it is unchecked by default (see Section 4.1.2).

4.2.13.1 Input parameters

Hydrogen demand: Generate data

Parameter	Description
Demand pattern [dropdown menu]	How the daily demand repeats during the week: *Same every day*, *Weekday/weekend* or *Daily*. The inputs below the dropdown menu change with the selected pattern. Default: Same every day.
Daily hydrogen demand [kg/day]	*Same every day*: mass of hydrogen consumed every day of the simulation.

Parameter	Description
Demand Monday-Friday [kg/day]	*Weekday/weekend*: mass of hydrogen consumed on each working day.
Demand Saturday-Sunday [kg/day]	*Weekday/weekend*: mass of hydrogen consumed on each weekend day.
Demand on Monday … **Demand on Sunday** [kg/day]	*Daily*: mass of hydrogen consumed on each day of the week (seven inputs).

The generated profile has one value per day, from the start date to the end date set in the Simulation parameters. The days of the week follow the real calendar of these dates, e.g. 1 January 2023 is a Sunday.

Hydrogen demand: Import data

Parameter	Description
File with the demand profile [CSV file]	A CSV file with a single column and no header. It has either **365 rows** (daily demand in kg/day) or **8760 rows** (hourly demand in kg/h). For a leap year, remove one day of data. The first value corresponds to the beginning of the year. Negative values are not allowed.

Other parameters

Parameter	Description
Use green hydrogen only? [checkbox]	If checked, only 100% green hydrogen can be used to cover the demand. The tool takes the energy sources into account. Otherwise any mix of green and non-green hydrogen is allowed. Default: checked.
Pressure level [radio button]	*Low pressure*: the consumer can take hydrogen directly from the sources at the pressure of the hydrogen bus, without compression, and also from the hydrogen tank. *High pressure*: the consumer takes hydrogen only from the hydrogen tank, i.e. the hydrogen is compressed before it is used. Default: High pressure.
Required pressure [bar]	Pressure at which the consumer needs the hydrogen. Default: 350 bar.

Parameter	Description
Price of hydrogen bought from outside [€/kg]	Price paid for hydrogen bought from outside the hub when the local production does not cover the demand, after tax. It is also used to value the consumption that existed before the hub. It must be greater than zero.

4.2.13.2 Interface options

- **Generate or import**: Select the *Generate data* tab to build the demand from a weekly pattern, or the *Import data* tab to load a custom profile with the *Browse…* button. The selected tab is used in the optimization.
- **Switching patterns**: Values entered for the other patterns are kept, so switching back to a pattern restores them.
- **Check demand data**: Plots the demand profile of the selected tab: daily values in kg/day for generated data and for 365-row files, or hourly values in kg/h for 8760-row files.

4.2.13.3 Optimization output

Variable	Description
Hydrogen used locally [kg/h]	Green and non-green hydrogen used by the consumer, taken from the hydrogen tank and, with low pressure, directly from the hydrogen bus. The use is constant within each sample of the demand profile (one day for generated data and 365-row files, or the hydrogen sampling time for 8760-row files, whichever is longer).
Hydrogen bought [kg/year]	Part of the demand that is not covered by the hub and is bought at the given price.
Operating cost [€/year]	Cost of the bought hydrogen. Before the investment, the whole demand was bought, so the savings count as profit of the investment.

The hydrogen tank is sized to hold at least one day of hydrogen used from the tank. In the results, hydrogen used locally is also counted in the levelized cost of hydrogen (LCOH).

4.2.13.4 Related components

- Electrolyser, which produces hydrogen on site.

- The dehydrogenation unit of LOHC purchase, which releases hydrogen from bought LOHC (LOHC import stream).
- H2 sales and Fuel cell stack, which are other possible uses of the hydrogen.

4.2.13.5 Troubleshooting

Common issues and solutions:

- **H2 cons. checkbox not visible**: Select the Electrolyser or LOHC import first. Unchecking the last selected hydrogen source also unchecks H2 cons.
- **"Hydrogen demand must be positive."**: Enter a demand greater than zero for the selected pattern (for *Weekday/weekend* and *Daily* at least one day must have a demand).
- **"Hydrogen demand profile is not imported."**: The *Import data* tab is selected but no file has been imported. Import a file or switch to *Generate data*.
- **"Price of hydrogen bought from outside must be positive."**: Enter a price greater than zero.
- **File is not imported**: Check that the file has one column, no header, and exactly 365 or 8760 rows of non-negative numbers.
- **"Required pressure ... is higher than the pressure of the hydrogen bus"**: With low pressure, the required pressure must not exceed the pressure of the hydrogen bus, i.e. the lowest outlet pressure of the electrolysers, the waste treatment reactors and the dehydrogenation unit (30 bar without them, see Section 4.2.12). Lower the required pressure or choose high pressure.
- **"Required pressure ... is higher than the rated pressure of the hydrogen tank"**: With high pressure, the required pressure must not exceed the rated pressure of the tank in Hydrogen storage.
- **No hydrogen is used locally**: Producing hydrogen in the hub is more expensive than the given price, or the consumer requires green hydrogen and there is no green electricity.

4.2.14 H2 sales

The **Hydrogen sales** component represents the interface for selling hydrogen to different customers and the gas grid. Hydrogen can be sold to different potential customers, each with different parameters. The customers are said to be "potential" because the model determines the optimal amounts to be sold, and the customers' demands are limits for the sales.

Every customer can demand purely green hydrogen. The tool then takes into account energy sources and makes sure that hydrogen produced is 100% green. Because hydrogen is usually transported at a pressure higher than for stationary application, additional hydrogen compressor is needed. However, when injecting hydrogen into the gas grid, the compressor is not needed

because the pressure of gas grids is lower. A predefined compressor is already entered in the tool. However, parameters of the gas grid connection and prices for gas injection are left to the user to enter.

The component consists of 3 main sections:

- **Customer management**: Add and manage multiple hydrogen customers with individual demands.
- **Gas grid injection**: Parameters for selling hydrogen by injecting it into the gas grid.
- **Hydrogen compressor**: Technical and financial parameters for the compressor used for hydrogen sales to customers (not needed for gas grid injection).

4.2.14.1 Input parameters

Customer parameters

Each customer has the following configurable parameters:

Parameter	Description
Add new customer [button]	Button to add a new hydrogen customer. Each customer has individual demand parameters and can be configured separately.
Allow only green hydrogen? [checkbox]	Whether the hydrogen sold to this customer must be 100% green or not. If checked, the tool takes into account energy sources and makes sure that hydrogen sold to this customer is 100% green.
Demand frequency [dropdown menu]	How often the customer requests hydrogen. Choose between daily, weekly, monthly, or yearly.
Demand mass [kg]	Maximum amount of hydrogen for the frequency defined above.
Price for hydrogen sold [€/kg]	Price that the renewable energy hub would realize, after tax.
Required pressure [bar]	Pressure of hydrogen that is required for the customer, expressed as difference from the atmospheric pressure (barg). It is the outlet pressure of the hydrogen compressor for this customer and is used to calculate its *Power consumption*.

Gas grid injection – *Gas grid connection*

Parameter	Description
Hydrogen allowed in the gas grid? [checkbox]	Whether or not you allow the optimization to sell hydrogen by injecting it into the gas grid. If enabled, the parameters below become visible.

Parameter	Description
Gas grid connection exists? [checkbox]	Whether or not there is an existing gas grid connection at the site. The connection must be intended for gas injection.
Existing connection capacity [kWh/h]	The capacity of the existing gas grid connection for injection, in kWh/h (kW). The connection is assumed to allow hydrogen injection.
Distance to the nearest connection [km]	When there is no existing gas grid connection at the site, a new one must be made. Also, a new pipeline must be put from the existing gas grid to the site. The total investment cost of the new gas grid connection depends on this distance.
Unit price of grid connection [€/kW/km]	Unit price for the installation of pipeline for the new grid connection, per unit of hydrogen flow (power), per unit of distance. It includes all necessary equipment, material, labor, administration, and financing costs. Since the maintenance of the connection is the responsibility of the grid operator, there are no operating and maintenance (O&M), and replacement costs.
Capacity price of grid connection [€/kW]	Capacity price for the installation of the new gas grid connection, per unit of hydrogen flow (power). It includes all necessary equipment, material, labor, administration, and financing costs. Since the maintenance of the connection is the responsibility of the grid operator, there are no operating and maintenance (O&M), and replacement costs.
Maximum hydrogen flow allowed [kg/h]	Maximum flow of hydrogen that is allowed to be injected into the gas grid.
Pressure of the gas grid [bar]	Pressure at which the gas grid operates.

Gas grid injection – *Hydrogen prices*

Parameter	Description
Allow only green hydrogen? [checkbox]	Whether the injected hydrogen must be 100% green or not. If checked, the tool takes into account energy sources and makes sure that hydrogen injected is 100% green.
Pricing frequency [dropdown menu]	Dynamics of pricing. It is possible to choose between: i) **constant price** throughout the year; ii) **winter and summer prices**; or iii) **monthly prices**. By selecting the frequency, corresponding prices are shown.
Start of winter period [dropdown menu]	The first month of winter period.
Start of summer period [dropdown menu]	The first month of summer period.

Gas grid injection – *Green H2 prices*

Parameter	Description
Price of green hydrogen - constant [€/MWh]	Price for injecting green hydrogen. This price is constant throughout the year.
Price of green hydrogen - winter [€/MWh]	Price for injecting green hydrogen during winter period.
Price of green hydrogen - summer [€/MWh]	Price for injecting green hydrogen during summer period.
Price of green hydrogen - [Month] [€/MWh]	Price for injecting green hydrogen during a specific month. Shown when *Pricing frequency* is set to monthly prices.

Gas grid injection – *Non-green H2 prices*

Parameter	Description
Price of non-green hydrogen - constant [€/MWh]	Price for injecting non-green hydrogen. This price is constant throughout the year.
Price of non-green hydrogen - winter [€/MWh]	Price for injecting non-green hydrogen during winter period.
Price of non-green hydrogen - summer [€/MWh]	Price for injecting non-green hydrogen during summer period.
Price of non-green hydrogen - [Month] [€/MWh]	Price for injecting non-green hydrogen during a specific month. Shown when *Pricing frequency* is set to monthly prices.
Grid fee for injection [€/MWh]	Grid fee that is paid to the grid operator. It is simply deducted from the hydrogen price.

Hydrogen compressor for H2 sales (advanced)

This section contains parameters for the hydrogen compressor used when selling hydrogen to customers (not needed for gas grid injection).

Parameter	Description
Hydrogen compressor unit [dropdown menu]	Choose a hydrogen compressor from the application's database and its parameters will be filled automatically. Only suitable compressors will be available. A suitable compressor can: i) handle hydrogen pressure of the electrolyser at the input, ii) provide at least the pressure of the storage tank at the output. Afterwards you can change any of its parameters.
Hydrogen compressor already exists? [checkbox]	Whether or not there is an existing hydrogen compressor at the site.
Existing compressor size [kW/(kg/h)]	Size of the existing hydrogen compressor.
Efficiency [%]	The tool first calculates the theoretical energy required to compress hydrogen from the pressure of the hydrogen storage tank to the required pressure of every customer, using the adiabatic compression model. This energy is then divided by the compressor efficiency to get the electrical *Power consumption*. When entering *Efficiency*, *Power consumption per customer* is recalculated.
Inlet pressure [bar]	Pressure of the hydrogen entering the compressor, which is the *Rated pressure* of the hydrogen storage tank in Hydrogen storage, expressed as difference from the atmospheric pressure (barg). This value is calculated by the tool and cannot be entered.
Power consumption per customer [kWh/kg]	Read-only table with the electrical energy required to compress 1 kg of hydrogen for every customer, from the *Inlet pressure* to the *Required pressure* of the customer (see the formula below the table). Every customer has its own value. If a customer requires a pressure that is not higher than the tank pressure, the value is 0, but the hydrogen still passes through the compressor and counts towards its size. The table is recalculated whenever *Efficiency*, the tank pressure, or a customer changes.
Minimum inlet pressure [bar]	Minimum inlet pressure required for proper compressor operation. Below this threshold, the compressor may be damaged or underperform.
Maximum discharge pressure [bar]	Maximum discharge pressure the compressor can generate safely. Exceeding this may cause system failure.
Investment price [€/(kg/h)]	Investment price per unit of gas flow which includes the compressor itself, additional equipment (e.g. metering devices), installation, commissioning, financing, and all other necessities until full operation. The unit of flow rate is [kg/h], but it can be easily converted into normal volumetric flow rate. For reference, 1 kg/h 11.207 Nm^3/h of hydrogen.

Parameter	Description
Operation and maintenance price [€/(kg/h)]	Estimated annual operation and maintenance (O&M) price per unit of gas flow. O&M includes (but does not limit to) inspections, regular services, administration costs, overhauls, and replacing broken components. O&M does not include degradation of the compressor. The unit of flow rate is [kg/h], but it can be easily converted into normal volumetric flow rate. For reference, 1 kg/h = 11.207 Nm3/h of hydrogen.
Replacement price [€/(kg/h)]	Replacement price per unit of gas flow includes all necessary elements, materials, and labour from dismantling the old compressor until the full operation of the new compressor. The unit of flow rate is [kg/h], but it can be easily converted into normal volumetric flow rate. For reference, 1 kg/h = 11.207 Nm3/h of hydrogen.
Lifetime [years]	Number of years after which the tank needs to be replaced.
Subsidy [%]	Percentage that is deductible from the investment cost. Does not apply to the replacement cost.

Calculation of the compressor power consumption

The compressor takes hydrogen from the hydrogen storage tank at its *Rated pressure* p^{htank} and compresses it to the *Required pressure* $p^{hsale}(a)$ of every customer a. Both are gauge pressures, so the atmospheric pressure $p_{atm} = 1.01325$ bar is added to each of them to get absolute pressures. Assuming adiabatic compression of a real gas, the power consumption $e_{el}(a)$ for customer a in [kWh/kg] is:

$$e_{el}(a) = \frac{1}{\eta_{el}} \frac{\kappa}{\kappa - 1} Z\, R\, T \left[\left(\frac{p^{hsale}(a) + p_{atm}}{p^{htank} + p_{atm}} \right)^{\frac{\kappa-1}{\kappa}} - 1 \right]$$

where η_{el} is the compressor *Efficiency* and the physical constants of hydrogen are:

- $\kappa = 1.41$ – ratio of specific heat at constant pressure to that at constant volume,
- $Z = 1.0059$ – compressibility factor, which accounts for the deviation of hydrogen from an ideal gas (taken as constant, which has negligible effect on accuracy),
- $R = 0.00114563$ kWh/(kg K) – individual gas constant,
- $T = 298$ K – temperature of hydrogen during compression.

If the required pressure is not higher than the tank pressure, the power consumption is 0. The electrical power of the compressor in each time step is the sum over all customers of $e_{el}(a)$ multiplied by the mass flow of hydrogen sold to customer a. Gas grid injection does not use this compressor.

4.2.14.2 Interface options

The Hydrogen sales component provides several interface options:

- **Adding customers**: Click the *Add new customer* button to create a new customer.
- **Managing customers**: Right-click on a customer ID to access options to rename or delete customers.
- **Predefined compressor**: Select a hydrogen compressor from the dropdown menu in the compressor section to automatically populate technical and financial parameters.
- **Pricing strategies**: Choose between constant, winter/summer, or monthly pricing for hydrogen injected into the gas grid.

4.2.14.3 Optimization output

After running the optimization, the tool calculates the optimal technical and economic values for hydrogen sales.

Variable	Description
Hydrogen sold to customers	For each customer, the optimization determines the optimal amount of hydrogen sold, at each time step. Results include: - **Hydrogen sales profile** [kg/h] - **Revenue from hydrogen sales** [€]
Hydrogen injected to grid	The optimization determines the optimal amount of hydrogen injected into the gas grid, at each time step. Results include: - **Hydrogen injection profile** [kg/h] - **Revenue from grid injection** [€]
Compressor operation	If a compressor is used for customer sales, the optimization determines the compressor's operating profile, at each time step: - **Compressor power consumption** [kW]
Gas grid connection investment	If a new gas grid connection is needed, the optimization determines: - **Required connection capacity** [kW] - **Investment cost** [€]

4.2.14.4 Related components

- Electrolyser, as hydrogen is produced by the electrolyser and sold through this component. Also, the hydrogen storage tank is defined with the electrolyser.
- The dehydrogenation unit of LOHC purchase, which supplies hydrogen released from bought LOHC when *LOHC import* is selected.

4.2.14.5 Troubleshooting

Common issues and solutions:

- **H2 sales checkbox not visible**: H2 sales needs a hydrogen source. Select the Electrolyser or LOHC import first.
- **No customers configured**: Ensure at least one customer or the gas grid connection is added with valid parameters before running optimization.
- **Compressor not suitable**: When selecting a predefined compressor, ensure it can handle the required pressure levels for your customers.
- **Green hydrogen requirement**: When *Allow only green hydrogen?* is enabled, ensure sufficient renewable energy sources are configured in the system.

4.2.15 O2 sales

The **Oxygen sales** component enables the renewable energy hub to generate revenue by selling oxygen produced as a by-product of water electrolysis. Oxygen can be sold to different potential customers, and each of them has different parameters. The customers are said to be "potential" because the model determines the optimal amounts to be sold, and the customers' demands are limits for the sales.

The component consists of 3 main sections:

- **Customer management**: Add and manage multiple oxygen customers with individual demands.
- **Oxygen storage tank**: Optional storage infrastructure for oxygen.
- **Oxygen compressor**: Optional compression equipment to fill the storage tank.

4.2.15.1 Input parameters

Customer parameters

Each customer represents a potential oxygen buyer with specific demand requirements. You can add multiple customers to model different sales opportunities. For each customer, you can define the demand frequency, amount, price, and required delivery pressure.

Parameter	Description
Add new customer [button]	Click the *Add new customer* button to create a new oxygen customer. A popup window will appear asking for the customer's name. After entering a unique name, the customer entry is created with default values. You can then configure the customer's demand parameters.
Customer tree node [tree node]	Each customer appears as a tree node in the interface. Right-click on a customer name to access options for renaming or deleting the customer. All demand parameters for that specific customer are configured within this tree node.
Demand frequency [dropdown menu]	Determines how the maximum oxygen sales amount is applied over time. Options include: - *Daily*: Maximum amount can be sold each day - *Weekly*: Maximum amount can be sold each week - *Monthly*: Maximum amount can be sold each month - *Yearly*: Maximum amount represents total annual sales limit
Demand mass [kg]	Mass of oxygen demanded by the customer. If you enter the demand in volume, the mass will be calculated automatically. It also works vice versa, if you enter the demand in mass the volume will be calculated automatically. For reference, 1 kg of oxygen has volume of 700.44 l at STP.
Price [€/kg]	Price that a customer would pay for oxygen.
Rated pressure [bar]	Pressure of oxygen that is required for the customer.

Oxygen storage tank parameters

The oxygen storage tank section allows you to configure an optional oxygen storage tank. This tank stores oxygen produced by the electrolyser, enabling flexible sales patterns and buffer capacity. This section is advanced and hidden by default in a collapsible tree node.

Parameter	Description
Oxygen storage tank type [dropdown menu]	Choose an oxygen storage tank from the application's database and its parameters will be filled automatically. Afterwards you can change any of its parameters, and save this changes for future use.
Oxygen tank already exists? [checkbox]	Enable this checkbox if there is existing oxygen storage infrastructure at the location. When enabled, the *Existing oxygen storage tank* parameter becomes visible.
Existing oxygen storage tank [kg]	If there is existing oxygen storage, input the size of existing storage in kg of oxygen that can be stored. This parameter is visible only if *Oxygen tank already exists?* is enabled.

Parameter	Description
Rated pressure [bar]	The maximum pressure allowed in the storage tank, expressed as difference from the atmospheric pressure (barg). It is also the outlet pressure of the oxygen compressor and is used to calculate its *Power consumption*.
Minimum capacity [kg]	Lower limit of the oxygen tank size. If this value is greater than zero, *Maximal payoff period* and *Maximum investment* parameters are disabled.
Maximum capacity [kg]	Upper limit of the oxygen tank size. The tank's size is considered unlimited if no value is entered.
Investment price [€/kg]	Investment price per unit of storage capacity which includes the storage tank itself, additional equipment (e.g. metering devices), installation, commissioning, financing, and all other necessities until full operation.
Operation and maintenance price [€/kg]	Estimated annual operation and maintenance (O&M) price per unit of storage capacity. O&M does not include degradation of the tank.
Replacement price [€/kg]	Replacement price per unit of storage capacity that includes all necessary elements, materials, and labour from dismantling the old tank until the full operation of the new tank.
Lifetime [years]	Number of years after which the tank needs to be replaced.
Subsidy [%]	Percentage that is deductible from the investment cost. Does not apply to the replacement cost.

Oxygen compressor parameters

The oxygen compressor section allows you to configure compression equipment that pressurizes oxygen before storing it in the tank. This section is advanced and hidden by default in a collapsible tree node.

Parameter	Description
Oxygen compressor unit [dropdown menu]	Choose a oxygen compressor from the application's database and its parameters will be filled automatically. Only suitable compressors will be available. A suitable compressor can: - handle oxygen pressure of the electrolyser at the input, - provide at least the pressure of the storage tank at the output. Afterwards you can change any of its parameters, and save this changes for future use.
Oxygen compressor already exists? [checkbox]	Enable this checkbox if there is existing oxygen compression equipment at the location. When enabled, the *Existing compressor size* parameter becomes visible.

Parameter	Description
Existing compressor size [kg/h]	Size of the existing oxygen compressor. This parameter is visible only if *Oxygen compressor already exists?* is enabled.
Efficiency [%]	The tool first calculates the theoretical energy required to compress oxygen from the oxygen outlet pressure of the electrolysers to the rated pressure of the oxygen storage tank, using the adiabatic compression model. This energy is then divided by the compressor efficiency to get the electrical *Power consumption*. When entering *Efficiency*, *Power consumption* is recalculated.
Inlet pressure [bar]	Pressure of the oxygen entering the compressor, expressed as difference from the atmospheric pressure (barg). The oxygen outputs of all electrolysers are joined, so it is the lowest *Oxygen outlet pressure* of the electrolysers. This value is calculated by the tool and cannot be entered.
Power consumption [kWh/kg]	Electrical energy required to compress 1 kg of oxygen from the *Inlet pressure* to the *Rated pressure* of the oxygen storage tank. This value is calculated by the tool and cannot be entered (see the formula below the table). It is recalculated whenever *Efficiency*, the oxygen outlet pressure of an electrolyser, or the tank pressure changes.
Minimum inlet pressure [bar]	Most compressors require a minimum inlet pressure to operate efficiently. This minimum pressure ensures that the compressor can maintain proper lubrication and cooling, especially for oil-lubricated compressors. If the inlet pressure drops below this limit, the compressor may not function correctly, leading to potential damage or reduced performance.
Maximum discharge pressure [bar]	Compressors have a maximum discharge pressure limit, which represents the highest pressure that the compressor can safely generate. Exceeding this limit can lead to overpressurization of the system, potentially causing leaks, ruptures, or other catastrophic failures. Manufacturers specify this limit based on the design and materials of the compressor components, as well as safety standards and regulations.
Investment price [€/(kg/h)]	Investment price per unit of gas flow which includes the compressor itself, additional equipment (e.g. metering devices), installation, commissioning, financing, and all other necessities until full operation. The unit of flow rate is [kg/h], but it can be easily converted into normal volumetric flow rate. For reference, 1 kg/h 0.7 Nm^3/h of oxygen.

Parameter	Description
Operation and maintenance price [€/(kg/h)]	Estimated annual operation and maintenance (O&M) price per unit of gas flow. O&M includes (but does not limit to) inspections, regular services, administration costs, overhauls, and replacing broken components. O&M does not include degradation of the compressor. The unit of flow rate is [kg/h], but it can be easily converted into normal volumetric flow rate. For reference, 1 kg/h 0.7 Nm³/h of oxygen.
Replacement price [€/(kg/h)]	Replacement price per unit of gas flow includes all necessary elements, materials, and labour from dismantling the old compressor until the full operation of the new compressor. The unit of flow rate is [kg/h], but it can be easily converted into normal volumetric flow rate. For reference, 1 kg/h 0.7 Nm³/h of oxygen.
Lifetime [years]	Number of years after which the compressor needs to be replaced.
Subsidy [%]	Percentage that is deductable from the investment cost. Does not apply to the replacement cost.

Calculation of the compressor power consumption

The compressor takes oxygen at the *Inlet pressure* p^{elys} (the lowest oxygen outlet pressure of the electrolysers) and compresses it to the *Rated pressure* p^{otank} of the oxygen storage tank. Both are gauge pressures, so the atmospheric pressure $p_{atm} = 1.01325$ bar is added to each of them to get absolute pressures. Assuming adiabatic compression of a real gas, the power consumption e_{el} in [kWh/kg] is:

$$e_{el} = \frac{1}{\eta_{el}} \frac{\kappa}{\kappa - 1} Z\, R\, T \left[\left(\frac{p^{otank} + p_{atm}}{p^{elys} + p_{atm}} \right)^{\frac{\kappa-1}{\kappa}} - 1 \right]$$

where η_{el} is the compressor *Efficiency* and the physical constants of oxygen are:

- $\kappa = 1.40$ – ratio of specific heat at constant pressure to that at constant volume,
- $Z = 0.99935$ – compressibility factor, which accounts for the deviation of oxygen from an ideal gas,
- $R = 7.21701 \cdot 10^{-5}$ kWh/(kg K) – individual gas constant,
- $T = 298$ K – temperature of oxygen during compression.

If the tank pressure is not higher than the inlet pressure, the power consumption is 0. The electrical power of the compressor in each time step is e_{el} multiplied by the mass flow of oxygen compressed into the tank.

4.2.15.2 Interface options

The Oxygen sales component provides several interface options:

- **Adding customers**: Click the *Add new customer* button to create a new oxygen customer.
- **Managing customers**: Right-click on a customer name to access options to rename or delete customers.
- **Predefined storage tank**: Select an oxygen storage tank from the dropdown menu to automatically populate technical and financial parameters.
- **Predefined compressor**: Select an oxygen compressor from the dropdown menu to automatically populate technical and financial parameters.

4.2.15.3 Optimization output

After running the optimization, the tool calculates the optimal technical and economic values for the defined scenario. This subsection shows what variables are tied to the *Oxygen sales* component and are shown in Results.

Variable	Description
Oxygen sales revenue	The optimization determines the optimal amount of oxygen to sell to each customer based on their demand limits and the product price. Results include: - **Oxygen sales profile** as mass flow rates for each time step [kg/h] - **Total oxygen sold** over the simulation period [kg] - **Revenue from oxygen sales** [€]
Oxygen storage tank	If the oxygen storage tank is enabled, the optimization determines the optimal tank size and operation: - **Optimal tank capacity** [kg] - **Storage level profile** over time [kg] - **Investment cost** for new tank or use of existing capacity [€]
Oxygen compressor	If the oxygen compressor is enabled, the optimization determines the optimal compressor size and power consumption: - **Optimal compressor size** [kg/h] - **Compressor power consumption profile** [kW] - **Investment cost** for new compressor or use of existing capacity [€] - **Operation and maintenance costs** over the simulation period [€]

4.2.15.4 Related components

- Electrolyser, as oxygen is a by-product of hydrogen production through water electrolysis.

4.2.15.5 Troubleshooting

Common issues and solutions:

- **At least one customer must be defined**: No oxygen customers have been added to the component. The solution: Click the *Add new customer* button to create at least one oxygen customer.

4.2.16 Fuel cell stack

The **fuel cell stack** is a component that produces electricity and heat from hydrogen. Normally, *fuel cell* refers to a single electrochemical cell, while *fuel cell stack* refers to a series of individual fuel cells connected together with all other necessary elements to make a functional electricity generation unit used in real applications.

It is considered a combined heat and power (CHP) technology by the EU in Directive 2012/27/EU. It combines hydrogen and oxygen molecules into water to produce electricity while producing heat due to inefficiencies in the conversion process. The hydrogen is supplied from the hydrogen storage tank while the oxygen comes from air. Therefore, it has a single input (hydrogen) and 3 outputs:

- **Electricity** – the main output of the fuel cell stack. It can be considered green or non-green depending on the type of hydrogen used.
- **Heat** – thermal energy resulting from conversion losses. The tool distinguishes between low-temperature (LT) and high-temperature (HT) heat. A fuel cell stack may produce either of these, both, or none, depending on the selected configuration.
- **Water** – in the form of steam. It can be extracted during electricity production, saved in the water tank and reused later.

The user can specify **multiple fuel cell stacks** in order to find the most suitable for the investment, each with its own set of parameters. Optimization will determine the optimal size of each stack as nominal electrical power. A fuel cell stack is added by clicking on **Add new fuel cell stack** button.

By **right-clicking** on a fuel cell stack **node**, a pop-up menu appears with the following options:

- **Rename fuel cell stack** – change the name of the fuel cell stack.
- **Delete fuel cell stack** – removes the fuel cell stack from the overall setup.

Hydrogen taken from (radio button above the stacks) selects where all fuel cell stacks take hydrogen from:

- **Hydrogen tank** (default): hydrogen is taken from the hydrogen storage tank (see Section 4.2.12), so it can be produced at one time and used later.
- **Hydrogen bus**: hydrogen is taken directly from the low-pressure hydrogen bus, without compression, so the fuel cells can only use hydrogen at the same time it is produced by the electrolysers or the dehydrogenation unit.

4.2.16.1 Input parameters

Basic parameters

Parameter	Description
Fuel cell stack type [dropdown menu]	A fuel cell stack from the tool catalog. Once selected, all required parameters are automatically populated according to the chosen type. Any parameter can be modified afterward.
Electrical efficiency [%]	The electrical efficiency of the fuel cell stack represents the percentage of chemical energy from hydrogen that is converted to electrical energy from higher heating value (HHV) perspective. Efficiency expressed for lower heating value (LHV) is 18.2% higher. This parameter is tied to *Electrical energy production*, where efficiency of 100% equals to: - 285.8 kJ/mol - 39.41 kWh/kg - 3.54 kWh/Sm3 - 3.36 kWh/Nm3 Sm3 = Standard cubic meter means temperature of 0 °C and pressure of 1.01325 bar (standard DIN 1343). Nm3 = Normal cubic meter means temperature of 15 °C and pressure of 1.01325 bar (standard ISO 2533).
Electrical energy production [kWh/kg]	Amount of electrical energy produced by the fuel cell stack from 1 kg of hydrogen. Theoretical maximum for energy production is 39.41 kWh/kg, corresponding to electrical efficiency of 100% (HHV limit). This parameter is tied to *Electrical efficiency*.
Investment price [€/kW]	Specific investment cost per unit of output electrical power. It includes all required equipment, materials, and labor (design, installation, commissioning, etc.) until full operation.
Fuel cell stack already exists? [checkbox]	Indicates whether a fuel cell stack is already installed at the site.
Existing fuel cell stack size [kW]	Nominal electrical power of the existing fuel cell stack. Visible only if *Fuel cell stack already exists?* is enabled. The parameters above apply to both the existing unit and any optimized expansion.

Advanced parameters

Parameter	Description
Recoverable high temperature heat? [checkbox]	Indicates whether heat from the fuel cell stack cooling circuit can be recovered for high-temperature heating applications.
Heating efficiency (high temperature) [%]	The *Electrical efficiency* determines hydrogen consumption per unit of electrical energy, while the remaining share represents total heat losses. *Heating efficiency* defines the fraction of this recoverable heat available at high temperature. The sum of high- and low-temperature heating efficiencies must not exceed 100%. This parameter is tied to *Usable heat output*.
Usable heat output (high temperature) [kWh/kg]	The amount of usable heat (at high temperature) produced by the fuel cell stack for every 1 kg of consumed hydrogen. This parameter is calculated as: 39.41 kWh/kg × (100% - *Electrical efficiency*) × *Heating efficiency (high temperature)*
Operating temperature (high temperature) [°C]	Temperature at which the high-temperature heat is released from the fuel cell stack at nominal conditions, i.e. nominal power.
Recoverable low temperature heat? [checkbox]	Indicates whether heat from the fuel cell stack cooling circuit can be recovered for low-temperature heating applications.
Heating efficiency (low temperature) [%]	The *Electrical efficiency* determines hydrogen consumption per unit of electrical energy, while the remaining share represents total heat losses. *Heating efficiency* defines the fraction of this recoverable heat available at low temperature. The sum of low- and high-temperature heating efficiencies must not exceed 100%. This parameter is tied to *Usable heat output*.
Usable heat output (low temperature) [kWh/kg]	The amount of usable heat (at low temperature) produced by the fuel cell stack for every 1 kg of consumed hydrogen. This parameter is calculated as: 39.41 kWh/kg × (100% - *Electrical efficiency*) × *Heating efficiency (low temperature)*
Operating temperature (low temperature) [°C]	Temperature at which the low-temperature heat is released from the fuel cell stack at nominal conditions, i.e. nominal power.
Water production [l/MWh]	It represents amount of usable water produced as a byproduct of electricity generation, since some water is lost as vapour. The value is expressed as liters of water for each MWh of electrical energy generated. Equivalent value is for the parameter in ml/kWh. The parameter is capped to the theoretical maximum. The theoretical maximum is determined as: 9 / (39.41 kWh/kg * *Electrical efficiency*)

Parameter	Description
Minimum stack size [kW]	Lower limit on the size (installed power in [kW]) of the fuel cell stack. If greater than zero, *Maximal payoff period* and *Maximum investment* in *Financial parameters* must be disabled.
Maximum stack size [kW]	Upper limit on the size (installed power in [kW]) of the fuel cell stack.
Maintenance price [€/kW]	Estimated annual operation and maintenance (O&M) price per installed power. O&M does not include degradation of the system.
Replacement price [€/kW]	Future specific investment cost applied when the fuel cell stack reaches end of life. The subsidy does not apply to replacement costs.
Lifetime [years]	Number of years after which the whole fuel cell stack must be replaced.
Subsidy [%]	Percentage that is deductible from the investment cost. Does not apply to the replacement price.

i Water extraction

Although there is parameter **Water production** that determines how much water can be extracted from the fuel cell stack for each 1 kg of hydrogen used, it is not used in the background calculation. All the water produced by the fuel cell stack is considered to be released as steam into the atmosphere.
Water extraction would introduce computational complexity while not contributing significantly to the result. Future versions of the tool might include the water extraction.

4.2.16.2 Optimization output

Variable	Description
Installed capacity [kW]	Optimized installed electrical power for each fuel cell stack and in total.
Power production [kW]	Electrical power production of each fuel cell stack in every time step, optimized by the model. Based on the power production, the following time series are derived: - **Hydrogen consumption** [kg/h] - **Low-temperature heat generation** [kW] - **High-temperature heat generation** [kW]
Investment cost [€]	Investment costs for each fuel cell stack and in total.
Maintenance cost [€]	Annual operation and maintenance costs for each fuel cell stack and in total.
Degradation cost [€]	Annual cost associated with system degradation over its lifetime, reported for each fuel cell stack and in total.

4.2.16.3 Interface options

Parameters of fuel cell stacks can be predefined by selecting one of the available types from the database. Any parameter can still be manually modified after selection. By default, the first type in the list is used.

4.2.16.4 Related Components

- Electrolyser – defines hydrogen storage tank, among other components
- The dehydrogenation unit of LOHC purchase – supplies hydrogen released from bought LOHC when *LOHC import* is selected
- Waste heat – waste heat utilization, where heat consumption is defined

4.2.16.5 Troubleshooting

- **Fuel cell checkbox not visible**: The fuel cell needs a hydrogen source. Select the Electrolyser or LOHC import first.
- **No fuel cell stack defined**: At least one fuel cell stack must be defined in the tool. Otherwise, the optimization will return an error.
- **Deleting fuel cell stack**: If a fuel cell stack is removed, all associated data are deleted as well. Once a fuel cell stack is deleted, it cannot be restored.
- **Efficiency and power consumption do not match**: When reviewing a product datasheet, the manufacturer may provide both the efficiency and the energy production of the fuel cell stack. If these values do not match those in the tool, the efficiency may be reported with respect to the lower heating value (LHV) instead of HHV, or it may refer only to the fuel cells and not the whole fuel cell stack as a system.

4.2.17 Waste heat

The **heat** generated from conversion losses of the electrolysers and/or the fuel cell stacks, i.e. from their cooling systems. This heat can be supplied to a local consumer within the renewable energy hub (e.g., a building, a greenhouse, a drying facility, etc.) and/or exported to a district heating (DH) network.

In the case of **local heat consumption**, any excess heat that cannot be absorbed by the consumer is considered wasted. If there is not enough heat produced by electrolysers and fuel cell stacks, additional heat must be purchased from the current heating source at its

regular price. Therefore, the waste heat is effectively lowering current spending on heating requirements.

When heat is sold to a **DH network**, the model assumes no export limit by default. However, a limit can be imposed if the DH network cannot accept large amounts of heat.

The **temperature level** of both the local consumption and DH connection is assumed to be the same.

For more flexible heat delivery, the system includes a **thermal energy storage (TES)** unit.

If low-temperature heat must be upgraded to supply a high-temperature demand, a **heat pump** is required, which operates using electricity.

Heat from the electrolyser(s) and fuel cell stack(s) is delivered through low-temperature and high-temperature **heat exchangers**. These represent the complete heating subsystem, including pipes, valves, pumps, and control equipment.

Heat flows are automatically balanced by the optimization model between local consumption, TES, and DH export.

4.2.17.1 Input parameters

Local consumption

The local heat demand can be provided as a generic profile from the tool catalogue and annual energy consumption, or it can be imported from a file. Each way of providing the demand data has its own tab.

Generate data tab

Parameter	Description
Building type [dropdown menu]	A type of building for which heating demand exists. The tool has a generic profile for each of the provided building types.
Annual heat consumption [MWh]	Amount of heat consumed annually. The profile defined with *Building type* is scaled to achieve this value.

Import data tab

Parameter	Description
File with the consumption profile [text field]	Defines the name of the CSV file containing the heat demand time series. It is not possible to write in this field – it is automatically populated after clicking on button *Browse…*.

Parameter	Description
Browse [button]	By clicking *Browse*, the user selects a CSV file from the local system. File format requirements: - File format: .csv - Single column - No header row Allowed number of rows: - 8,760 (hourly resolution) - 17,520 (30-minute resolution) - 35,040 (15-minute resolution) Each row must contain a numerical value representing the heat consumption in kW. The first value in the file corresponds to the beginning of the year (January 1st, 00:00), and the last value corresponds to the end of the year (December 31st, 23:00 for hourly resolution). If the dataset originates from a leap year (8,784 hourly values), one day of data must be removed to match one of the supported row counts. After selecting a file, a message is displayed indicating whether the data import was successful or if errors were detected.

Common parameters

Parameter	Description
Check consumption data [button]	Opens a pop-up window displaying a chart of the heat consumption profile. The chart is dynamic, meaning it can be zoomed in or out and moved around. This allows the user to visually verify data continuity, seasonal trends, extreme values, and possible data inconsistencies. The chart is for verification purposes only and does not modify the data. This button is applicable to both tabs (generic data and custom data).
Current heating price [€/MWh]	The current price of heat paid by the local consumer.
Temperature level [dropdown menu]	The temperature level required for the local consumer – it could be either low- or high-temperature. It matches the temperature level of the DH network, if enabled. By changing this value, the temperature level of the DH changes to the same value, and vice versa.

District heating

Parameter	Description
DH demand has limit? [checkbox]	Indicates whether the DH network has an export limit or the renewable energy hub can supply unlimited heat. If there is a limit, it should be imported as a profile. There is no option to generate the data.

Parameter	Description
File with the demand profile [textbox]	Defines the name of the CSV file containing the DH demand limit time series. It is not possible to write in this field – it is automatically populated after clicking on button *Browse…*.
Browse [button]	By clicking *Browse*, the user selects a CSV file from the local system. File format requirements: - File format: .csv - Single column - No header row Allowed number of rows: - 365 (daily resolution) - 8,760 (hourly resolution) - 17,520 (30-minute resolution) - 35,040 (15-minute resolution) Each row must contain a numerical value representing the heat demand limit as mean powers in kW. The first value in the file corresponds to the beginning of the year (January 1st, 00:00), and the last value corresponds to the end of the year (December 31st, 23:00 for hourly resolution). If the dataset originates from a leap year (8,784 hourly values), one day of data must be removed to match one of the supported row counts. After selecting a file, a message is displayed indicating whether the data import was successful or if errors were detected.
Check demand data [button]	Opens a pop-up window displaying a chart of the heat demand profile. The chart is dynamic, meaning it can be zoomed in or out and moved around. This allows the user to visually verify data continuity, seasonal trends, extreme values, and possible data inconsistencies. The chart is for verification purposes only and does not modify the data.
DH connection already exists? [checkbox]	Indicates whether the connection to the DH network already exists or not.
Size of the existing connection [kW]	Nominal heating power of the existing connection to the DH network. It is visible only if *DH connection already exists?* is activated.
Temperature level [dropdown menu]	The temperature level required for the DH network – it could be either low- or high-temperature. It matches the temperature level of the Local consumption, if enabled. By changing this value, the temperature level of the Local consumption changes to the same value, and vice versa.
Price for provided heat [€/MWh]	The price that the renewable energy hub earns from selling heat to the DH network, after taxes.
Investment price [€/kW]	The price of the investment into the DH connection per unit of power. It is assumed that the DH provider takes care of the connection and provides necessary maintenance. Therefore, the DH connection does not have maintenance nor replacement price.
Subsidy [%]	Percentage that is deductible from the investment cost.

Heat exchangers (advanced)

The tool assumes there are two heat exchangers: one for each temperature level. It does not represent only the heat exchanger itself, but rather a heating (sub)system that consists of pipes, valves, pumps, etc. In the table below there are parameters listed for a single heat exchanger, but it applies to both of them.

Parameter	Description
Heat transfer efficiency [%]	Total heat transfer efficiency of the heat exchanger.
Heat exchanger already exists? [checkbox]	Indicates whether the heat exchanger already exists at the site or not.
Existing heat exchanger power [kW]	The nominal power of the existing heat exchanger. It is visible only if *Heat exchanger already exists?* is enabled.
Investment price [€/kW]	The price of the investment into the heat exchanger per unit of power. It includes the exchanger itself, all necessary material and auxiliary equipment (pipes, valves, pumps, etc.), and labour until the full operation.
Maintenance price [€/kW]	Estimated annual operation and maintenance (O&M) price per installed power. O&M does not include degradation of the system.
Replacement price [€/kW]	Same as investment price, but after the lifetime of the system, i.e. the future investment price. The main difference from the investment price is that the subsidy does not apply to the replacement price.
Lifetime [years]	Number of years after which the heat exchanger needs to be replaced.
Subsidy [%]	Percentage that is deductible from the investment cost. Does not apply to the replacement price.

Thermal energy storage (advanced)

Parameter	Description
Storage efficiency [%/day]	Effectiveness of a system to store and reuse heat, minimizing energy loss and maximizing beneficial output, expressed as percentage of heat retained during 24 hours (100% - self-discharge percentage).
Storage already exists? [checkbox]	Indicates whether a thermal energy storage already exists at the site or not.
Existing storage capacity [kWh]	The capacity of the existing thermal energy storage. It is visible only if *Storage already exists?* is enabled.

Parameter	Description
Investment price [€/kWh]	The price of the investment into the thermal energy storage per unit of energy. It includes the storage tank itself, all necessary material and auxiliary equipment (pipes, valves, pumps, etc.), and labour until the full operation.
Maintenance price [€/kWh]	Estimated annual operation and maintenance (O&M) price per installed capacity. O&M does not include degradation of the system.
Replacement price [€/kWh]	Same as investment price, but after the lifetime of the system, i.e. the future investment price. The main difference from the investment price is that the subsidy does not apply to the replacement price.
Lifetime [years]	Number of years after which the thermal energy storage needs to be replaced.
Subsidy [%]	Percentage that is deductible from the investment cost. Does not apply to the replacement price.

Heat pump (advanced)

The heat pump is visible only when the sources (electrolysers or fuel cell stacks) provide low-temperature heat while the demand (Local consumption or DH network) requires high-temperature heat.

Parameter	Description
Heat pump COP [-]	Coefficient of performance of the heat pump. It represents a ratio of thermal power to electrical power required.
Heat pump already exists? [checkbox]	Indicates whether a heat pump already exists at the site or not.
Existing heat pump power [kW]	The nominal thermal power of the existing heat pump. It is visible only if *Heat pump already exists?* is enabled.
Investment price [€/kW]	The price of the investment into the heat pump per unit of thermal power output. It includes the heat pump itself, all necessary material, auxiliary equipment, and labour until the full operation.
Maintenance price [€/kW]	Estimated annual operation and maintenance (O&M) price per installed power. O&M does not include degradation of the system.
Replacement price [€/kW]	Same as investment price, but after the lifetime of the system, i.e. the future investment price. The main difference from the investment price is that the subsidy does not apply to the replacement price.
Lifetime [years]	Number of years after which the heat pump needs to be replaced.
Subsidy [%]	Percentage that is deductible from the investment cost. Does not apply to the replacement price.

4.2.17.2 Optimization output

The optimization provides detailed results describing how recovered heat is utilized, stored, exported, or wasted within the renewable energy hub. All time-series outputs are reported with the same temporal resolution as the input profiles.

Variable	Description
Local heat utilization	- **Locally consumed heat – profile** in [kW]: Amount of recovered heat supplied to the local consumer at each timestep. This represents useful heat demand covered by waste heat from electrolysers and/or fuel cell stacks. - **Heat from current source – profile** in [kW]: Heat supplied by the existing (conventional) heating source when recovered heat is insufficient to meet local demand. This value is non-zero only when waste heat and thermal storage cannot fully satisfy the local consumption. - **Savings by recovering waste heat** in [€]: Annual cost savings achieved by replacing heat from the current heating source with recovered waste heat. The savings are calculated using the specified current heating price.
District heating (DH) export	- **Heat exported to DH network – profile** in [kW]: Heat delivered from the renewable energy hub to the DH network at each timestep. Export respects any user-defined DH limit. - **Revenue from DH export** in [€]: Annual revenue generated by selling heat to the DH network, calculated using the specified DH selling price.
Heat losses	Amount of usable heat that could not be utilized locally, stored in the TES, or exported to the DH network. This typically occurs when: - local demand is fully satisfied, - TES is full, and - DH export is limited or disabled.
Additional profiles	- **TES state of charge (SoC) – profile** in [kWh]: Thermal energy stored in the TES at each timestep, expressed relative to the storage capacity. The profile reflects charging, discharging, and standing losses according to the specified storage efficiency. - **Heat pump electricity consumption – profile** in [kW]: Electrical energy consumed by the heat pump at each timestep when temperature upgrading is required.
Economic results	The optimization reports investment and operational costs associated with heat recovery infrastructure: - **DH connection**: investment cost - **Heat exchanger**: investment, maintenance, and degradation cost - **Thermal energy storage**: investment, maintenance, and degradation cost - **Heat pump**: investment, maintenance, and degradation cost

4.2.17.3 Interface options

The **Waste heat** section allows the user to configure how recovered heat from electrolysers and fuel cell stacks is utilized, upgraded, stored, or exported. The user can select either **Local consumption** or **District heating**, or both, for waste heat utilization.

- **Local consumption**: The profile of consumption can be either generated from the generic data in the tool background, or imported from a CSV file.
- **District heating**: If the exported heat should be limited, the profile of limits must be imported from a CSV file.
- **Heat pump**: It is visible only if the temperature level of local consumption and/or DH is set to high-temperature.

4.2.17.4 Related components

- Electrolyser – a possible source of heat.
- Fuel cell stack – a possible source of heat.

4.2.17.5 Troubleshooting

Common issues and solutions:

- **Waste heat checkbox not visible**: The Waste heat checkbox is shown only while the Electrolyser or the Fuel cell is selected.
- **No heat utilization selected**: If neither *Local consumption* nor *District heating* is enabled, the application will display an error message when attempting to run the optimization.
- **No heat sources defined**: The application will display an error message if none of the configured electrolysers or fuel cell stacks have recoverable waste heat.

4.2.18 Waste treatment plant

The **Waste treatment plant** (WT) converts waste into hydrogen, electrical energy and heat. The plant consists of one or more **reactors**, each with its own process, technical and financial parameters. The tool determines the optimal size of every reactor, expressed as the amount of waste it can treat per day [t/day].

Besides hydrogen, the waste treatment process produces several by-products:

- **Electrical energy** – the reactor can produce or consume electricity, depending on the process. Net production is fed to the electrical bus of the hub, while net consumption is supplied by the other electricity sources.
- **Heat** – high- and low-temperature heat that can be used by the Waste heat component.
- **Carbon black** – solid carbon that is sold to the market and generates revenue.
- **Tar** – a by-product that must be disposed of, which causes an operating cost.
- **CO_2** – emissions that are paid for at the CO_2 emission price.

The waste for the reactors comes from the Waste source and storage component, which is enabled automatically together with the waste treatment plant.

> **!** Important
>
> Hydrogen from the waste treatment plant is **not green** hydrogen. It is always counted as non-green hydrogen in the hydrogen balance and cannot be used to satisfy green hydrogen demand. In the same way, the electricity consumed by the reactors is supplied by non-green sources unless green electricity is available.

Operation of reactors

The waste treatment process is modelled with a daily time resolution. Reactors are assumed to operate continuously at their full installed capacity, i.e. every reactor treats an amount of waste equal to its size in every hour it is available. Availability is reduced only by planned maintenance and the following heating (start-up) period:

1. At the **maintenance start date**, the reactor stops and treats no waste for the **maintenance duration** (in days).
2. On the first day after the maintenance, at the **start time of the reactor** (hour of the day), the reactor starts preheating.
3. During the **starting duration** (in hours), the reactor still treats no waste, but it consumes additional electricity – the **starting electricity consumption surge**. The surge is proportional to the reactor size and is modelled with the electrical time step, so it appears as a short peak in electricity consumption.
4. After the preheating, the reactor returns to full operation.

The maintenance happens on the same date every year of the simulation. If a day is only partially affected by the maintenance (e.g. the day when the preheating ends), the reactor treats proportionally less waste in that day. If the maintenance duration is 0, there is no maintenance and no starting surge.

The outputs of a reactor are defined with the **process matrix**. Each output (electricity, heat, hydrogen, carbon black, tar, CO_2) is calculated from the treated amount of waste and its composition (mass fractions of hydrogen, carbon and water defined in the Waste source and storage component), e.g. for electricity:

$$P_{\text{el}} = \dot{m}_{\text{waste}} \cdot \left(\beta_{\text{H2,el}}\,\omega_{\text{H2}} + \beta_{\text{C,el}}\,\omega_{\text{C}} + \beta_{\text{H2O,el}}\,\omega_{\text{H2O}}\right)$$

where β are the process coefficients of the reactor and ω are the mass fractions of the waste.

Use of produced hydrogen

Hydrogen from the reactors is compressed and stored in the hydrogen tank, from where it can be used by the Fuel cell stack or sold through H2 sales. If neither of these components is enabled, the hydrogen produced by waste treatment is released into the atmosphere.

4.2.18.1 Input parameters

By-product prices

These prices are common to all reactors.

Parameter	Description
Tar treatment price [€/kg]	Price per unit of produced tar. It includes transport and fees to dispose of all tar produced during operation.
CO2 emissions price [€/t]	Price paid for each tonne of emitted CO_2 (carbon price).
Carbon black price [€/t]	Price of carbon black sold to the market.

Reactors

A new reactor is added by clicking the button *Add new waste treatment reactor*. Each reactor is shown as a separate node, which can be renamed or deleted by right-clicking on it. At least one reactor must be defined.

Parameter	Description
Reactor model [dropdown menu]	Choose one of the predefined reactor models. Changing the model updates all parameters in the advanced nodes.
Investment price [€/(t/day)]	Investment price per unit of reactor size. It includes all necessary elements, materials, labour (design, mounting, commissioning, etc.), and financing until full operation.
Reactor already exists? [checkbox]	Indicates whether the reactor already exists at the site or not.
Existing reactor size [t/day]	Size of the existing reactor. It is visible only if *Reactor already exists?* is enabled. All other parameters of the reactor are used for both the existing reactor and the new, optimized addition.

Process parameters (advanced)

The process matrix consists of coefficients β for every output and every mass fraction of the waste (hydrogen H_2, carbon C, and water H_2O). The outputs are grouped into separate nodes:

Node	Description
El. energy production/usage [kWh/t]	Coefficients $\beta_{H2,el}$, $\beta_{C,el}$, $\beta_{H2O,el}$. A positive value denotes production, while a negative value denotes consumption of electrical energy.
Heat energy production [kWh/t]	Coefficients for high-temperature heat ($\beta_{H2,HTH}$, $\beta_{C,HTH}$, $\beta_{H2O,HTH}$) and low-temperature heat ($\beta_{H2,LTH}$, $\beta_{C,LTH}$, $\beta_{H2O,LTH}$).
Hydrogen production [t/t]	Coefficients $\beta_{H2,H2}$, $\beta_{C,H2}$, $\beta_{H2O,H2}$.
Carbon black production [t/t]	Coefficients $\beta_{H2,C}$, $\beta_{C,C}$, $\beta_{H2O,C}$.
Tar production [t/t]	Coefficients $\beta_{H2,tar}$, $\beta_{C,tar}$, $\beta_{H2O,tar}$.
CO2 production [t/t]	Coefficients $\beta_{H2,CO2}$, $\beta_{C,CO2}$, $\beta_{H2O,CO2}$.

Technical parameters (advanced)

Parameter	Description
Operating temperature (high temperature) [°C]	Operating temperature of the high-temperature heat circuit at nominal conditions. High-temperature heat is usable only if this temperature is high enough for high-temperature heating applications.
Operating temperature (low temperature) [°C]	Operating temperature of the low-temperature heat circuit at nominal conditions. Low-temperature heat is usable only if this temperature is high enough for low-temperature heating applications.
Hydrogen outlet pressure [bar]	Pressure of the hydrogen exiting the reactor at nominal conditions, expressed as the difference from atmospheric pressure (barg). It determines the work needed by the hydrogen compressor.
Minimum reactor size [t/day]	Lower limit on the size of the reactor. If this value is greater than zero, *Maximal payoff period* and *Maximum investment* parameters are disabled.
Maximum reactor size [t/day]	Upper limit on the size of the reactor.
Maintenance start date [date]	The first day of the maintenance period, in the format MM-DD. The maintenance starts on the same day every year.

Parameter	Description
Maintenance duration [day]	Number of days the reactor is shut down for maintenance. It must be between 0 and 365 days. If it is 0, there is no maintenance and no starting surge.
Starting electricity consumption surge [kW/(t/day)]	Additional electrical power needed to preheat the reactor after maintenance, per unit of reactor size.
Starting duration [h]	Number of hours of preheating during which the starting surge occurs and the reactor treats no waste.
Start time of reactor [hour]	Hour of the day (0–23) when the reactor starts preheating on the first day after the maintenance. Since it causes a peak in electricity consumption, it can be moved to hours with cheaper or more available electricity.

Financial parameters (advanced)

Parameter	Description
Maintenance price [€/(t/day)]	Estimated annual operation and maintenance (O&M) price per unit of reactor size. It should include operator salaries (e.g. 3 shifts × 2 operators + a 4th cover shift = 8 FTE). O&M does not include degradation of the system.
Replacement price [€/(t/day)]	Same as investment price, but after the lifetime of the reactor, i.e. the future investment price. The main difference from the investment price is that the subsidy does not apply to the replacement price.
Lifetime [years]	Number of years after which the whole reactor needs to be replaced.
Subsidy [%]	Percentage that is deductible from the investment cost. Does not apply to the replacement price.

4.2.18.2 Optimization output

Variable	Description
Reactor size	Optimal size of every reactor (existing + new) in [t/day].
Treated waste	Amount of waste treated by every reactor and by all reactors together in [t/year].
Hydrogen production	Hydrogen produced by every reactor in [kg/year]. It is always non-green.
Electricity production	Net electrical energy of every reactor in [MWh/year], including the starting surge after maintenance. A negative value means that the reactor consumes more electricity than it produces.

Variable	Description
Heat production	Usable high- and low-temperature heat in [MWh/year].
Economic results	- **Investment cost** of new reactor capacity - **Maintenance and degradation cost** of all reactors - **Operating cost** of tar treatment and CO_2 emissions - **Revenue** from selling carbon black

The results are shown in the Waste treatment section of the results.

4.2.18.3 Interface options

- **Add new waste treatment reactor**: adds a new reactor with default parameters.
- **Right-click on a reactor node**: rename or delete the reactor.
- **Reactor model**: fills all advanced parameters with the values of a predefined reactor model.

4.2.18.4 Related components

- Waste source and storage – supplies waste to the reactors and defines its composition.
- Waste heat – uses heat produced by the reactors.
- Fuel cell stack and H2 sales – use hydrogen produced by the reactors.
- Electrical grid connection – supplies electricity for the starting surge and net consumption of the reactors.

4.2.18.5 Troubleshooting

Common issues and solutions:

- **No reactor defined**: at least one reactor must be added, otherwise the optimization cannot start.
- **Invalid maintenance start date**: the date must be a valid day of the year in the format MM-DD (29 February is moved to 28 February in non-leap years).
- **Infeasible optimization**: reactors always operate at their full available capacity, so the available waste (and the waste storage) must be large enough to supply them. If *Treat all waste* is selected, or the minimum reactor size is large, the problem can become infeasible. In that case, reduce the minimum reactor size, increase the maximum storage capacity, or deselect *Treat all waste*.

- **Not enough demand for hydrogen**: hydrogen produced by the reactors cannot be released into the atmosphere, so all of it must be used in every time step. Reactors always operate at their full available capacity, so the demand for non-green hydrogen (*H2 sales* customers and gas grid injection that accept non-green hydrogen, *Fuel cell*, *Fixed hydrogen consumption* and *LOHC export* that accept non-green hydrogen) must be large enough to take the hydrogen of the existing and minimal reactor sizes. Otherwise, the validation reports an error or the optimization becomes infeasible. In that case, increase the demand for non-green hydrogen, or decrease the existing or minimum reactor size.

4.2.19 Waste source and storage

The **Waste source and storage** (WSS) component defines how much waste is available to the Waste treatment plant, what its composition is, and how it can be stored before treatment. It does not have its own checkbox – it is enabled automatically together with the waste treatment plant.

Waste source

The waste source is defined by the amount of waste available over the year. The waste can be given as monthly amounts or imported as a profile with daily, weekly or monthly values. Since the waste treatment is modelled with a daily resolution, the tool splits the available waste into days of the simulation. Monthly and weekly amounts are considered to arrive evenly through the whole month or week.

The energy hub is paid a **waste treatment remuneration** for every tonne of waste it takes. By default, the optimization takes only the amount of waste that pays off. If *Treat all waste* is selected, all available waste must be taken and treated.

Waste storage

The waste storage decouples the arrival of waste from its treatment. Waste that is taken but not treated in a day is stored and can be treated in later days, e.g. during periods with less available waste or after the maintenance of the reactors. The optimization determines the optimal storage capacity. The amount of stored waste at the end of the simulation must be the same as at its beginning, so the operation can be repeated every year.

4.2.19.1 Input parameters

Waste source parameters

The available waste is defined in one of two tabs.

Monthly amounts tab

Parameter	Description
Available waste in January ... A December [t/month]	mount of waste available in each month. It is considered to arrive evenly through the whole month.

Import profile tab

Parameter	Description
Import profile [button]	By clicking the button, the user selects a CSV file from the local system. File format requirements: - File format: .csv - Single column - No header row Allowed number of rows: - 12 (monthly values in t/month) - 52 or 53 (weekly values in t/week, starting on January 1st; the 53rd week contains only the last day of the year, or the last 2 days in a leap year) - 365 or 366 (daily values in t/day) Monthly and weekly values are split evenly into days of the simulation. Daily profiles are adjusted to leap and non-leap years automatically (29 February is added or removed). After selecting a file, a message is displayed indicating whether the data import was successful or if errors were detected.
Check imported data [button]	Opens a pop-up window with an interactive chart showing the imported available waste profile.

Common parameters

Parameter	Description
Treat all waste [checkbox]	If selected, all available waste must be taken and treated. Otherwise, only the amount that pays off is taken. It cannot be combined with *Maximal payoff period* or *Maximum investment* limits. It can be selected only if the maximal size of all reactors is enough to treat all waste available in a year (maintenance included) and the maximal waste storage capacity is enough to store all available waste at once. Make sure there is also enough demand for non-green hydrogen (H2 sales or Fuel cell), because hydrogen produced from waste is not green.
Waste treatment remuneration [€/t]	Price the energy hub is paid for taking waste from someone else. It can be negative if the hub has to pay for the waste.
Mass ratio of hydrogen [%]	Mass fraction of hydrogen in the waste (ω_{H2}). It significantly influences the efficiency of the waste treatment process.

Parameter	Description
Mass ratio of carbon [%]	Mass fraction of carbon in the waste (ω_{C}).
Mass ratio of water [%]	Mass fraction of water in the waste (ω_{H2O}). The sum of all three mass ratios cannot be greater than 100 %.

Waste storage parameters

Parameter	Description
Investment price [€/t]	Investment price per tonne of storage capacity. It includes all necessary materials, labour (design, construction, commissioning, etc.), and financing until full operation.
Operation and maintenance price [€/t]	Estimated annual O&M price per tonne of storage capacity. It includes cleaning, insurance and administration costs, but not degradation of the storage.
Replacement price [€/t]	Same as investment price, but after the lifetime of the storage, i.e. the future investment price.
Existing installed capacity [t]	Storage capacity that already exists at the hub site.
Minimum storage capacity [t]	Lower limit on the storage capacity. It can be used for any custom requirement, but it can make the optimization infeasible or prevent reaching the target ROI.
Maximum storage capacity [t]	Upper limit on the storage capacity.
Lifetime [years]	Number of years after which the storage needs to be replaced.

4.2.19.2 Optimization output

Variable	Description
Taken waste	Amount of waste taken from the source in every day [t]. It is lower than or equal to the available waste (equal if *Treat all waste* is selected).
Treated share of available waste	Share of all available waste that is treated by the reactors [%].
Waste storage size	Optimal storage capacity (existing + new) [t].
Stored waste	Amount of waste in the storage at the end of every day [t].
Economic results	- **Revenue** from waste treatment remuneration - **Investment, maintenance and degradation cost** of the waste storage

4.2.19.3 Interface options

- **Monthly amounts** or **Import profile** tab: two ways to define the available waste.
- **Check imported data**: shows the imported profile in an interactive chart.

4.2.19.4 Related components

- Waste treatment plant – treats the waste and uses its composition in the process matrix.
- Financial parameters – *Maximal payoff period* and *Maximum investment* cannot be used together with *Treat all waste*.

4.2.19.5 Troubleshooting

Common issues and solutions:

- **Waste profile is not imported**: if the *Import profile* tab is used, a valid CSV file must be imported.
- **Wrong number of rows**: the imported file must have 12, 52, 53, 365 or 366 rows.
- **Mass ratios greater than 100 %**: the sum of the mass ratios of hydrogen, carbon and water cannot exceed 100 %.
- **Treat all waste with financial limits**: *Treat all waste* cannot be combined with the payoff period or investment limit. Disable one of them.
- **Infeasible optimization**: if the available waste is not enough for the minimum size of the reactors, or the storage is too small to store waste that must be treated, increase the maximum storage capacity or reduce the minimum reactor size.

4.2.20 LOHC components

Liquid organic hydrogen carriers (LOHC) allow hydrogen to be stored and transported in liquid form. The LOHC components form two streams. The **export stream** (LOHC sales with its hydrogenation unit → LOHC tank) loads hydrogen produced on site into the carrier and sells the loaded LOHC; it requires the Electrolyser. The **import stream** (LOHC purchase with its dehydrogenation unit → LOHC tank) buys loaded LOHC and releases its hydrogen on site; it is always available. The released hydrogen can be used by H2 sales, the Fuel cell or the fixed hydrogen consumption, so LOHC import can be used without the Electrolyser. Only one stream can be active at a time. The stream is selected with the **LOHC export** / **LOHC import** checkboxes: checking one unchecks the other. The LOHC components are shown under three headers: LOHC sales (including the hydrogenation unit), LOHC purchase (including the dehydrogenation unit) and LOHC tank, which is used by both streams.

Hydrogen produced or released on site goes through the low-pressure hydrogen bus (see Section 4.2.12). The hydrogenation unit takes hydrogen from the bus, and the dehydrogenation unit delivers hydrogen to it.

4.2.20.1 LOHC sales

The **LOHC sales** component represents selling loaded LOHC to one or more customers. Each customer has a daily demand and a price, and may require green hydrogen. When loaded LOHC is delivered, the customer returns the same amount of unloaded LOHC. LOHC is sold only during the daily sales window, and the demand is satisfied only if it is economically viable. All quantities are expressed as mass of bound hydrogen.

The hydrogen is bound to the LOHC on site by the **hydrogenation unit**, which is defined in the *Hydrogenation unit (advanced)* section at the bottom of this component.

The component is part of the LOHC export stream and is shown only when the **LOHC export** checkbox is checked (see Section 4.2.20).

4.2.20.1.1 Input parameters

Basic parameters

Parameter	Description
Sales window start [h]	Hour of the day when the sales window opens, i.e. when trucks start being loaded at the site. To allow sales during the whole day, set the start to 0 and the end to 24. Default: 8.
Sales window end [h]	Hour of the day when the sales window closes. Must be greater than the start. Default: 20.
Add new customer [button]	Add a new LOHC customer. Right-click a customer to rename or delete it.

Customer parameters

Each customer has the following configurable parameters:

Parameter	Description
Green hydrogen only [checkbox]	If checked, the customer accepts only LOHC loaded with green hydrogen. Otherwise only non-green hydrogen is sold to this customer. Checked by default.
Daily demand [kg/day]	Maximum amount of hydrogen bound in LOHC that the customer buys per day.

Parameter	Description
Selling price [€/kg]	Selling price per kg of hydrogen bound in LOHC.

The carrier and its price are defined in the LOHC tank.

Hydrogenation unit (advanced)

The hydrogenation unit binds hydrogen produced on site to the LOHC by an exothermic catalytic reaction. It includes the reactor(s), pumps, separator, heat exchangers and a small compressor. Unloaded LOHC is taken from the LOHC tank and the loaded LOHC is returned to it. The size is expressed as hydrogen mass flow fed to the unit [kg/h].

Parameter	Description
Hydrogenation unit type [dropdown menu]	Choose a hydrogenation unit from the application's database to fill in its parameters automatically. Afterwards you can change any of them. The reaction heat is not changed, because it depends on the carrier chosen in the LOHC tank.
Hydrogenation unit already exists? [checkbox]	Whether there is an existing hydrogenation unit at the site. If checked, the existing size becomes visible.
Existing size [kg/h]	Capacity of the existing hydrogenation unit, expressed as the maximum hydrogen mass flow fed to the unit.
Minimum size [kg/h]	Lower limit on the size of the newly installed hydrogenation unit. If greater than zero, the payoff period limit in Financial parameters cannot be used.
Maximum size [kg/h]	Upper limit on the size of the newly installed hydrogenation unit.

Hydrogenation unit (advanced) – *Technical parameters*

Parameter	Description
Specific electricity consumption [kWh/kg]	Electricity consumed per kg of hydrogen fed to the unit, including the compressor, pumps, agitators and other auxiliary equipment. Default: 0.38 kWh/kg.
Hydrogen utilization [%]	Share of the fed hydrogen that is bound to the LOHC. The rest is lost in the hydrogenation module. Default: 97 %.
Reaction heat [kWh/kg]	Heat released per kg of hydrogen bound to the LOHC (exothermic reaction, approx. 150–200 °C). It is set automatically when the carrier is chosen in the LOHC tank. Default: 8.11 kWh/kg (3,5-dibenzyl toluene).

Parameter	Description
Recoverable share of reaction heat [%]	Share of the reaction heat that can be recovered as high-temperature heat. Heat that is not recovered is removed by cooling, whose cost is neglected. Default: 97.5 %.
LOHC loss per hydrogenation [%]	Share of the carrier that degrades during one hydrogenation. Lost carrier is replaced by buying make-up LOHC at the price given in the LOHC tank. Default: 0.02 %, i.e. 20 % of a total degradation of 0.1 % per cycle (range 0–0.04 %).

Hydrogenation unit (advanced) – *Financial parameters*

Parameter	Description
Investment price [€/(kg/h)]	Investment price per hydrogen feed capacity, including all equipment of the unit. Default: 8775 €/(kg/h).
Maintenance price [€/(kg/h)/year]	Yearly maintenance price per hydrogen feed capacity, including catalyst. Default: 522 €/(kg/h)/year.
Replacement price [€/(kg/h)]	Price of replacing the unit at the end of its lifetime, per hydrogen feed capacity. Default: 8775 €/(kg/h).
Lifetime [years]	Lifetime of the hydrogenation unit. Default: 15 years.
Subsidy [%]	Share of the investment covered by a subsidy. Does not apply to the replacement price. Overridden by the subsidy in Financial parameters if that one is applied.

4.2.20.1.2 Interface options

- **Adding customers**: Click the *Add new customer* button to create a new customer.
- **Managing customers**: Right-click on a customer to rename or delete it. Customer names must be unique; a duplicate name is rejected.
- **Existing hydrogenation unit**: Check *Hydrogenation unit already exists?* to enter the size of an existing unit.
- **Predefined hydrogenation unit**: Select a unit from the *Hydrogenation unit type* dropdown menu in the *Hydrogenation unit (advanced)* section to fill in its technical and financial parameters automatically. The reaction heat is not changed by this selection.
- **Reaction heat from the carrier**: Choosing a carrier in the LOHC tank automatically sets the reaction heat of the hydrogenation unit to the value of that carrier. You can change it afterwards.

4.2.20.1.3 Optimization output

Variable	Description
LOHC sold [kg/h]	Bound hydrogen sold to each customer in every time step, separately for green and non-green customers. Sales are zero outside the sales window, and the amount sold during a day is limited by the daily demand of the customer. The time steps of the optimization follow the hydrogen sampling time. A time step belongs to the window if its interval overlaps the window, e.g. with a 24 h sampling time every day is in the window, and with a 4 h sampling time and a window from 8:00 until 20:00, the time steps starting at 8:00, 12:00 and 16:00 are in it.
Revenue [€/year]	Yearly revenue from LOHC sales.
Hydrogenation unit: installed capacity [kg/h]	Optimized size of the hydrogenation unit, as mass flow of hydrogen fed.
Hydrogenation unit: hydrogen fed [kg/h]	Green and non-green hydrogen taken from the hydrogen bus in every time step. The hydrogen bound to the LOHC is the fed hydrogen multiplied by the hydrogen utilization.
Hydrogenation unit: electricity and heat [kW]	Electricity consumption (included in the electricity supply) and recoverable high temperature heat (available to the high temperature heat exchanger in Waste heat).
Hydrogenation unit: costs [€]	Investment, yearly maintenance and degradation costs, and the yearly cost of the LOHC make-up (carrier lost in hydrogenation).

4.2.20.1.4 Related components

- LOHC tank, which supplies the unloaded LOHC to the hydrogenation unit, stores the loaded LOHC until it is sold, receives the returned unloaded LOHC and defines the carrier (and with it the reaction heat).
- Electrolyser or the waste treatment plant (WT), which produce the hydrogen fed to the hydrogenation unit. LOHC export is available only when at least one of them is selected. Hydrogen from waste treatment is non-green, so it can only be sold to customers that accept non-green hydrogen.

4.2.20.1.5 Troubleshooting

Common issues and solutions:

- **LOHC export checkbox not visible**: LOHC export requires the Electrolyser or WT. Select one of them first. Unchecking the last of them also unchecks LOHC export.
- **No customers configured**: Add at least one customer before running the optimization.
- **Duplicate customer names**: Each customer must have a unique name.

- **Invalid daily demand**: The daily demand of every customer must be greater than zero.
- **Invalid price**: The selling price must be greater than or equal to zero.
- **Invalid sales window**: The start and end must satisfy 0 start < end 24.
- **Green hydrogen requirement**: When *Green hydrogen only* is checked, ensure sufficient renewable energy sources are configured in the system.
- **Invalid hydrogenation unit sizes**: The existing, minimum and maximum sizes of the hydrogenation unit must be greater than or equal to zero, and the minimum size must not be greater than the maximum size.
- **Invalid hydrogenation unit technical parameters**: The specific electricity consumption and the reaction heat must be greater than or equal to zero. The hydrogen utilization must be greater than 0 % and at most 100 %, the recoverable share of reaction heat must be between 0 % and 100 %, and the LOHC loss must be at least 0 % and less than 100 %.
- **Invalid hydrogenation unit financial parameters**: All prices must be greater than or equal to zero, the lifetime must be at least 1 year, and the subsidy must be between 0 % and 100 %.
- **Nothing is sold**: Check that the selling price is higher than the cost of producing and loading the hydrogen, and that the customer's daily demand is greater than zero. Green customers can only be served with green hydrogen.

4.2.20.2 LOHC purchase

The **LOHC purchase** component represents buying loaded LOHC from one or more suppliers. Each supplier has a daily availability and a price, and may supply green hydrogen. When loaded LOHC is delivered, the same amount of unloaded LOHC is returned to the supplier. Bought LOHC is stored in the LOHC tank and dehydrogenated on site. LOHC is bought only during the daily purchase window, and only if it is economically viable. All quantities are expressed as mass of bound hydrogen.

The hydrogen is released from the LOHC on site by the **dehydrogenation unit**, which is defined in the *Dehydrogenation unit (advanced)* section at the bottom of this component.

The component is part of the LOHC import stream and is shown only when the **LOHC import** checkbox is checked (see Section 4.2.20).

4.2.20.2.1 Input parameters

Basic parameters

Parameter	Description
Purchase window start [h]	Hour of the day when the purchase window opens, i.e. when trucks start being unloaded at the site. To allow purchases during the whole day, set the start to 0 and the end to 24. Default: 8.
Purchase window end [h]	Hour of the day when the purchase window closes. Must be greater than the start. Default: 20.
Add new supplier [button]	Add a new LOHC supplier. Right-click a supplier to rename or delete it.

Supplier parameters

Each supplier has the following configurable parameters:

Parameter	Description
Green hydrogen [checkbox]	If checked, the supplier delivers LOHC loaded with green hydrogen. Otherwise the bought hydrogen is treated as non-green. Checked by default.
Daily availability [kg/day]	Maximum amount of hydrogen bound in LOHC that can be bought from the supplier per day.
Purchase price [€/kg]	Purchase price per kg of hydrogen bound in LOHC.

The carrier and its price are defined in the LOHC tank.

Dehydrogenation unit (advanced)

The dehydrogenation unit releases hydrogen bound to the LOHC by an endothermic catalytic reaction at approx. 270–320 °C. The reaction heat is supplied by an electric heater, which is part of the unit. Loaded LOHC is taken from the LOHC tank, and the released hydrogen is delivered to the low-pressure hydrogen bus at the *Hydrogen outlet pressure* of the unit. The size is expressed as hydrogen mass flow released from the LOHC [kg/h].

Parameter	Description
Dehydrogenation unit type [dropdown menu]	Choose a dehydrogenation unit from the application's database to fill in its parameters automatically. Afterwards you can change any of them.
Dehydrogenation unit already exists? [checkbox]	Whether there is an existing dehydrogenation unit at the site. If checked, the existing size becomes visible.
Existing size [kg/h]	Capacity of the existing dehydrogenation unit, expressed as the maximum hydrogen mass flow released from the LOHC.

Parameter	Description
Minimum size [kg/h]	Lower limit on the size of the newly installed dehydrogenation unit. If greater than zero, the payoff period limit in Financial parameters cannot be used.
Maximum size [kg/h]	Upper limit on the size of the newly installed dehydrogenation unit.

Dehydrogenation unit (advanced) – *Technical parameters*

Parameter	Description
Specific electricity consumption [kWh/kg]	Electricity consumed per kg of hydrogen released from the LOHC: the electric heater (reaction heat and heat losses, reduced by internal heat recovery) plus auxiliary equipment (pumps, purification, etc.). Default: 10.4 kWh/kg, which is the value for 3,5-dibenzyl toluene.
Hydrogen yield [%]	Share of the hydrogen released from the LOHC that is delivered to the hydrogen bus. The rest is lost in purification and purge. Default: 99 %.
LOHC loss per dehydrogenation [%]	Share of the carrier that degrades during one dehydrogenation. Lost carrier is replaced by buying make-up LOHC at the price given in the LOHC tank. Default: 0.08 %, i.e. 80 % of a total degradation of 0.1 % per cycle (range 0–0.16 %).
Hydrogen outlet pressure [bar]	Pressure of the hydrogen delivered to the hydrogen bus, expressed as difference from the atmospheric pressure (barg). The reaction runs at approx. 1–5 bar, so the default is 2 bar. The lowest outlet pressure of all hydrogen sources is the pressure of the hydrogen bus, which is the inlet pressure of the hydrogen compressor in Hydrogen storage and the highest pressure available to the fixed hydrogen consumption at low pressure. Default: 2 bar.

Dehydrogenation unit (advanced) – *Financial parameters*

Parameter	Description
Investment price [€/(kg/h)]	Investment price per released hydrogen capacity, including the electric heater and all other equipment of the unit. Default: 12270 €/(kg/h).
Maintenance price [€/(kg/h)/year]	Yearly maintenance price per released hydrogen capacity. Default: 491 €/(kg/h)/year.

Parameter	Description
Replacement price [€/(kg/h)]	Price of replacing the unit at the end of its lifetime, per released hydrogen capacity. Default: 12270 €/(kg/h).
Lifetime [years]	Lifetime of the dehydrogenation unit. Default: 15 years.
Subsidy [%]	Share of the investment covered by a subsidy. Does not apply to the replacement price.

4.2.20.2.2 Interface options

- **Adding suppliers**: Click the *Add new supplier* button to create a new supplier.
- **Managing suppliers**: Right-click on a supplier to rename or delete it. Supplier names must be unique; a duplicate name is rejected.
- **Existing dehydrogenation unit**: Check *Dehydrogenation unit already exists?* to enter the size of an existing unit.
- **Predefined dehydrogenation unit**: Select a unit from the *Dehydrogenation unit type* dropdown menu in the *Dehydrogenation unit (advanced)* section to fill in its technical and financial parameters automatically.
- **Carrier-specific consumption**: The specific electricity consumption of the dehydrogenation unit is not changed when a different carrier is chosen in the LOHC tank. The default value applies to 3,5-dibenzyl toluene; adjust it manually if you use another carrier.

4.2.20.2.3 Optimization output

Variable	Description
LOHC bought [kg/h]	Bound hydrogen bought from each supplier in every time step, separately for green and non-green suppliers. Purchases are zero outside the purchase window, and the amount bought during a day is limited by the daily availability of the supplier. The time steps of the optimization follow the hydrogen sampling time. A time step belongs to the window if its interval overlaps the window, e.g. with a 24 h sampling time every day is in the window, and with a 4 h sampling time and a window from 8:00 until 20:00, the time steps starting at 8:00, 12:00 and 16:00 are in it.
Cost of bought LOHC [€/year]	Yearly cost of LOHC purchases (operating cost).
Dehydrogenation unit: installed capacity [kg/h]	Optimized size of the dehydrogenation unit, as mass flow of hydrogen released from the LOHC.

Variable	Description
Dehydrogenation unit: hydrogen released [kg/h]	Green and non-green hydrogen released from the LOHC in every time step. The hydrogen delivered to the hydrogen bus is the released hydrogen multiplied by the hydrogen yield.
Dehydrogenation unit: electricity [kW]	Electricity consumption, including the electric heater (included in the electricity supply).
Dehydrogenation unit: costs [€]	Investment, yearly maintenance and degradation costs, and the yearly cost of the LOHC make-up (carrier lost in dehydrogenation).

4.2.20.2.4 Related components

- LOHC tank, where the bought LOHC is stored before dehydrogenation, from which the unloaded LOHC is returned to the supplier, and which defines the carrier.
- Electrical grid connection and other electricity sources, which supply the electric heater of the dehydrogenation unit.
- Fuel cell stack, H2 sales and Fixed hydrogen consumption, which can use the released hydrogen. At least one of them must be selected together with LOHC import; they do not require the Electrolyser.

4.2.20.2.5 Troubleshooting

Common issues and solutions:

- **Last hydrogen use cannot be unchecked**: While LOHC import is selected, at least one of H2 cons., Fuel cell or H2 sales must stay selected. Checking LOHC import selects H2 sales automatically if none of them is selected yet.
- **No suppliers configured**: Add at least one supplier before running the optimization.
- **Duplicate supplier names**: Each supplier must have a unique name.
- **Invalid daily availability**: The daily availability of every supplier must be greater than zero.
- **Invalid price**: The purchase price must be greater than or equal to zero.
- **Invalid purchase window**: The start and end must satisfy 0 start < end 24.
- **Invalid dehydrogenation unit sizes**: The existing, minimum and maximum sizes of the dehydrogenation unit must be greater than or equal to zero, and the minimum size must not be greater than the maximum size.
- **Invalid dehydrogenation unit technical parameters**: The specific electricity consumption must be greater than or equal to zero. The hydrogen yield must be greater than 0 % and at most 100 %, and the LOHC loss must be at least 0 % and less than 100 %.

- **Invalid dehydrogenation unit financial parameters**: All prices must be greater than or equal to zero, the lifetime must be at least 1 year, and the subsidy must be between 0 % and 100 %.
- **Nothing is bought**: Check that the hydrogen released from the LOHC is worth more to H2 sales, the fuel cells or the fixed hydrogen consumption than the purchase price plus the cost of dehydrogenation. Hydrogen from non-green suppliers cannot be used where green hydrogen is required.

4.2.20.3 LOHC tank

The **LOHC tank** stores the liquid organic hydrogen carrier (LOHC). A single tank stores both unloaded and loaded LOHC, separated by a flexible membrane, so the amount of carrier in the tank is constant. The tank size is the mass of hydrogen it holds when all of its LOHC is loaded [kg]. The investment includes the tank and the first fill of the carrier.

The LOHC tank is also where the carrier is chosen. The carrier defines the hydrogen weight capacity, the carrier price and the reaction heat of the hydrogenation unit of LOHC sales.

The component is used by both LOHC streams and is shown when either the **LOHC export** or the **LOHC import** checkbox is checked (see Section 4.2.20).

4.2.20.3.1 Input parameters

Basic parameters

Parameter	Description
Carrier [dropdown menu]	Liquid organic hydrogen carrier used in the tank. Choosing a carrier fills in its hydrogen weight capacity and price, and the reaction heat of the hydrogenation unit. Afterwards you can change any of them. Note: 2-methyl indole and N-ethylcarbazole are solid at ambient temperature (melting points 58 °C and 70 °C), and heating of the storage is not modelled. Default: 3,5-dibenzyl toluene.
Hydrogen weight capacity [wt%]	Mass of hydrogen that can be bound per mass of carrier. 6.2 wt% means that 1 kg of LOHC holds 0.062 kg of hydrogen.
Carrier price [€/kg]	Price of the carrier (unloaded LOHC) per kg of LOHC. Used for the first fill of a new tank and for make-up of carrier lost in hydrogenation and dehydrogenation.

Parameter	Description
Carrier price per kg of stored hydrogen [€/kg]	Calculated as carrier price / hydrogen weight capacity. Read-only. For 3,5-dibenzyl toluene: 0.90 / 0.062 = 14.52 €/kg.
LOHC tank already exists? [checkbox]	Whether there is an existing LOHC tank at the site. An existing tank is assumed to be already filled with the carrier. If checked, the existing size becomes visible.
Existing size [kg]	Size of the existing tank, expressed as the mass of hydrogen it holds when all of its LOHC is loaded.

Advanced parameters

Parameter	Description
Minimum size [kg]	Lower limit on the size of the new tank. If greater than zero, the payoff period limit in Financial parameters cannot be used.
Maximum size [kg]	Upper limit on the size of the new tank.
Investment price [€/kg]	Investment price of the tank itself (without the carrier and the membrane) per kg of stored hydrogen. Default: 5 €/kg.
Maintenance price [€/kg/year]	Yearly maintenance price per kg of stored hydrogen. Default: 0.10 €/kg/year.
Replacement price [€/kg]	Price of replacing the tank at the end of its lifetime, per kg of stored hydrogen. Default: 5 €/kg.
Lifetime [years]	Lifetime of the LOHC tank. Default: 20 years.
Subsidy [%]	Share of the investment covered by a subsidy. Does not apply to the replacement price.

Predefined carriers

The application's database contains the following carriers. Hydrogen weight capacities, prices and melting points are taken from (García et al. 2024). The reaction heat is calculated from the reaction enthalpy and stoichiometry given in the same source, per kg of bound hydrogen.

Carrier	Hydrogen weight capacity [wt%]	Carrier price [€/kg LOHC]	Reaction heat [kWh/kg H2]	Melting point [°C]
3,5-dibenzyl toluene	6.2	0.90	8.11	-39
2-methyl indole	5.8	0.86	7.10	58
N-ethylcarbazole	5.6	1.64	7.85	70

4.2.20.3.2 Interface options

- **Choosing a carrier**: Selecting a carrier from the *Carrier* dropdown menu fills in its hydrogen weight capacity and carrier price. It **also** sets the reaction heat of the hydrogenation unit of LOHC sales to the value of the chosen carrier.
- **Custom carrier**: You can change the hydrogen weight capacity and the carrier price after choosing a carrier. The dropdown menu keeps its value, and the read-only *Carrier price per kg of stored hydrogen* is updated automatically.
- **Existing tank**: Check *LOHC tank already exists?* to enter the size of an existing tank.

4.2.20.3.3 Optimization output

Variable	Description
Installed capacity [kg]	Optimized size of the tank, as mass of hydrogen it holds when all of its LOHC is loaded. The tank is at least as large as one day of LOHC sales or purchases, since LOHC is exchanged with customers and suppliers once a day.
Loaded LOHC [kg]	Amount of loaded LOHC (as bound hydrogen) in every time step, separately for green and non-green hydrogen. The rest of the tank holds unloaded LOHC. The level at the end of the simulation equals the level at the beginning.
Investment cost [€]	Investment in the tank and in the first fill of the carrier (carrier price per kg of stored hydrogen times the new size). The subsidy applies to both.
Maintenance and degradation costs [€/year]	Yearly costs of the tank itself. Carrier losses are paid as make-up by the hydrogenation and dehydrogenation units.

4.2.20.3.4 Related components

- The hydrogenation unit of LOHC sales, which loads the LOHC from the tank (export stream). Its reaction heat is set by the carrier chosen here.
- LOHC sales, which sells loaded LOHC from the tank (export stream).
- LOHC purchase, which delivers loaded LOHC to the tank (import stream).
- The dehydrogenation unit of LOHC purchase, which unloads the LOHC from the tank (import stream).

4.2.20.3.5 Troubleshooting

Common issues and solutions:

- **Invalid hydrogen weight capacity**: The hydrogen weight capacity must be greater than 0 wt% and less than 100 wt%.
- **Invalid carrier price**: The carrier price must be greater than or equal to zero.
- **Invalid sizes**: The existing, minimum and maximum sizes must be greater than or equal to zero, and the minimum size must not be greater than the maximum size.
- **Invalid financial parameters**: All prices must be greater than or equal to zero, the lifetime must be at least 1 year, and the subsidy must be between 0 % and 100 %.
- **Reaction heat changed unexpectedly**: Choosing a carrier overwrites the reaction heat of the hydrogenation unit. If you entered a custom reaction heat, enter it again after choosing the carrier.
- **Solid carriers**: 2-methyl indole and N-ethylcarbazole are solid at ambient temperature. Heating of the storage is not modelled.

4.2.21 Financial parameters

The **Financial parameters** component defines the economic constraints and incentives that apply to the entire hydrogen infrastructure project. These global financial settings affect the optimization of all components in the system.

This component is always present in the system and does not require any component selection checkbox. It contains settings for investment constraints, subsidies, and project lifetime.

4.2.21.1 Input parameters

Investment constraints

The investment constraints section allows you to set limits on the project's payback period and total investment amount. These constraints help guide the optimization towards economically viable solutions.

Parameter	Description
Limit the payoff period? [checkbox]	If enabled, the optimization will constrain the payback period to be within the specified maximum. If a lower bound on the size of any component is set to > 0, then this should be automatically turned off to ensure the feasibility of the optimization problem.
Maximum payback period [years]	The longest acceptable timeframe (in years) for the project's cumulative savings to offset the initial investment. This parameter is only visible if *Limit the payoff period?* is enabled.

Parameter	Description
Limit the investment amount? [checkbox]	If enabled, the optimization will constrain the total investment to be within the specified maximum. If a lower bound on the size of any component is set to > 0, then this should be automatically turned off to ensure the feasibility of the LP solution.
Maximum investment [€]	The highest total amount (in €) you're willing to invest in the project. This parameter is only visible if *Limit the investment amount?* is enabled.

Investment incentives

The investment incentives section allows you to define subsidies that can reduce the effective cost of the project. These subsidies are applied to component investments during the optimization.

Parameter	Description
Apply subsidy to all components? [checkbox]	If checked, the subsidy will be applied to the cost of all microgrid components (e.g., electrolyzer, heat exchanger). If unchecked, you can specify which components receive the subsidy individually.
Investment subsidy [%]	The percentage of the total project cost covered by government or other financial incentives. This parameter is only visible if *Apply subsidy to all components?* is enabled.

Project lifetime

The project lifetime settings define the planning horizon for the optimization. This affects the calculation of annualized costs and the payback period analysis.

Parameter	Description
Design lifetime infinite [checkbox]	Check this box if you want your system to work indefinitely. When enabled, the *Design lifetime* parameter is hidden.
Design lifetime [years]	Lifetime of the whole system. This parameter determines the number of years used for calculating annualized costs and the payback period. This parameter is hidden if *Design lifetime infinite* is enabled.

4.2.21.2 Interface options

The Financial parameters component provides the following interface options:

- **Payoff period constraint**: Enable or disable the payback period constraint based on your investment criteria.
- **Investment limit**: Set a maximum investment amount to keep the project within budget.
- **Subsidy configuration**: Apply subsidies uniformly across all components or configure them individually per component.
- **Lifetime settings**: Choose between a finite project lifetime or an infinite planning horizon.

4.2.21.3 Troubleshooting

Common issues and solutions:

- **Minimum component size set**: If size of any component is constraint in a way that a minimum size is set, you should unselect *Limit the payoff period* and *Limit the investment amount*.

4.2.22 Simulation parameters

Simulation parameters control the simulation period, time resolution and solver settings used in the mathematical model. The simulation parameters are always present and must be configured for each scenario.

4.2.22.1 Input parameters

Time settings

Parameter	Description
Simulation start date [date picker]	Starting day of the simulation period. If the simulation period is shorter than a year, all profiles (demands, consumptions, prices, etc.) are repeated until the whole year is fulfilled. The maximum duration is 365 days.
Simulation end date [date picker]	The last day of the simulation period. If the simulation period is shorter than a year, all profiles (demands, consumptions, prices, etc.) are repeated until the whole year is fulfilled. The maximum duration is 365 days.

Sampling settings

Parameter	Description
Sampling time - electrical part [h] (dropdown)	The sampling time for the purely electrical components of the hub: electrical grid connection, LREP, PV system, Wind farm, BESS, and electrical consumption. This sampling time must be less than or equal to the sampling time for hydrogen and heat. Common values are 1 hour (default), 2 hours, or 4 hours. With lower sampling time comes greater accuracy, but the price is greater computational complexity.
Sampling time - hydrogen and heat [h] (dropdown)	The sampling time for H2 and heat components of the hub: water supply, electrolyser, fuel cell, waste heat. This sampling time must be greater than or equal to the sampling time for electrical part. Common values are 1 hour, 2 hours, 4 hours, or 24 hours. H2 and O2 sales have a fixed sampling time of 24 hours. With lower sampling time comes greater accuracy, but the price is greater computational complexity.

Optimization settings

Parameter	Description
Solver optimality tolerance [-]	Both primal and dual feasibility tolerance. A lower value means more precise optimization but may require longer computation time. Default value is 1e-07.
Optimization timeout [min]	The maximum time within which the optimization can be performed. If an optimal solution can be found within the solver optimality tolerance before the timeout is reached, the Results screen will be shown. Otherwise, a timeout error message will be displayed. Default value is 60 minutes.

4.2.22.2 Understanding time sampling

The sampling time determines the time resolution of the optimization model. There is a trade-off between accuracy and computational complexity:

- **Hourly sampling (1 hour)**: Provides the highest accuracy for modeling daily fluctuations in electricity prices, renewable generation, and demand. This is the default and recommended setting for most analyses.
- **2-4 hour sampling**: Reduces computational time but may miss some short-term dynamics. Suitable for preliminary analyses or when investigating seasonal patterns.

- **Daily sampling (24 hours)**: Very fast but may not capture important intra-day effects. Typically used only for hydrogen part components which are modeled with daily resolution.

The electrical sampling time should always be less than or equal to the hydrogen/heat sampling time. The tool will automatically enforce this constraint.

4.2.22.3 Simulation period

The simulation period defines the time horizon for the optimization:

- **Full year (default)**: January 1 to December 31. All hourly profiles are used directly.
- **Partial year**: Any period up to 365 days. If the period is shorter than a year, the profiles are repeated to fill the year. For example, if you simulate January to March, the model will repeat Q1 data for the remaining months.

The default simulation period is one full year (365 days).

4.2.22.4 Troubleshooting

- **Simulation duration too long**: If the simulation period exceeds 365 days, an error message will be displayed. Reduce the simulation duration to a maximum of one (non-leap) year.
- **Optimization timeout**: If the optimization does not complete within the specified timeout, consider:
  - Increasing the optimization timeout
  - Increasing the sampling time (e.g., from 1 hour to 2 hours)
  - Reducing the simulation period
  - Simplifying the system configuration by removing some components
- **Sampling time mismatch**: If the electrical sampling time is greater than the hydrogen sampling time, the tool will automatically adjust the hydrogen sampling time to match.

4.3 Run optimization

After configuring all system components and their parameters, click **Run optimization** to generate the results. Once clicked, the tool performs the following sequence of operations:

1. **Project save prompt**

: The tool first asks whether you want to **save the project** before starting the optimization. You can choose to: - save the project,
- continue without saving, or
- cancel the optimization.

```
Saving the project is recommended in case an error occurs and the application closes unexpec
```

2. **Data validation**

: The tool validates all parameters and their combinations to ensure they are within allowed ranges. If an invalid value is detected, a pop-up window titled **Input validation failed** appears. The message: - identifies the component where the issue occurs, and
- explains the reason for the failure.

```
Only the first detected issue is shown. After resolving it, run the optimization again to re
```

3. **Data processing**

: The tool prepares all required inputs for the optimization, including: - obtaining profiles from online sources (PVGIS, renewables.ninja, Open-Meteo),
- scaling profiles according to user-defined values,
- generating and converting additional parameters (e.g., unit adjustments).

```
During this phase, additional validation is performed on the processed data.
If validation fails here, the cause is usually:
- an incompatible combination of parameters, or
- a temporary internet connectivity issue.
```

4. **Problem generation**

: After successful data preparation, the tool formulates the optimization problem and passes it to the solver.

5. **Solving the problem**

: The optimization problem is solved using the open-source HiGHS solver. When the solver finishes, one of the following status messages is displayed: - **Successful optimization** – the optimal solution was found.
- **Time limit reached** – a feasible but suboptimal solution was obtained.
- **Unexpected error** – the solver failed to complete.

6. **Results post-processing**

: After a successful run, the tool processes the results for presentation. A new window (Results) is created under the scenario tree node and automatically opened in the main workspace.

Waiting During the entire procedure, a message *Working, please wait…* is displayed together with a loading indicator. While optimization is running:

- the user interface is temporarily disabled,
- no interaction is possible, and
- the process cannot be stopped from within the application.

Stopping the optimization is planned for future releases. Currently, it can only be terminated via the system task manager.

Saving the data Results are **not saved automatically** after optimization. Make sure to save the project manually to preserve the results.

5 Results

After running the optimization, the **Results** window displays the complete output of the optimization process. This chapter explains each section of the Results view.

The Results window is organized into several collapsible sections, each presenting different aspects of the optimization output. The following sections provide detailed explanations of each component of the results.

5.1 Opening results

To view the results of a scenario:

1. In the **Sidebar**, expand the scenario you want to analyze
2. Click on **Results**
3. The Main window will display results

Results are only available after a successful optimization run.

5.2 Quick summary

At the top of the Results window, a quick summary provides immediate insight into the investment viability:

- **Total profit**: The amount of money the recommended investment could generate or save over the defined project lifetime compared to the no-investment case
- **Payback period**: The number of years required for the investment to pay for itself through generated savings or revenue

5.3 Key recommendations (optimal sizes)

The **Key recommendations** section presents the optimal sizes of all major components in your energy system. These are the capacities that the optimization algorithm has determined will maximize the economic performance while meeting all technical constraints.

5.3.1 Components displayed

The following components are shown with their optimal sizes:

Component	Unit	Description
PV capacity	kW	Total photovoltaic system capacity
Wind farm capacity	kW	Installed wind turbine capacity
Leonitus WEC power	kW	Installed power of the Leonitus wave energy converter
Battery storage	kWh	Total battery energy storage capacity
Battery converter	kW	Power conversion system capacity for battery
Electrolyser capacity	kW	Total electrolyser capacity for hydrogen production
Fuel cell capacity	kW	Total fuel cell capacity for power generation
H2 storage	kg	Hydrogen storage tank capacity
O2 storage	kg	Oxygen storage tank capacity
Hydrogenation unit	kg/h	Hydrogenation unit capacity (hydrogen fed), LOHC export
Dehydrogenation unit	kg/h	Dehydrogenation unit capacity (hydrogen released), LOHC import
LOHC tank	kg	LOHC tank capacity, as mass of hydrogen it holds when all LOHC is loaded
Grid connection	kW	Electrical grid connection capacity
Waste treatment plant capacity	t/day	Total capacity of all waste treatment reactors, existing and new

5.3.2 How to interpret

- Only components selected in the System structure are shown.
- Components with **zero values** are not recommended for installation in the optimal configuration.
- These sizes represent the economically optimal solution based on your inputs.

5.4 Financial results

The **Financial Results** section provides a comprehensive economic analysis of the optimized system.

5.4.1 Costs, revenues, and profits

This table presents the basic financial metrics:

Indicator	Description
Total investment cost (CAPEX)	The total upfront capital expenditure required to build the system, including all components and installation costs
Yearly operational cost (OPEX)	Annual operating expenses including maintenance, consumables, and variable costs
Yearly revenue	Annual income generated from selling hydrogen, oxygen, heat, or electricity
Yearly profit	Annual net profit defined as $REVENUE - OPEX$

5.4.2 Financial indicators

The financial indicators table provides advanced metrics for investment analysis.

5.4.2.1 Payback period

The **payback period**, also refered to as **payoff period** represents the time required for the cumulative cash flows from the investment to equal the initial capital expenditure.

Formula:

$$N_{payoff} = \frac{CAPEX}{Profit_{total} - Profit_{exist}}$$

Where:

- $CAPEX$ = Total investment cost
- $Profit_{total}$ = Annual profit with the new system
- $Profit_{exist}$ = Annual profit from existing infrastructure only

5.4.2.2 Net present value (NPV)

NPV calculates the present value of all cash flows over the project lifetime, discounted at the specified rate. A positive NPV indicates that the investment is financially attractive.

Formula:

$$NPV = -CAPEX + (Profit_{total} - Profit_{exist}) \times \frac{1 - (1 + R)^{-N_{DL}}}{R}$$

Where:

- R = Discount rate
- N_{DL} = Project lifetime in years

5.4.2.3 Return on investment (ROI)

ROI expresses the total return on investment as a percentage over the project lifetime.

Formula:

$$ROI = N_{DL} \times \frac{Profit_{total} - Profit_{exist}}{CAPEX} \times 100\%$$

5.4.2.4 Internal rate of return (IRR)

IRR is the discount rate that makes the NPV equal to zero. It represents the effective annual return on the investment. If the IRR exceeds your required rate of return, the investment is worthwhile.

5.4.2.5 Levelized cost of electricity (LCOE)

LCOE represents the average cost per unit of electricity produced by the system over its lifetime. It allows comparison with grid electricity prices or other generation technologies.

Formula:

$$LCOE = \frac{PROFIT_{total_without_en} - PROFIT_{exist} - \frac{CAPEX}{N_{DL}}}{E_{consumed}} \times 1000$$

Unit: €/MWh

- $PROFIT_{total_without_en}$ stands for annual profit without cost of electricity in [€].
- $E_{consumed}$ stands for annual amount of electricity imported into the hydrogen hub in [kWh].

5.4.2.6 Levelized cost of hydrogen (LCOH)

LCOH represents the average cost per kilogram of hydrogen delivered by the hub: sold (H2 sales and gas grid injection), sold bound to LOHC (LOHC sales), and used locally by the fixed hydrogen consumption. This is particularly useful for comparing hydrogen production costs from different technologies or against market prices.

Formula:

$$LCOH = \frac{\frac{CAPEX}{N_{DL}} + PROFIT_{exist} - PROFIT_{total_without_H2}}{m_{H2,sold}}$$

Unit: €/kg

- $PROFIT_{total_without_H2}$ stands for annual profit without revenue from selling hydrogen (H2 sales, gas grid injection, LOHC sales) and without the savings of the fixed hydrogen consumption [€].
- $m_{H2,sold}$ stands for annual amount of hydrogen sold on the market, sold bound to LOHC, and used locally by the fixed hydrogen consumption in [kg].

5.5 Operational overview

The **Operational Overview** section shows how the optimized system operates over time.

5.5.1 Yearly energy flows

This table summarizes the annual energy flows for each component, expressed in MWh/year:

Component	Description
PV generation	Total electricity produced by photovoltaic system
Wind farm generation	Total electricity produced by wind turbines
Leonitus WEC generation	Total electricity produced by the Leonitus wave energy converter
Local REP generation	Local Renewable Energy Plant generation
RESu utilization	Renewable energy system utilization
Electrolyser consumption	Electricity consumed by electrolyser for hydrogen production
Fuel cell generation	Electricity generated by fuel cell
BESS charging	Energy that went in battery
BESS discharging	Energy discharged from battery
Grid import	Electricity bought from the grid
Grid export	Electricity sold to the grid

Component	Description
Fixed electricity consumption	Electricity used by the fixed electricity consumption
WT el. en. production/consumption	Net electrical energy of all waste treatment reactors. A positive value is production, a negative value is consumption
Total production	Sum of all electricity produced on site

Only rows of the selected components with non-zero values are shown.

5.5.2 Production and sales profiles

The operational overview includes interactive charts showing:

1. **Electrical Power Flows**: A time-series chart showing electricity generation and consumption over the simulation period: renewable sources (PV, wind farm, Leonitus WEC, LREP, RESu), fuel cell, net power of the waste treatment plant (**WT net el. power**), electrolyser, dehydrogenation unit, battery charging and discharging, fixed electricity consumption and grid power.
2. **Non-Electrical Flows**: Shows hydrogen production, treated waste, hydrogen sales (including gas grid injection), oxygen sales, local hydrogen consumption, LOHC sales and purchase, and heat sales over time. **Hydrogen production** includes hydrogen from electrolysers, hydrogen released by the dehydrogenation unit and hydrogen produced by the waste treatment reactors.
3. **Tank States (Mass)**: Shows the hydrogen (green and non-green), oxygen, loaded LOHC and waste storage levels over time.

> **Note**
>
> The daily electrical chart does not contain all consumers, e.g. compressors, the water demineralizer or the heat pump. The complete electricity balance is shown in the Detailed el. energy flows section.

5.6 Detailed el. energy flows

This section contains the **Detailed Electrical Power Flows** chart with every electrical power flow that enters the electricity balance of the site in each time step:

- **Production** is shown as **positive**: PV, wind farm, Leonitus WEC, LREP, RESu, battery discharging, fuel cell and net power of the waste treatment plant (when it produces electricity).
- **Consumption** is shown as **negative**: battery charging, electrolyser, water demineralizer, H2 compressor 1 and 2, hydrogenation unit, dehydrogenation unit, water pump, O2 compressor, heat pump and fixed electricity consumption.
- **Grid (+ import)**: electricity exchanged with the grid. A positive value is import from the grid and a negative value is export to the grid.

The sum of all flows in every time step is zero. Use this chart to find which consumers use the electricity that is not visible in the daily chart of the operational overview. For example, H2 compressor 1 can consume a significant share of electricity when hydrogen is compressed from a low outlet pressure to the storage pressure.

Only the components selected in the System structure are shown.

5.7 Detailed investment specification

The **Detailed investment specification** section provides a breakdown of each component's optimal size and corresponding investment cost.

5.7.1 Components included

The table includes all system components with a new (optimized) size larger than zero:

Component	Unit	Description
PV system	kWp	Solar photovoltaic installation
Wind farm	kWp	Wind turbine installation
Leonitus WEC	kW	Leonitus wave energy converter power
Battery storage	kWh	Battery capacity
Battery converter	kW	Power electronics for battery
Electrolyser	kW	Hydrogen production equipment, one row per electrolyser
Fuel cell	kW	Power generation from hydrogen, one row per fuel cell
Hydrogen tank	kg	Hydrogen storage vessel
Oxygen tank	kg	Oxygen storage vessel
Hydrogenation unit	kg/h	Loading of hydrogen into LOHC
Dehydrogenation unit	kg/h	Release of hydrogen from LOHC
LOHC tank (incl. carrier)	kg	LOHC storage, including the carrier

Component	Unit	Description
Grid connection	kW	Electrical grid interface
H2 compressor 1/2	kg/h	Hydrogen compression equipment
O2 compressor	kg/h	Oxygen compression equipment
Water demineralizer	kW	Water treatment for electrolysis
Water pump	kW	Water supply pumping
Water grid connection	kg/h	Municipal water connection
Body of water	kg/h	Groundwater extraction
Rainwater harvesting	m^2	Rainwater collection area
Water tank	m^3	Water storage
Low temp heat exchanger	kW	Low-temperature heat recovery
High temp heat exchanger	kW	High-temperature heat recovery
Heat pump	kW	Heat pump for temperature upgrade
Thermal storage	kWh	Thermal energy storage
District heating connection	kW	District heating network interface
H2 grid connection	kg/h	Hydrogen pipeline or delivery connection
Waste treatment reactor	t/day	Waste treatment reactor, one row per reactor
Waste storage	t	Storage of waste before treatment

5.7.2 How to read this table

- **Optimal Size**: The capacity recommended by the optimization
- **Investment Cost**: The calculated capital cost based on the component's cost parameters
- Components with zero size are not recommended for the optimal configuration

5.8 Yearly operation summary

This section provides a detailed breakdown of annual production and consumption quantities for all resources:

5.8.1 Hydrogen

Metric	Description
Hydrogen production	Total hydrogen produced by the electrolyser(s) in kg/year

Metric	Description
Hydrogen sales (green)	Green hydrogen sold (produced from renewable sources) in kg/year
Hydrogen sales (non-green)	Non-green hydrogen sold (from grid electricity) in kg/year
Hydrogen to grid	Hydrogen delivered to external grid/pipeline in kg/year
Local hydrogen consumption (green / non-green)	Hydrogen used by the fixed hydrogen consumption in kg/year
Hydrogen bought for local consumption	Part of the fixed hydrogen consumption bought from outside the hub in kg/year
Hydrogen fed to hydrogenation	Hydrogen loaded into the LOHC by the hydrogenation unit in kg/year
LOHC sales (green / non-green, bound H2)	Loaded LOHC sold, as bound hydrogen, in kg/year
LOHC purchase (bound H2)	Loaded LOHC bought, as bound hydrogen, in kg/year
Hydrogen released from LOHC	Hydrogen released by the dehydrogenation unit in kg/year
Hydrogen production (waste treatment)	Hydrogen produced by all waste treatment reactors in kg/year

5.8.2 Oxygen

Metric	Description
Oxygen production	Total oxygen produced as a by-product of electrolysis in kg/year
Oxygen sales	Oxygen sold to external customers in kg/year

5.8.3 Heat

Metric	Description
Low temperature heat generation	Waste heat recovered at lower temperatures (e.g., from electrolyser) in MWh/year
High temperature heat generation	Waste heat recovered at higher temperatures (electrolyser and hydrogenation unit) in MWh/year

5.8.4 Water

Metric	Description
Water consumption	Water consumed by the electrolyser for hydrogen production in kg/year

5.8.5 Waste treatment

Metric	Description
Treated waste	Waste treated by all waste treatment reactors in t/year
Tar production	Tar produced by the reactors in kg/year
CO2 emissions	CO_2 emitted by the reactors in kg/year
Carbon black production	Carbon black produced by the reactors in kg/year

Only resources with values larger than zero are shown.

5.9 Electrolyser details

When multiple electrolysers are configured, this section provides a detailed breakdown of each electrolyser stack:

5.9.1 Information provided

Column	Description
Electrolyser	Name/identifier of the electrolyser
Size (kW)	Installed capacity of this specific electrolyser
Investment (€)	Investment cost allocated to this electrolyser
H2 production (kg/year)	Annual hydrogen production from this electrolyser

This detailed view helps you understand:

- Which electrolyser types are recommended
- How the total electrolyser capacity is distributed
- Individual electrolyser utilization rates

5.10 Waste treatment

This section is shown only when the Waste treatment plant is selected. It contains one table with one row per waste treatment reactor, named as the reactor in the System structure, and the **Total** row with the sum over all reactors. All values include the existing and the new (optimized) capacity together.

Column	Unit	Description
Capacity	t/day	Size of the reactor, i.e. the amount of waste it can treat per day
Treated waste	t/year	Amount of waste treated by the reactor
Hyd. prod.	kg/year	Hydrogen produced by the reactor. It is always counted as non-green hydrogen
El. en. prod.	MWh/year	Net electrical energy of the reactor, including the starting surge after maintenance. A negative value means that the reactor consumes more electricity than it produces
Ht. en. prod.	MWh/year	Usable high- and low-temperature heat produced by the reactor

A value is shown as **N/A** if it could not be calculated from the optimization results.

Costs and revenues of the waste treatment plant, the waste source and the waste storage are shown in the Detailed financial specification section.

> **Note**
>
> Annual values are calculated from the simulation period and scaled to one year.

5.11 Green vs non-green energy balance

This section provides insight into the origin of energy used and hydrogen produced. This is particularly important for:

- Green hydrogen certification
- Carbon footprint analysis
- Sustainability reporting

5.11.1 Hydrogen production by source

This table shows the origin of all hydrogen that enters the hub:

Type	kg/year	Percentage
Produced green	Amount	%
Produced non-green	Amount	%
Imported green	Amount	%
Imported non-green	Amount	%
Total	Amount	100%

- **Produced green**: Hydrogen produced by electrolysers using renewable energy sources (PV, wind, Leonitus WEC, etc.)
- **Produced non-green**: Hydrogen produced by electrolysers using grid electricity, and all hydrogen produced by the waste treatment reactors
- **Imported green**: Hydrogen released by the dehydrogenation unit from bought green LOHC
- **Imported non-green**: Hydrogen released by the dehydrogenation unit from bought non-green LOHC

Only rows with values larger than zero are shown.

5.11.2 BESS energy flows

For battery storage systems, this table shows the source of energy used for charging:

Flow	Green (MWh)	Non-green (MWh)	Total (MWh)
Charging	From RES	From grid	Total
Discharging	-	-	Total

This helps you understand the environmental characteristics of your battery operation.

5.12 Capacity factors

Capacity factors indicate how intensively each component is being used relative to its maximum potential.

5.12.1 Understanding capacity factors

The capacity factor is calculated as:

$$CF = \frac{Average\ Output}{Installed\ Capacity} \times 100\%$$

The table has columns **Installed Capacity**, **Avg Output** and **Capacity Factor**, and contains the following components, if they are selected and their installed capacity is larger than zero:

Component	Unit	Output
PV	kW	Electricity production
Wind Farm	kW	Electricity production
Leonitus WEC	kW	Electricity production
Electrolyser (Total) and each electrolyser	kW	Electricity consumption
Fuel Cell	kW	Electricity production
BESS Converter	kW	Mean of the average charging and the average discharging power
Hydrogenation Unit	kg/h	Hydrogen fed into the LOHC
Dehydrogenation Unit	kg/h	Hydrogen released from the LOHC

5.13 Detailed financial specification

The **Detailed Financial Specification** section shows the costs and revenues of every component selected in the System structure and of the components it depends on (e.g. water supply and compressors of the electrolyser). One row is shown per component, and the **Total** row is the sum of all rows. Values include the existing and the new parts together.

Column	Unit	Description
CAPEX	€	Investment cost of the new capacity of the component
Revenue	€/year	Yearly revenue of the component, e.g. selling hydrogen, oxygen, LOHC, heat or carbon black, or remuneration for taking waste
Operation cost	€/year	Yearly operating cost of the component, e.g. grid connection fee, water, LOHC purchase, local hydrogen or heat bought from outside the hub, tar treatment and CO_2 emissions

Column	Unit	Description
Degradation cost	€/year	Yearly cost of degradation (wear) of the component

Notes:

- **Electricity supply (purchase - sale)** is the yearly cost of electricity bought from the grid minus revenue from electricity sold to the grid. It is negative if more electricity is sold than bought.
- **Battery (storage and converter)** combines the battery storage and its power converter.
- Waste treatment results are split into **Waste treatment plant** (all reactors), **Waste source** (remuneration for taking waste) and **Waste storage**.
- Maintenance costs are not included in this table.

5.14 Economic viability assessment

When evaluating the results, consider the following key indicators:

5.14.1 Primary decision metrics

1. **Payback Period**: How long until the investment pays for itself.
2. **NPV**: Whether the investment creates value
    - Positive NPV: Investment creates value
    - Negative NPV: Investment destroys value
3. **IRR**: The effective return on investment
    - Compare against your required rate of return or cost of capital

5.14.2 Secondary considerations

- **LCOE/LCOH**: Compare against current market prices
- **Capacity factors**: Higher utilization generally means better returns
- **Green hydrogen percentage**: Important for sustainability goals and green hydrogen certifications

5.15 Exporting results

Results can be exported to an HTML file by clicking the **Export results to HTML** button at the top of the Results window. A dialog opens where the name and location of the file are chosen; by default, the file is saved in the project folder and named as the scenario.

The exported file can be opened in any web browser and contains:

- all result tables shown in the Results window (charts are not exported)
- **User inputs** – the input parameters of all components selected in the scenario

Version history

2026-11-09 v1.1.0 build 002

2026-11-09 v1.1.0 build 002 - feature: Leonitus wave energy converter (Leonitus WEC) replaces the classic wave energy converter; it is sized by installed power and priced by the wave potential of the site - feature: power consumption of the compressor before the H2 tank (HCOMP1), of the H2 sales compressor (HCOMP2, per customer) and of the O2 sales compressor (OCOMP) is calculated instead of entered - feature: compressors keep only the minimum inlet and maximum outlet pressure - feature: battery lifetime input in BESS, separate from the power converter - feature: “Restore defaults” buttons in waste heat sections - feature: HTML report exports all inputs, the system diagram, profiles and charts; chart data can be downloaded as CSV - feature: the system diagram is shown on the results page - feature: rename dialogs are preloaded with the current name - bug: PVGIS and renewables.ninja profiles converted to local time; fixed time of plots - bug: H2 sales compressor tooltips were offset by one row - bug: waste heat error - bug: wind farm error

2026-11-02 v1.0.0 build 001

2026-11-02 v1.0.0 build 001 - feature: waste treatment plant (WT) and waste source and storage merged from NACHIP-dev; hydrogen from waste is non-green and goes to the low-pressure hydrogen bus - feature: LOHC export is available with the electrolyser or WT - feature: alternative H2 products (Alt. H2) removed - feature: LOHC components (hydrogenation, LOHC sales, LOHC purchase, dehydrogenation, LOHC tank) and fixed hydrogen consumption included in the optimization and in the results - feature: low-pressure hydrogen bus; hydrogen tank and its compressor moved to a new Hydrogen storage header, available also without the electrolyser - feature: fuel cells can take hydrogen from the hydrogen tank or directly from the hydrogen bus - feature: LCOH includes hydrogen sold as LOHC and hydrogen used locally - bug: yearly limit of non-green hydrogen sales customers caused an error - bug: heat from fuel cells was overestimated with more than one stack - feature: fixed hydrogen consumption (H2 cons.) parameters: demand generated from a weekly pattern or imported, green-only option, pressure level, required pressure and price of hydrogen bought from outside; not yet included in the optimization - feature: fixed hydrogen consumption (H2 cons.) checkbox added; parameters and optimization to follow - feature: component checkboxes arranged in rows: “Purely electrical”, “H2 sources”, “H2 usage” and “H2 byproducts” - feature: H2 cons., H2

sales and Fuel cell are available with the Electrolyser or LOHC import; Waste heat is available with the Electrolyser or Fuel cell - bug: renaming a customer in H2 sales and O2 sales no longer disables the right-click menus - feature: LOHC components (LOHC sales with hydrogenation unit, LOHC purchase with dehydrogenation unit, LOHC tank) added to the interface; not yet included in the optimization - bug: default price for water tank - bug: water tank financials - bug: fixed excessive memory use in heat modelling - feature: RESu can be used when electricity import is not possible (import from the supplier is blocked) - new solver parameters for faster solving - improved preprocessing and postprocessing times - based on HyEfRe 2026-06-11 v0.3.2 build 011

References

Date	Title	Type	Details
02/2025	Joint optimal sizing and operation scheduling of a power-to-gas hub based on a linear program	Journal article	*Applied Energy*
06/2023	Aggregated Representation of Electric Vehicles Population on Charging Points for Demand Response Scheduling	Journal article	*IEEE Transactions on Intelligent Transportation Systems*
06/2023	Photovoltaic and Wind Power Plants Production Profiles Generation from Scarce Data	Conference paper	*2023 IEEE ISIE*
09/2020	Parameterization of a Photovoltaic and a Battery System Add-On for a Consumer Based on a Sequential Linear Program	Journal article	*Engineering Power – Bulletin of the Croatian Academy of Engineering*
09/2020	Optimal parameterization of a PV and a battery system add-on for a consumer	Conference paper	*2020 IEEE PEDG*
09/2020	Price-optimal Electrical and Thermal Energy Flow Control within Microgrid – Smart Grid Interaction	Conference paper	*2020 IEEE PEDG*

García, A., P. Marín, and S. Ordóñez. 2024. "Hydrogenation of Liquid Organic Hydrogen Carriers: Process Scale-up, Economic Analysis and Optimization." *International Journal of Hydrogen Energy* 52: 1113–23.